

Mathematical Notation Catalogue for Analysis/FabiusFunction

Canonical symbols, exact definitions, local namespaces, and historical aliases

Encyclopedic companion to Vladimir Reshetnikov’s ProveIt formalization

Notation contract 2026-08-31

Active corpus at the validation snapshot: 162 recursively inventoried `.tex` sources

Abstract

This catalogue is the normative dictionary for mathematical notation in the active Analysis/FabiusFunction documentation corpus. It distinguishes the bounded Fabius function, the signed global extension, and Rvachev’s up-function; fixes number-system, probability, transform, binary, q -series, asymptotic, fractional-calculus, approximation, and operator conventions; and records every recurring collision discovered by a recursive audit of the current 162-source active `.tex` corpus. Each entry states its type, domain, codomain, indices, parameters, normalization, exceptional values, and formalization status. No phrase such as “with the usual convention” supplies missing data: when two usual conventions exist, both are defined and their conversion is printed. Historical papers and frontier reports remain mathematically interpretable through explicit alias and local-namespace crosswalks rather than through overloaded bare letters.

Normative scope. The command definitions in `docs/fabius-notation.tex`, version 2026-08-31, are normative for active project-authored LaTeX. This catalogue defines their mathematical meaning. The directory `docs/archive/` is not an active migration target and is used only as provenance. A historical source under `docs/papers/` may display its original notation only inside an explicit historical scope. A source-faithful whole-paper transcription opens that scope with the exact preamble marker `HISTORICAL-NOTATION-SCOPE`; a shorter quotation labels the historical convention at the quotation itself. The marker never exempts surrounding project prose, which uses the canonical symbols. Included tables and generated fragments inherit the namespace of their standalone parent document.

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1 Scope contract and how to use this catalogue

1.1 What counts as notation

The inventory includes a glyph or compound expression whenever its interpretation requires one or more mathematical choices: a domain, codomain, index set, endpoint rule, transform normalization, branch, sign, scaling, convergence mode, or proof-status boundary. Thus it includes not only named functions but also interval brackets, empty sums, 0^0 , arrows of convergence, subscripts on transforms, and superscripts denoting derivatives or powers. Typography that changes no mathematical meaning—for example the insertion of braces in $\mathbb{R}$ —is not a separate notation.

The catalogue has two levels.

1. A *global semantic symbol* has one meaning throughout the active corpus and, where available, one command in `fabius-notation.tex`.
2. A *local symbol* is scoped to a definition, theorem, table, or explicitly named document part. A local symbol must be declared where its scope begins and must not be cited outside that scope without qualification.

The local level is essential: a corpus this broad legitimately uses A , D , I , P , Q , and T for many unrelated temporary objects. Unification means making those scopes and meanings explicit, not forcing every matrix and polynomial to share a single name.

1.2 The no-implicit-parts checklist

Every new or revised notation declaration answers all applicable questions below.

1. What mathematical object is denoted: scalar, set, map, measure, sequence, matrix, operator, equivalence class, or proposition?
2. What are its domain and codomain? If it is indexed, what is the complete index set?
3. What is the exact defining formula, including all summation and product bounds?
4. Which parameters are fixed, which vary, and what inequalities or nonvanishing conditions do they satisfy?
5. Which normalization, sign, Fourier frequency, logarithm, complex-power branch, and measure convention is used?
6. What happens at endpoints, at an empty index set, at zero, and at a removable singularity?
7. Does equality mean pointwise equality, equality almost everywhere, equality of measures, equality in distribution, or asymptotic equivalence?
8. Is the assertion formally proved, a standard definition, derived only in a frontier report, conjectural, refuted, numerical, or historical?

1.3 Status vocabulary

Status	Exact meaning in this catalogue
Definition	A definition or typography contract; no theorem is asserted.
Lean-proved	The stated mathematical content has a named proved Lean declaration in the module identified by the crosswalk.
Classical	A standard mathematical convention used by the prose; its presence in the catalogue is not a claim that the project has formalized the general theorem.

Status	Exact meaning in this catalogue
Frontier-derived	Derived in a research-frontier document but not certified here as having an exact current Lean counterpart.
Conjectural	Proposed and explicitly unproved.
Refuted	A historical formula or normalization for which the project records a counterexample or negative theorem.
Historical alias	A symbol needed to read a source artifact; it is not canonical for new project prose.

2 Canonical quick reference

The first column is the literal shared command name. The second column is its rendering; the third fixes its semantics. Commands taking arguments show metavariables in braces.

Shared command	Rendering	Normative meaning
<code>\FabiusNotationVersion</code>	2026-08-31	Date-valued version of the shared notation contract; it is metadata, not a mathematical variable.
<code>\NaturalNumbers</code>	$\mathbb{N}_0$	Nonnegative integers $\{0, 1, 2, \dots\}$.
<code>\PositiveIntegers</code>	$\mathbb{N}_{>0}$	Strictly positive integers $\{1, 2, 3, \dots\}$.
<code>\IntegerNumbers</code>	$\mathbb{Z}$	Integers.
<code>\RationalNumbers</code>	$\mathbb{Q}$	Rational numbers.
<code>\RealNumbers</code>	$\mathbb{R}$	Real numbers.
<code>\ComplexNumbers</code>	$\mathbb{C}$	Complex numbers.
<code>\ExtendedRealNumbers</code>	$\overline{\mathbb{R}}$	Extended reals $\mathbb{R} \cup \{-\infty, +\infty\}$.
<code>\FabiusBounded</code>	F_{bd}	Bounded, CDF-normalized Fabius function.
<code>\FabiusGlobal</code>	F_{gl}	Signed global Fabius extension.
<code>\FabiusQuantile</code>	Q_F	Ordinary inverse of $F_{\text{bd}} _{[0,1]}$.
<code>\FabiusClampedQuantile</code>	$Q_{F,\text{cl}}$	Totalized clamped inverse on all real inputs.
<code>\RvachevUp</code>	up	Rvachev's even compactly supported up-function.
<code>\EulerE</code>	e	Base of the natural logarithm.
<code>\ImaginaryUnit</code>	i	The complex number satisfying $i^2 = -1$.
<code>\Differential</code>	dx	Upright differential in an integral or differential form.
<code>\AbsoluteValue{x}</code>	$ x $	Absolute value or, when a field is explicitly complex, complex modulus.
<code>\Norm{x}</code>	$\ x\ $	The norm supplied by the explicitly named normed space.
<code>\Floor{x}</code>	$\lfloor x \rfloor$	Greatest integer not exceeding x .
<code>\Ceiling{x}</code>	$\lceil x \rceil$	Least integer not below x .
<code>\FractionalPart{x}</code>	$\{x\}$	$x - \lfloor x \rfloor$ for $x \in \mathbb{R}$.
<code>\PositivePart{x}</code>	$(x)_+$	$\max\{x, 0\}$ for an ordered scalar x .
<code>\IndicatorOf{A}</code>	$\mathbf{1}_A$	Indicator of a set A , equal to one on A and zero off A .
<code>\SetOf{x:P(x)}</code>	$\{x : P(x)\}$	Delimiters for a set specified by membership data or a predicate; the ambient type is stated locally.
<code>\InnerProduct{x,y}</code>	$\langle x, y \rangle$	Inner-product delimiters; the scalar field, space, and linear-slot convention are stated locally.
<code>\CardinalityOf{A}</code>	$ A $	Finite cardinality or cardinal number according to the declared type; never measure or absolute value.
<code>\DefinitionEquals</code>	$a := b$	Definitional assignment: the left-hand symbol is introduced to mean exactly the right-hand object.
<code>\Sign</code>	sgn	Real sign function with $\text{sgn}(0) = 0$.
<code>\IdentityOperator</code>	id	Identity map on the domain stated at the point of use.
<code>\SpanOperator</code>	span	Linear span over the stated scalar field.
<code>\TraceOperator</code>	tr	Matrix or trace-class operator trace.
<code>\RankOperator</code>	rank	Dimension of an image, over the stated field.
<code>\DiagonalOperator</code>	diag	Diagonal matrix/operator constructor.
<code>\ResidueOperator</code>	Res	Complex residue at a stated point.

Shared command	Rendering	Normative meaning
<code>\LeastCommonMultiple</code>	lcm	Nonnegative least common multiple; finite-family and empty-family rules must be stated locally.
<code>\OrderOperator</code>	ord	An order invariant whose species (vanishing, pole, multiplicative, group-element, or recurrence order) is printed.
<code>\PrincipalLogarithm</code>	Log	Principal complex logarithm with argument in $(-\pi, \pi]$ and domain $\mathbb{C} \setminus (-\infty, 0]$.
<code>\FallingFactorial{x}{n}</code>	$(x)^{\overline{n}}$	Falling factorial $\prod_{j=0}^{n-1} (x - j)$, equal to one when $n = 0$.
<code>\RisingFactorial{x}{n}</code>	$(x)^{\overline{n}}$	Rising factorial $\prod_{j=0}^{n-1} (x + j)$, equal to one when $n = 0$.
<code>\SupportOperator</code>	supp	Pointwise-support operator token; its operand and ambient space must follow.
<code>\SupportOf{f}</code>	$\text{supp}(f)$	Pointwise support $\{x : f(x) \neq 0\}$.
<code>\TopologicalSupportOf{f}</code>	$\text{tsupp}(f)$	Closure of the pointwise support.
<code>\Convolution</code>	$f * g$	Convolution, with the underlying group and measure stated locally.
<code>\MetricDistance</code>	$\text{dist}(x, y)$	Distance of the named metric space.
<code>\TotalVariationOf{a}</code>	$\text{TV}(a)$	Total variation of one typed function, signed measure, or other locally declared object.
<code>\TotalVariationDistance</code>	$d_{\text{TV}}(\mu, \nu)$	Total variation distance with the probability convention $\sup_A \mu(A) - \nu(A) $.
<code>\Probability</code>	$\mathbb{P}$	Probability measure.
<code>\Expectation</code>	$\mathbb{E}$	Expectation with respect to the stated law.
<code>\Variance</code>	Var	Variance $\mathbb{E}[(X - \mathbb{E}X)^2]$.
<code>\Covariance</code>	Cov	Covariance of explicitly stated random variables.
<code>\LawOperator</code>	Law	Pushforward-law operator token; it is not a metric, density, or samplewise map.
<code>\LawOf{X}</code>	$\text{Law}(X)$	Pushforward probability law of X .
<code>\EqualInLaw</code>	$X \stackrel{\text{law}}{=} Y$	Equality of probability laws, not pointwise equality.
<code>\ConvergesInLaw</code>	$X_n \xrightarrow{\text{law}} X$	Weak convergence of laws.
<code>\DyadicSigmaField</code>	$\mathcal{A}_n$	Level- n dyadic filtration σ -field; the generating random coordinates and base probability space are printed.
<code>\DecayOptimizationObjective{n}</code>	$\text{Obj}_{\text{decay}, n}$	Exact level- n nonlinear objective in the Fourier-decay saddle problem, with variables and feasible set declared locally.
<code>\LinearizedDecayObjective{n}</code>	$\text{Obj}_{\text{decay}, n}^{\text{lin}}$	The explicitly linearized level- n decay objective; it is not identified with the nonlinear objective without an equality theorem.
<code>\FourierTwoPiOperator</code>	$\mathcal{F}_{2\pi}$	Forward Fourier operator with kernel $e^{-2\pi i x \xi}$.
<code>\FourierAngularOperator</code>	$\mathcal{F}_{\text{ang}}$	Forward Fourier operator with kernel $e^{-i x \omega}$.
<code>\InverseFourierTwoPiOperator</code>	$\mathcal{F}_{2\pi}^{-1}$	Inverse of $\mathcal{F}_{2\pi}$ on a named inversion domain.
<code>\InverseFourierAngularOperator</code>	$\mathcal{F}_{\text{ang}}^{-1}$	Inverse of $\mathcal{F}_{\text{ang}}$, including the $1/(2\pi)$ measure factor.
<code>\FourierTwoPiTransformOf{f}</code>	$\mathcal{F}_{2\pi}[f]$	Operator-application form of the 2π -frequency Fourier transform.
<code>\FourierAngularTransformOf{f}</code>	$\mathcal{F}_{\text{ang}}[f]$	Operator-application form of the angular-frequency Fourier transform.
<code>\FourierTwoPi{f}</code>	$\widehat{f}_{2\pi}$	Cycles-per-unit Fourier transform with kernel $e^{-2\pi i x \xi}$.
<code>\FourierAngular{f}</code>	$\widehat{f}_{\text{ang}}$	Angular-frequency Fourier transform with kernel $e^{-i x \omega}$.

Shared command	Rendering	Normative meaning
<code>\FourierSeriesCoefficient{h}{k}</code>	$\widehat{h}_{\text{FS}}[k]$	The k th coefficient of a one-periodic function, with kernel $e^{-2\pi i k x}$ on $[0, 1]$.
<code>\FourierSeriesCoefficientAtPeriod{h}{k}{T}</code>	$\widehat{h}_{\text{FS}; T}[k]$	The normalized k th coefficient of an explicitly T -periodic function; the period is part of the symbol.
<code>\LaplaceTransformSymbol</code>	$\mathcal{L}$	One-sided Laplace operator token unless a bilateral subscript is printed.
<code>\MellinTransformSymbol</code>	$\mathcal{M}$	Mellin-transform operator token on a stated strip of convergence.
<code>\LaplaceTransformOf{f}</code>	$\mathcal{L}[f]$	Laplace transform; one-sided or bilateral domain must be printed.
<code>\MellinTransformOf{f}</code>	$\mathcal{M}[f]$	Mellin transform $\int_0^\infty x^{s-1} f(x) \, dx$ on its stated strip.
<code>\MomentGeneratingFunction{X}</code>	M_X	$z \mapsto \mathbb{E}[e^{zX}]$ on its stated domain.
<code>\CharacteristicFunction{X}</code>	φ_X	$t \mapsto \mathbb{E}[e^{itX}]$; a declared probability measure μ may replace X , meaning its positive-sign Fourier–Stieltjes transform.
<code>\SincRad</code>	sinc_{rad}	Radian sinc, $\sin t/t$, totalized to one at zero.
<code>\SincPi</code>	sinc_π	Normalized sinc, $\sin(\pi x)/(\pi x)$, totalized to one at zero.
<code>\BinaryDigitSum</code>	wt_2	Number of one-bits in a nonnegative integer.
<code>\ThueMorseSignSymbol</code>	ε	The symbol reserved for the Thue–Morse sign sequence; use the argument-taking command in formulas.
<code>\ThueMorseSign{n}</code>	ε_n	$(-1)^{\text{wt}_2(n)}$.
<code>\PadicValuation{p}</code>	ν_p	p -adic valuation; the argument and zero convention are stated locally.
<code>\TwoAdicValuation</code>	ν_2	Two-adic valuation.
<code>\QPochhammer{a}{q}{n}</code>	$(a; q)_n$	Finite or infinite q -shifted factorial with explicit base and length.
<code>\GaussianBinomial{n}{k}{q}</code>	$\begin{bmatrix} n \\ k \end{bmatrix}_q$	Gaussian binomial coefficient with explicit base.
<code>\QInteger{n}{q}</code>	$[n]_q$	q -integer with explicit base.
<code>\Polylogarithm</code>	Li	Polylogarithm with order supplied as a subscript.
<code>\LambertW</code>	W	Lambert W ; a branch subscript is mandatory when more than one branch is possible.
<code>\GammaFunction</code>	Γ	Euler Gamma function.
<code>\RiemannZeta</code>	ζ	Riemann zeta function or its meromorphic continuation, as stated.
<code>\BigO (operator token; 0 TeX arguments)</code>	$\mathcal{O}g$	Big- $\mathcal{O}$ applied to the following mathematical operand g , in a limit declared in the same sentence.
<code>\LittleO (operator token; 0 TeX arguments)</code>	og	Little- o applied to the following mathematical operand g , in a limit declared in the same sentence.
<code>\BigOAt{g}{a}</code>	$\mathcal{O}_a(g)$	Big- $\mathcal{O}$ with limiting regime a printed.
<code>\LittleOAt{g}{a}</code>	$o_a(g)$	Little- o with limiting regime a printed.

3 Foundational mathematical grammar

Shared notation contract version

Command. `\FabiusNotationVersion` expands to the ISO date 2026-08-31.

Meaning. It identifies the semantic API described by this catalogue.

Contract file. The corresponding source is `docs/fabius-notation.tex`. The version is not a publication date, source revision, theorem date, numeric parameter, or claim that every historical source was authored on that date.

Update rule. Any addition, removal, arity change, or mathematical change to a shared command updates this date and the quick reference, alphabetical index, rendered symbol index, retired-alias mapping, and relevant detailed entry in the same change.

The exponential base and imaginary unit

Exponential base. $e \in \mathbb{R}$ is the unique positive number with $\log e = 1$. The expression e^z is the complex exponential for $z \in \mathbb{C}$, not a local error variable raised to a power.

Imaginary unit. $i \in \mathbb{C}$ is the root of $z^2 = -1$ with positive imaginary part. Thus $i^2 = -1$ and $\bar{i} = -i$. The other root is $-i$.

Typography. Both constants are upright. An italic e or i remains a local variable or index and acquires no constant meaning from context alone.

Number systems and inclusions

Canonical symbols. $\mathbb{N}_0 = \{0, 1, 2, \dots\}$, $\mathbb{N}_{>0} = \{1, 2, 3, \dots\}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$, and $\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, +\infty\}$.

Inclusions. $\mathbb{N}_0 \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$. These are the standard injective embeddings; a formula involving coercion into a larger system is interpreted only after the target has been named.

Endpoint convention. Neither $+\infty$ nor $-\infty$ is a real number. They belong only to $\overline{\mathbb{R}}$. Arithmetic involving them follows extended-order conventions only when explicitly stated.

Retired ambiguity. A bare $\mathbb{N}$ is forbidden in normative prose because authors disagree about whether it contains zero.

Default metavariables and mandatory overrides

Discrete indices. Unless an entry says otherwise, $j, k, m, n, r, N \in \mathbb{N}_0$. A statement requiring positivity writes $n \in \mathbb{N}_{>0}$ or $n \geq 1$. The letter p denotes a prime only when the same sentence says “prime”.

Real variables. The letters $x, y, t, u, s, \xi, \omega$ have no global type. Each definition declares whether they lie in $\mathbb{R}$, $\mathbb{C}$, an interval, or a measure space.

Parameters. The letters $a, b, q, \alpha, \beta, \lambda$ are never assigned a default domain. Their admissible set and all inequalities, such as $|q| < 1$ or $\alpha > 0$, are part of the declaration.

Functions and measures. Lower-case f, g, h denote local functions only after a domain and codomain are supplied. Greek μ, ν denote local measures only after a measurable space is supplied. No letter acquires mathematical meaning merely from its font.

Definition, equality, equivalence, and identification

Definition. The symbol $a := b$ introduces a as an abbreviation or new object whose value is exactly b . It is not an asymptotic statement.

Equality. The symbol $=$ means equality in the ambient type: equality of scalars, functions, sets, measures, matrices, or equivalence classes as determined by the declared type. Function equality is extensional; equality almost everywhere is written separately.

Almost-everywhere equality. For a measure μ , $f = g$ μ -a.e. means $\mu(\{x : f(x) \neq g(x)\}) = 0$. It does not license substitution into a pointwise endpoint formula without a continuous-representative theorem.

Law equality. For random variables X and Y , $X \stackrel{\text{law}}{=} Y$ abbreviates $\text{Law}(X) = \text{Law}(Y)$ and does not imply $X(\omega) = Y(\omega)$ on a shared sample space.

Isomorphism. Symbols $\cong$ and $\simeq$ require the category—sets, groups, rings, topological spaces, normed spaces, or probability spaces—to be stated locally.

Sets, set operations, and intervals

Set grammar. For a universe U , $x \in A$ means membership in $A \subseteq U$; $A \subseteq B$ permits equality and $A \subsetneq B$ does not. Union, intersection, complement, and set difference are $A \cup B$, $A \cap B$, $U \setminus A$, and $A \setminus B$.

Intervals. For $a, b \in \mathbb{R}$ with $a \leq b$, $[a, b] = \{x : a \leq x \leq b\}$, $(a, b) = \{x : a < x < b\}$, $[a, b) = \{x : a \leq x < b\}$, and $(a, b] = \{x : a < x \leq b\}$.

Empty reversed intervals. If $b < a$, each ordered interval above is empty. An oriented integral $\int_a^b$ is not an integral over that empty set; it changes sign under endpoint reversal.

Lean aliases. The Lean set constructors are `Set . Icc`, `Set . Ioo`, `Set . Ico`, and `Set . Ioc`, in the same order as the four printed intervals.

Maps, restrictions, composition, images, and preimages

Typed map. The declaration $f : A \rightarrow B$ includes domain A and codomain B . The restriction to $S \subseteq A$ is $f|_S : S \rightarrow B$.

Composition. For $f : A \rightarrow B$ and $g : B \rightarrow C$, $(g \circ f)(x) = g(f(x))$. Composition is evaluated right to left.

Identity map. For a declared set or type X , $\text{id}_X : X \rightarrow X$ is the map $x \mapsto x$. The zero-argument command `\IdentityOperator` supplies only the operator token; the domain subscript X is source notation and must be printed whenever the surrounding typed equation does not determine it uniquely. Thus $f \circ \text{id}_X = f$ and $\text{id}_Y \circ f = f$ for every $f : X \rightarrow Y$.

Image and preimage. For $S \subseteq A$ and $T \subseteq B$, $f(S) = \{f(x) : x \in S\}$ and $f^{-1}(T) = \{x \in A : f(x) \in T\}$. The latter does not require f to be invertible.

Inverse map. The symbol $f^{-1} : B \rightarrow A$ denotes an inverse function only after f has been declared bijective. A superscript -1 on a set image is always the preimage, not reciprocal arithmetic.

Set, inner-product, cardinality, and definition delimiters

Set constructor. The command `\SetOf{body}` supplies only extensible braces. The body supplies either an explicit finite list or a typed predicate, for example $\{n \in \mathbb{N}_0 : n < N\}$. A colon and a vertical bar have the same set-builder meaning only after the ambient type has been printed. The command does not deduplicate a list, choose a universe, or turn a class into a set.

Inner product. The command `\InnerProduct {x,y}` supplies angle delimiters. In this catalogue $\langle x, y \rangle_H$ is linear in x and conjugate-linear in y on the complex Hilbert space H ; a source adopting the opposite convention must say so at the first use and conjugate all adjoint formulas consistently. A duality pairing is instead labelled $\langle \phi, x \rangle_{X^*, X}$.

Cardinality. For a finite set A , $|A| \in \mathbb{N}_0$ is its number of elements and $|\emptyset| = 0$. For an infinite set it may denote a cardinal number only when the ambient set theory is explicit. Length, Lebesgue measure, probability, determinant, absolute value, and norm are never inferred from the same vertical bars.

Definitional equality. $a := b$ introduces the symbol a with type and value supplied by b . Reversing the direction uses $b =: a$; a theorem already requiring proof uses ordinary equality. Definitional equality in prose is not a claim of Lean kernel reduction unless an exact Lean definition is cited.

Empty and singleton cases. $\{\} = \emptyset$ when the body is an explicit empty list, whereas $\{x\}$ is a singleton. A displayed set-builder with an unsatisfiable predicate is empty by its predicate, not by typography.

Sequences, families, ranges, and finite indexing

Sequence. A sequence $(a_n)_{n \in \mathbb{N}_0}$ is a map $a : \mathbb{N}_0 \rightarrow A$ into a declared codomain A . A finite row $(a_{n,k})_{0 \leq k \leq n}$ is defined only for the printed triangular range.

Half-open range. The project and Lean frequently use $j < n$ to mean $j \in \{0, 1, \dots, n-1\}$. Thus $\sum_{j < n}$ and $\prod_{j < n}$ are half-open at n .

Inclusive range. The notation $0 \leq j \leq n$ includes both endpoints and has $n+1$ indices. It is never silently identified with $j < n$.

Indexed family. For a possibly infinite set I , $(a_i)_{i \in I}$ is a family indexed by I . Summability over I is unconditional unless an order is separately specified.

Sums, products, and neutral-element conventions

Finite operations. For a finite set S , $\sum_{i \in S} a_i$ and $\prod_{i \in S} a_i$ are taken in the stated additive and multiplicative structures.

Empty cases. The empty sum is 0 and the empty product is 1. Consequently $\sum_{j < n}$ is zero and $\prod_{j < n}$ is one when $n = 0$.

Infinite series. The notation $\sum_{n=0}^{\infty} a_n$ includes an ordering by $\mathbb{N}_0$. For a family over an arbitrary type, the project uses unconditional summability; rearrangement requires absolute or unconditional convergence.

Infinite products. The product $\prod_{n=0}^{\infty} a_n$ is the limit of the ordered partial products $\prod_{n < N} a_n$ as $N \rightarrow \infty$. Local uniform convergence is stated when the factors depend on a complex variable.

Powers, factorials, and binomial coefficients

Natural powers. For an element x of a monoid and $n \in \mathbb{N}_0$, $x^0 = 1$ and $x^{n+1} = x^n x$. In particular, the natural-power convention is $0^0 = 1$.

Integer powers. For $n \in \mathbb{Z}$, x^n requires an invertible nonzero x when $n < 0$; the ambient field may totalize 0^{-1} algebraically, but an analytic formula never relies on that totalization without saying so.

Complex powers. For z^α with noninteger α , the chosen branch of $\log z$ and its cut are mandatory. The principal convention is $z^\alpha = \exp(\alpha \operatorname{Log} z)$ with $\arg z \in (-\pi, \pi]$.

Factorial. For $n \in \mathbb{N}_0$, $0! = 1$ and $n! = \prod_{1 \leq j \leq n} j$. The odd double factorial satisfies $(-1)!! = 1$ and $(2n+1)!! = \prod_{j=0}^n (2j+1)$.

Binomial coefficient. For $n, k \in \mathbb{N}_0$, $\binom{n}{k} = n!/(k!(n-k)!)$ when $k \leq n$ and $\binom{n}{k} = 0$ when $k > n$.

Falling factorial. For a scalar x and $n \in \mathbb{N}_0$, $(x)^{\underline{n}} := \prod_{j=0}^{n-1} (x-j)$; the empty product gives $(x)^{\underline{0}} = 1$. Thus $(x)^{\underline{n}} = \Gamma(x+1)/\Gamma(x-n+1)$ only where the quotient is defined or after a separately justified continuation.

Rising factorial. For a scalar x and $n \in \mathbb{N}_0$, $(x)^{\overline{n}} := \prod_{j=0}^{n-1} (x+j)$ and $(x)^{\overline{0}} = 1$. It equals the ordinary Pochhammer symbol $(x)_n$ in this catalogue; the base- q analogue is always `\QPochhammer` and never shares this two-argument syntax.

The principal logarithm and order operators

Principal logarithm. The command `\PrincipalLogarithm` denotes the analytic branch

$$\operatorname{Log} z = \log |z| + i \operatorname{Arg} z, \quad -\pi < \operatorname{Arg} z \leq \pi,$$

on $\mathbb{C} \setminus (-\infty, 0]$. In particular it is undefined at zero and has distinct boundary values on the two sides of the negative real axis. The real symbol $\log x$ for $x > 0$ is its restriction and carries no branch ambiguity.

Complex powers. The principal power is $z^\alpha = \exp(\alpha \operatorname{Log} z)$ on the same cut domain. Algebraic laws such as $(zw)^\alpha = z^\alpha w^\alpha$ are not used across a branch crossing without an explicit argument calculation.

Vanishing order. For a nonzero holomorphic f near z_0 , $\operatorname{ord}_{z_0}(f) = m \in \mathbb{N}_0$ when $f(z) = (z - z_0)^m g(z)$ with $g(z_0) \neq 0$. For a pole, a signed divisor order is written $\operatorname{ord}_{z_0}^{\operatorname{div}}(f) < 0$ so that it cannot be mistaken for the nonnegative zero order.

Multiplicative order. For a group element a , put $S_G(a) = \{n \in \mathbb{N}_{>0} : a^n = e_G\}$. When this set is nonempty,

$$\operatorname{ord}_G(a) = \min S_G(a).$$

When it is empty, $\operatorname{ord}_G(a) = \infty$. The group subscript is mandatory whenever an analytic order also occurs.

Other orders. Differential-equation order, recurrence order, matrix order, and asymptotic truncation order are prose-qualified quantities. The shared operator provides typography, not permission to leave the species or base point implicit.

Absolute value, norms, and distance

Absolute value. For $x \in \mathbb{R}$, $|x| = \max\{x, -x\}$. For $z \in \mathbb{C}$, $|z| = \sqrt{z\bar{z}}$.

Norm. The expression $\|x\|$ is meaningful only after a normed space has been named. For a function, $\|f\|_\infty$ means an essential supremum or a pointwise supremum exactly as the surrounding declaration states.

Distance. In a metric space (X, d) , $\operatorname{dist}(x, y) = d(x, y)$. Probability metrics such as total variation and Wasserstein distance are given their own decorated symbols and never use a bare `dist` without the metric name.

Floor, ceiling, fractional part, sign, and positive part

Definitions. For $x \in \mathbb{R}$, $\lfloor x \rfloor = \max\{n \in \mathbb{Z} : n \leq x\}$, $\lceil x \rceil = \min\{n \in \mathbb{Z} : x \leq n\}$, $\{x\} = x - \lfloor x \rfloor \in [0, 1)$, $(x)_+ = \max\{x, 0\}$, and $\operatorname{sgn}(x) = -1, 0, 1$ according as $x < 0$, $x = 0$, or $x > 0$.

Nearest integer. A nearest-integer or nearest-dyadic operation must separately state its tie-breaking rule; neither floor nor ceiling implies one.

Use in powers. For $\alpha > 0$, $(x)_+^\alpha$ is zero when $x \leq 0$. For $\alpha = 0$, the natural-power value at $x \leq 0$ is one only if a formula explicitly uses natural exponentiation; real powers require a separate endpoint rule.

Indicator and half-weighted interval indicator

Ordinary indicator. For a set A in a universe U , $\mathbf{1}_A : U \rightarrow \{0, 1\}$ is one on A and zero on $U \setminus A$.

Half-weighted indicator. When adjacent closed intervals meet, the project uses a separately named half-endpoint indicator taking value $1/2$ at each boundary and 1 in the interior. It is not denoted by the ordinary $\mathbf{1}_A$.

Measure-theoretic caveat. Changing endpoint values does not change a Lebesgue integral, but it does change pointwise finite-spline identities and exact values at grid points.

Pointwise support, topological support, and essential support

Pointwise support. For a function $f : X \rightarrow A$ with zero in A , $\text{supp}(f) = \{x \in X : f(x) \neq 0\}$. The bare command `\SupportOperator` is the same operator token before an explicitly supplied operand; `\SupportOf` supplies the parentheses and is preferred in prose.

Topological support. For topological X , $\text{tsupp}(f) = \overline{\text{supp}(f)}$. Thus a boundary point can belong to the topological support even when f and all of its derivatives vanish there.

Measure support. For a Borel measure μ , $\text{supp } \mu$ is the complement of the largest open μ -null set. This is not obtained by evaluating a measure as a pointwise function.

Essential support. Essential support depends on a background measure and ignores null changes; it must be labeled “essential”.

Derivatives, iterated derivatives, and integrals

Derivative. For $f : \mathbb{R} \rightarrow E$ into a real normed space, $f'(x)$ denotes the Fréchet derivative evaluated on 1 when it exists. Higher derivatives are $f^{(n)}$, with $f^{(0)} = f$.

Partial derivatives. For a multivariable function, $\partial_j f$ differentiates in the j th coordinate. A gradient, Jacobian, Hessian, or directional derivative is named explicitly and includes its inner-product or coordinate convention.

Differential. In $\int_a^b f(x) \, dx$, the upright dx identifies the variable of integration and is not a free multiplicative variable.

Oriented interval integral. For integrable f , $\int_b^a f(x) \, dx = -\int_a^b f(x) \, dx$ and $\int_a^a f(x) \, dx = 0$.

Whole-space integral. An integral $\int_{\mathbb{R}}$ is with respect to Lebesgue measure unless another measure is printed. An expectation is written $\mathbb{E}$, not as an unlabeled integral.

Convolution of functions, measures, and sequences

Functions. For integrable functions $f, g : \mathbb{R} \rightarrow \mathbb{C}$, $(f * g)(x) = \int_{\mathbb{R}} f(x - y)g(y) \, dy$.

Measures. For finite Borel measures μ, ν on $\mathbb{R}$, $(\mu * \nu)(A) = \int \int \mathbf{1}_A(x + y) \mu(dx) \nu(dy)$. It is the law of $X + Y$ for independent X, Y with laws μ, ν .

Sequences. For sequences indexed by $\mathbb{N}_0$, the Cauchy convolution is $(a * b)_n = \sum_{k=0}^n a_k b_{n-k}$. It is distinct from bilateral convolution over $\mathbb{Z}$.

Power. The notation f^{*n} means n -fold convolution, with $f^{*0} = \delta_0$ for measures and the appropriate convolution unit for the ambient algebra.

4 The semantic Fabius–Rvachev objects

The bounded Fabius function F_{bd}

Command. `\FabiusBounded`.

Type. A smooth function $F_{\text{bd}} : \mathbb{R} \rightarrow [0, 1] \subset \mathbb{R}$.

Normalization. $F_{\text{bd}}(x) = 0$ for $x \leq 0$, $F_{\text{bd}}(x) = 1$ for $x \geq 1$, $F_{\text{bd}}(1 - x) = 1 - F_{\text{bd}}(x)$ for every real x , and $F'_{\text{bd}}(x) = 2F_{\text{bd}}(2x)$ on the left half of the unit interval.

Shape. Its restriction $F_{\text{bd}}|_{[0,1]}$ is continuous and strictly increasing from zero to one. Its constant tails are part of the object, not an informal extension.

Lean crosswalk. The real-valued object is `fabiusReal`; the abstract specification is `IsFabius`; construction and uniqueness are exposed through `fabius` and `fabius_spec`. Status: **Lean-proved**.

Forbidden aliases. The bare capital letter, the former calligraphic glyph, and the former decorated-capital glyph are not canonical because the active historical corpus also uses them for transforms, families, and the signed global extension. This catalogue names those aliases textually rather than re-emitting their retired control sequences.

Rvachev’s up-function up

Command. `\RvachevUp`.

Type and definition. The function $\text{up} : \mathbb{R} \rightarrow \mathbb{R}$ is defined by

$$\text{up}(x) = F_{\text{bd}}(1 - |x|).$$

This single formula uses the constant tails of F_{bd} and therefore gives $\text{up}(x) = 0$ whenever $|x| \geq 1$.

Normalization. It is even, nonnegative, smooth, has integral one, satisfies $\text{up}(0) = 1$, and has topological support $[-1, 1]$. Every derivative vanishes at $x = \pm 1$.

Differential law. For every $x \in \mathbb{R}$, $\text{up}'(x) = 2(\text{up}(2x + 1) - \text{up}(2x - 1))$.

Lean crosswalk. `rvachevUp` in `Basic.lean`, with differential, support, Fourier, and shape theorems in their dedicated modules. Status: **Lean-proved**.

Historical aliases. The source papers print an upright word “up”, use legacy commands named `\up` or `\Up`, or use the local glyph φ . These spellings are historical evidence, not commands to be re-emitted in project-authored formulas; normative prose uses only `up`.

The signed global extension F_{gl}

Command. `\FabiusGlobal`.

Type. A smooth function $F_{\text{gl}} : \mathbb{R} \rightarrow \mathbb{R}$. It is not a cumulative distribution function and is not constrained to $[0, 1]$.

Canonical translate formula. For $x \in \mathbb{R}$,

$$F_{\text{gl}}(x) = \sum_{n=0}^{\infty} \varepsilon_n \text{up}(x - 2n - 1).$$

At each fixed x , compact support leaves only finitely many nonzero summands; hence this formula is pointwise locally finite even when written as an infinite sum.

Identifications. For $x \in [0, 1]$, $F_{\text{gl}}(x) = F_{\text{bd}}(x)$. Globally, $F'_{\text{gl}}(x) = 2F_{\text{gl}}(2x)$. Repeated differentiation gives $F_{\text{gl}}^{(m)}(x) = 2^{\binom{m+1}{2}} F_{\text{gl}}(2^m x)$ for $m \in \mathbb{N}_0$.

Lean declaration. `extendedFabius`.

Lean module and status. Translate-cell and calculus results are in `GlobalExtension.lean`. Status: **Lean-proved**.

Retired aliases. The former calligraphic glyph, the former decorated-capital glyph, and the plain capital glyph of the 2017 arithmetic paper are historical aliases. They must be translated semantically, never by a blind global replacement.

The ordinary and clamped Fabius inverses

Commands. `\FabiusQuantile` and `\FabiusClampedQuantile`.

Ordinary inverse. Because $F_{\text{bd}}|_{[0,1]} : [0, 1] \rightarrow [0, 1]$ is a strictly increasing bijection,

$$Q_F : [0, 1] \rightarrow [0, 1], \quad Q_F(y) = (F_{\text{bd}}|_{[0,1]})^{-1}(y).$$

Thus $F_{\text{bd}}(Q_F(y)) = y$ for $y \in [0, 1]$ and $Q_F(F_{\text{bd}}(x)) = x$ for $x \in [0, 1]$.

Clamped inverse. For $y \in \mathbb{R}$, define $\text{clamp}(y) = \min\{1, \max\{0, y\}\}$ and

$$Q_{F,\text{cl}}(y) = Q_F(\text{clamp}(y)).$$

Consequently it is zero on $(-\infty, 0]$ and one on $[1, \infty)$.

Lean crosswalk. `fabiusInv` and `fabiusOrderIso` in `FabiusInverse.lean`. Status: **Lean-proved**.

Retired aliases. Bare I , G , J , the formal inverse expression, and a legacy inverse macro occur in older prose. They are local or historical only.

The abstract predicate `IsFabius`

Kind. A proposition about a candidate bounded function, not a fourth Fabius function.

Role. It packages the exact tails, reflection, smoothness/differentiability, and left-half differential law required by the formal development. A theorem quantified over an arbitrary f with `IsFabiusf` is more general than a theorem stated only for F_{bd} .

Authority. The Lean declaration's type is authoritative for its precise fields, coercions, and domains. A prose summary is not permission to omit a hypothesis.

Status. **Lean-proved**: existence supplies the canonical witness and uniqueness identifies every witness with it.

Canonical relations among the three functions

Fold. For every $x \in \mathbb{R}$, $\text{up}(x) = F_{\text{bd}}(1 - |x|)$.

Density relation. For $x \in \mathbb{R}$, $F'_{\text{bd}}(x) = 2 \text{up}(2x - 1)$. Both sides vanish outside the unit interval.

Unit-interval translate. For $x \in [0, 1]$, $F_{\text{bd}}(x) = \text{up}(x - 1)$. This is not a global identity.

Global agreement. For $x \in [0, 1]$, $F_{\text{gl}}(x) = F_{\text{bd}}(x)$. Outside that interval the left side is signed and the right side has constant tails.

Audit rule. Every use of a differential, translation, or scaling identity states whether its subject is F_{bd} , up , or F_{gl} and prints the domain on which it holds.

Dyadic points, reduced levels, and derivative cells

Dyadic point. A dyadic rational is $x = m/2^n$ with $m \in \mathbb{Z}$ and $n \in \mathbb{N}_0$. A unit-interval dyadic additionally has $0 \leq m \leq 2^n$.

Reduced positive level. For an interior dyadic, “reduced at level n ” means $0 < m < 2^n$ and m is odd. The oddness prevents the same rational from being represented at a smaller positive denominator level.

Cell. A level- n dyadic cell is an interval between consecutive multiples of 2^{-n} ; open, closed, or half-open endpoints are printed because derivative and spline formulas can distinguish them.

Derivative scaling. At a reduced interior dyadic $m/2^n$, derivatives above a finite order vanish according to the exact global-scaling theorem. The catalogue does not infer local analyticity from the terminating Taylor jet.

Representation independence. An exact evaluator accepting (m, n) must prove invariance under $(m, n) \mapsto (2m, n + 1)$ rather than assuming a reduced input.

5 Probability, measures, and distributional notation

Probability spaces and random variables

Probability space. $(\Omega, \mathcal{A}, \mathbb{P})$ denotes a probability space: Ω is the sample set, $\mathcal{A}$ is a σ -algebra on Ω , and $\mathbb{P} : \mathcal{A} \rightarrow [0, 1]$ is countably additive with $\mathbb{P}(\Omega) = 1$.

Random variable. A real random variable is an $\mathcal{A}/\mathcal{B}(\mathbb{R})$ -measurable map $X : \Omega \rightarrow \mathbb{R}$. A random vector $X : \Omega \rightarrow \mathbb{R}^d$ is measurable for the Borel σ -algebra of $\mathbb{R}^d$.

Events. Braces inside probability are inverse images: $\mathbb{P}\{X \leq x\}$ abbreviates $\mathbb{P}(X^{-1}((-\infty, x]))$. They do not imply that X has a density.

Almost surely. $X = Y$ almost surely, written $X = Y$ a.s., means $\mathbb{P}\{\omega : X(\omega) = Y(\omega)\} = 1$. This is weaker than pointwise equality and different from [equality in law](#).

Independence. Random variables $(X_j)_{j \in J}$ are independent when

$$\mathbb{P}\left(\bigcap_{j \in K} \{X_j \in B_j\}\right) = \prod_{j \in K} \mathbb{P}\{X_j \in B_j\}$$

for every finite $K \subseteq J$ and every family of Borel sets B_j . *i.i.d.* means independent and identically distributed.

Law, pushforward, equality in law, and weak convergence

Law. For a measurable $X : \Omega \rightarrow S$,

$$\text{Law}(X) := \text{Law}(X) := X_{\#}\mathbb{P}, \quad \text{Law}(X)(B) = \mathbb{P}(X^{-1}(B)),$$

where S is a measurable space, B is measurable, and $X_{\#}$ denotes pushforward. Thus `\LawOperator` is the operator token and `\LawOf` is its argument-delimited form; both take a random variable and return a probability measure.

Equality in law. $X \stackrel{\text{law}}{=} Y$ means $\text{Law}(X) = \text{Law}(Y)$. It neither asserts that X and Y share a sample space nor that they are equal almost surely.

Convergence in law. $X_n \xrightarrow{\text{law}} X$ means

$$\int_S h \, d\text{Law}(X_n) \longrightarrow \int_S h \, d\text{Law}(X)$$

for every bounded continuous $h : S \rightarrow \mathbb{R}$, with S a metric space. For real-valued variables this is equivalent to convergence of distribution functions at every continuity point of the limiting distribution function.

Measure notation. $\mu_n \Rightarrow \mu$ is allowed for weak convergence of measures; it must not be reused for implication in the same displayed formula.

Lean crosswalk. Probability measures and pushforwards use Mathlib's `ProbabilityMeasure`, `Measure.map`, and distribution APIs. A prose probabilistic representation is labelled **prose-derived** unless an exact Lean theorem is cited.

The canonical dyadic filtration

Command. `\DyadicSigmaField`. The printed base glyph is $\mathcal{A}$; it is never a generic event, matrix, Appell family, or algebra when the command is used.

Probability space. Fix a probability space $(\Omega, \mathcal{A}, \mathbb{P})$ and an explicitly ordered coordinate family $(U_j)_{j \in \mathbb{N}_{>0}}$. For $n \in \mathbb{N}_0$ define

$$\mathcal{A}_n := \sigma(U_1, \dots, U_n), \quad \mathcal{A}_0 := \{\emptyset, \Omega\}.$$

If a document instead generates from binary digits, Rademacher variables, or dyadic partition cells, it replaces the displayed generators while retaining the level-zero and monotonicity clauses.

Filtration law. $\mathcal{A}_n \subseteq \mathcal{A}_{n+1} \subseteq \mathcal{A}$ for every n . Conditional expectation $\mathbb{E}[X \mid \mathcal{A}_n]$ is defined only for an integrable X (or the appropriate nonnegative extended variant) and is an equivalence class modulo almost sure equality.

Limit field. $\mathcal{A}_\infty := \sigma(\bigcup_{n \in \mathbb{N}_0} \mathcal{A}_n)$. Equality with the ambient $\mathcal{A}$ is an additional generation hypothesis, not part of the notation.

Collision policy. An unqualified calligraphic A remains local. The shared command is reserved for this filtration, and every subscript is a level index in $\mathbb{N}_0 \cup \{\infty\}$ rather than a matrix dimension or event label.

Distribution functions, densities, and atoms

Distribution function. For a real random variable X ,

$$F_X(x) := \mathbb{P}\{X \leq x\}, \quad x \in \mathbb{R}.$$

It is right-continuous, nondecreasing, and has limits 0 and 1 at negative and positive infinity. The subscript is compulsory unless the distribution is uniquely fixed in the surrounding entry.

Density. A Borel function $f_X : \mathbb{R} \rightarrow [0, \infty)$ is a density when $\text{Law}(X)(B) = \int_B f_X(x) \, dx$ for every Borel set B . Then $F_X(x) = \int_{-\infty}^x f_X(t) \, dt$. A distribution need not possess a density.

Atom. The mass at a is $\mathbb{P}\{X = a\} = F_X(a) - F_X(a^-)$, where $F_X(a^-) = \lim_{x \uparrow a} F_X(x)$. A Dirac measure at a is $\delta_a(B) = \mathbf{1}_B(a)$.

Signed measures. A signed measure ν is not a probability law. Its total variation measure is $|\nu|$; the scalar $\nu(\mathbb{R})$ is its total mass and may be negative.

Endpoints. A uniform law on $[a, b]$, $a < b$, has density $(b - a)^{-1} \mathbf{1}_{[a, b]}(x)$. Changing the values at the two endpoints changes neither the associated absolutely continuous measure nor any Lebesgue integral.

Expectation, moments, variance, and covariance

Expectation. For an integrable real random variable,

$$\mathbb{E}[X] := \int_{\Omega} X(\omega) \, d\mathbb{P}(\omega) = \int_{\mathbb{R}} x \, d\text{Law}(X)(x).$$

For a nonnegative X , the extended integral is permitted and may equal $+\infty$.

Complex expectation. For an integrable $X : \Omega \rightarrow \mathbb{C}$, meaning $\mathbb{E}[|X|] < \infty$,

$$\mathbb{E}[X] := \mathbb{E}[\Re X] + i\mathbb{E}[\Im X] = \int_{\Omega} X \, d\mathbb{P},$$

where the last integral is the complex Bochner integral. This is the interpretation used when a characteristic-function integrand is complex-valued.

Raw moments. $m_n(X) := \mathbb{E}[X^n]$ for $n \in \mathbb{N}_0$ whenever the expectation exists; $m_0(X) = 1$, including on the event $X = 0$, by the polynomial convention $x^0 = 1$ for every x .

Central moments. $\mu_n(X) := \mathbb{E}[(X - \mathbb{E}X)^n]$, assuming the needed integrability. Raw m_n and central μ_n must not be interchanged.

Variance and covariance. For square-integrable real variables,

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}X)^2], \quad \text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)].$$

Thus $\text{Var}(X) = \text{Cov}(X, X) \geq 0$.

Cumulants. When $\log M_X(t)$ exists near 0, $\kappa_n(X)$ is the coefficient defined by $\log M_X(t) = \sum_{n \geq 1} \kappa_n(X) t^n / n!$. The logarithm is the real logarithm on the positive MGF, not an unstated complex branch.

Canonical probabilistic model of the Fabius distribution

Input variables. Let $(U_k)_{k \in \mathbb{N}_{>0}}$ be independent with $U_k \sim \text{Uniform}[0, 1]$. The index begins at 1.

Random series. Define

$$X := \sum_{k=1}^{\infty} 2^{-k} U_k.$$

The series converges absolutely for every sample point because $0 \leq U_k \leq 1$ and $\sum_{k \geq 1} 2^{-k} = 1$; hence $X \in [0, 1]$ almost surely.

Distribution. $F_{\text{bd}}(x) = \mathbb{P}\{X \leq x\}$ for $x \in \mathbb{R}$. The density is $f_X(x) = 2 \text{up}(2x - 1)$, so the distribution identity reproduces $F'_{\text{bd}}(x) = 2 \text{up}(2x - 1)$.

Centered variable. For $Y = 2X - 1$ one has $Y \in [-1, 1]$ almost surely and Y has density up . Equivalently, $\text{up}(t) = \frac{1}{2} f_X((t + 1)/2)$.

Finite approximants. $X_N = \sum_{k=1}^N 2^{-k} U_k$ has support $[0, 1 - 2^{-N}]$ and $X_N \rightarrow X$ almost surely, in every L^p for $1 \leq p < \infty$, and in law. These distinct modes are printed when a bound depends on one of them.

Status. The random-series representation is a semantic bridge used in the corpus. Each downstream theorem retains its own proof-status label; the representation alone does not formalize a probabilistic argument in Lean.

Geometrically weighted uniform series

Parameters. Unless a complex analytic continuation is explicitly declared, $q \in (0, 1)$. Let $U_0, U_1, \dots$ be independent and uniformly distributed on $[0, 1]$.

Normalized series. The probability-normalized variable is

$$X_q := (1 - q) \sum_{n=0}^{\infty} q^n U_n.$$

The weights sum to 1, so $X_q \in [0, 1]$ almost surely. At $q = \frac{1}{2}$ its law agrees with the canonical Fabius law after the index shift $U_n \leftrightarrow U_{n+1}$.

Unnormalized series. Some frontier documents instead study $S_q = \sum_{n \geq 0} q^n U_n = X_q / (1 - q)$ with support $[0, (1 - q)^{-1}]$. The letter S_q and its support must be printed whenever this normalization is used.

Characteristic function. Independence gives

$$\varphi_{X_q}(t) = \prod_{n=0}^{\infty} \frac{e^{it(1-q)q^n} - 1}{it(1-q)q^n},$$

with every factor assigned its continuous value 1 at $t = 0$.

Complex q . For $q \in \mathbb{C}$, the same product may define an analytic function for $|q| < 1$, but it is no longer the characteristic function of a real probability variable unless additional reality hypotheses are given.

Total variation, Wasserstein, and Kolmogorov distances

One-object total variation. The command `\TotalVariationOf{a}` denotes the total variation of one object a . Its type must be stated locally. For a real function on $[u, v]$,

$$\text{TV}(f) := \sup_{u=x_0 < \dots < x_n=v} \sum_{j=1}^n |f(x_j) - f(x_{j-1})|.$$

For a finite signed measure σ , it denotes $|\sigma|(X)$, equivalently the total-variation norm under the convention stated in the source. These are one-object functionals, not distances between two probability measures.

Total variation. For probability measures μ, ν on the same measurable space,

$$d_{\text{TV}}(\mu, \nu) := \sup_A |\mu(A) - \nu(A)| = \frac{1}{2} \int |f - g| \, d\lambda$$

when both have densities f, g with respect to λ . Some literature omits the factor $1/2$; that alternative is never silently substituted.

Kolmogorov distance. For real laws with CDFs F_X, F_Y , $d_K(F_X, F_Y) = \sup_{x \in \mathbb{R}} |F_X(x) - F_Y(x)|$.

Wasserstein distance. For $p \in [1, \infty)$ and laws with finite p th moments,

$$W_p(\mu, \nu) = \left(\inf_{\pi \in \Pi(\mu, \nu)} \int |x - y|^p \, d\pi(x, y) \right)^{1/p},$$

where $\Pi(\mu, \nu)$ is the set of couplings. The product coupling makes $\Pi(\mu, \nu)$ nonempty for probability measures on standard Borel spaces.

Type rule. A metric symbol takes two laws or measures, not two random variables, unless the law operation is explicitly stated in the local scope.

6 Fourier, Laplace, Mellin, and generating transforms

6.1 The two Fourier conventions

The 2π -frequency Fourier transform

Commands. `\FourierTwoPiOperator` is the operator token, `\FourierTwoPiTransformOf{f}` is its bracketed application, and `\FourierTwoPi{f}` is its normalization-labelled hat. These are three presentations of one forward transform, not three transforms.

Forward transform. For $f \in L^1(\mathbb{R})$,

$$\widehat{f}_{2\pi}(\xi) := \mathcal{F}_{2\pi}[f](\xi) := \mathcal{F}_{2\pi}[f](\xi) := \int_{\mathbb{R}} f(x) e^{-2\pi i x \xi} dx, \quad \xi \in \mathbb{R}.$$

The transform variable is ξ and carries cycles per unit.

Inverse. Under a hypothesis guaranteeing inversion,

$$f(x) = \int_{\mathbb{R}} \widehat{f}_{2\pi}(\xi) e^{2\pi i x \xi} d\xi.$$

There is no scalar prefactor in either direction. The operator implementing this inverse is `\InverseFourierTwoPiOperator`. It is written only on a named inversion domain; it does not assert pointwise inversion for arbitrary L^1 data. For L^2 , both formulas are interpreted through Plancherel extension.

Rules. With justified derivatives and integrals,

$$\widehat{f'}_{2\pi}(\xi) = 2\pi i \xi \widehat{f}_{2\pi}(\xi), \quad \widehat{f * g}_{2\pi} = \widehat{f}_{2\pi} \widehat{g}_{2\pi}.$$

Translation by a contributes $e^{-2\pi i a \xi}$; dilation obeys $\widehat{f(a \cdot)}_{2\pi}(\xi) = |a|^{-1} \widehat{f}_{2\pi}(\xi/a)$ for real $a \neq 0$.

Measures. For a finite Borel measure μ , $\widehat{\mu}_{2\pi}(\xi) = \int e^{-2\pi i x \xi} d\mu(x)$. This has no density assumption.

The angular-frequency Fourier transform

Operator token. `\FourierAngularOperator` denotes the transform operator without supplying its operand.

Bracketed application. `\FourierAngularTransformOf{f}` denotes the transform applied to f and prints the operand in brackets.

Normalization-labelled hat. `\FourierAngular{f}` denotes the same transform of f with the angular-frequency normalization printed on the hat.

Forward transform. For $f \in L^1(\mathbb{R})$,

$$\widehat{f}_{\text{ang}}(\omega) := \mathcal{F}_{\text{ang}}[f](\omega) := \mathcal{F}_{\text{ang}}[f](\omega) := \int_{\mathbb{R}} f(x) e^{-i\omega x} dx, \quad \omega \in \mathbb{R}.$$

Measures. For a finite Borel measure μ ,

$$\widehat{\mu}_{\text{ang}}(\omega) := \int_{\mathbb{R}} e^{-i\omega x} d\mu(x), \quad \omega \in \mathbb{R}.$$

This is a Fourier–Stieltjes transform and requires no density.

Inverse. Under an inversion hypothesis,

$$f(x) = \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{f}_{\text{ang}}(\omega) e^{i\omega x} d\omega.$$

The forward direction has no prefactor and the inverse has $1/(2\pi)$.

Inverse command. `\InverseFourierAngularOperator` denotes precisely this inverse on a stated inversion domain.

Rules. $\widehat{f'}_{\text{ang}}(\omega) = i\omega \widehat{f}_{\text{ang}}(\omega)$ and $\widehat{f * g}_{\text{ang}} = \widehat{f}_{\text{ang}} \widehat{g}_{\text{ang}}$.

Conversion. The two catalogued transforms satisfy

$$\widehat{f}_{2\pi}(\xi) = \widehat{f}_{\text{ang}}(2\pi\xi), \quad \widehat{f}_{\text{ang}}(\omega) = \widehat{f}_{2\pi}(\omega/(2\pi)).$$

The same identities hold, with f replaced throughout by a finite Borel measure μ , for the two Fourier–Stieltjes transforms just defined. A formula may change variables between the conventions only by displaying this rescaling.

Characteristic functions and Fourier signs

Definition. For a real random variable X with probability law μ ,

$$\varphi_X(t) := \mathbb{E}[e^{itX}] = \varphi_\mu(t) := \int_{\mathbb{R}} e^{itx} d\mu(x), \quad t \in \mathbb{R}.$$

Thus the command accepts either a declared random variable or a declared probability measure. The argument's type fixes the interpretation, and $\varphi_X = \varphi_{\text{Law}(X)}$ pointwise. It is not used for an arbitrary signed or complex measure without a local declaration. The probabilistic sign is positive.

Fourier conversion. Because both catalogued Fourier transforms use a negative sign,

$$\varphi_X(t) = \widehat{\text{Law}(X)}_{\text{ang}}(-t) = \widehat{\text{Law}(X)}_{2\pi}\left(-\frac{t}{2\pi}\right).$$

Moments. If $\mathbb{E}|X|^n < \infty$, then $\varphi_X^{(n)}(0) = i^n \mathbb{E}[X^n]$. Analyticity at zero requires stronger hypotheses than the existence of all moments.

Independent sums. For independent X, Y , $\varphi_{X+Y} = \varphi_X \varphi_Y$. The equality is pointwise in t .

The two sinc normalizations

Radian sinc. For $z \in \mathbb{C}$,

$$\text{sinc}_{\text{rad}}(z) := \begin{cases} \sin z/z, & z \neq 0, \\ 1, & z = 0. \end{cases}$$

The value at zero fills the removable singularity, and the result is entire.

Normalized sinc. For $z \in \mathbb{C}$,

$$\text{sinc}_\pi(z) := \begin{cases} \sin(\pi z)/(\pi z), & z \neq 0, \\ 1, & z = 0. \end{cases}$$

Conversion. $\text{sinc}_\pi(z) = \text{sinc}_{\text{rad}}(\pi z)$ and $\text{sinc}_{\text{rad}}(z) = \text{sinc}_\pi(z/\pi)$.

Zeros. sinc_{rad} vanishes at πk , while sinc_π vanishes at k , for $k \in \mathbb{Z} \setminus \{0\}$. This distinction is essential in zero-set and sampling arguments.

Retired alias. A bare sinc operator is forbidden in normative prose. Historical occurrences must be annotated with one of the two definitions.

Fourier products for Rvachev's up-function

2 π -frequency form. With $\int \text{up} = 1$ and the [2 \$\pi\$ convention](#),

$$\widehat{\text{up}}_{2\pi}(\xi) = \prod_{n=0}^{\infty} \text{sinc}_{\text{rad}}\left(\frac{\pi\xi}{2^n}\right) = \prod_{n=0}^{\infty} \text{sinc}_{\pi}\left(\frac{\xi}{2^n}\right).$$

Angular form. Equivalently,

$$\widehat{\text{up}}_{\text{ang}}(\omega) = \prod_{j=1}^{\infty} \text{sinc}_{\text{rad}}\left(\frac{\omega}{2^j}\right).$$

The angular product begins at $j = 1$; the 2π product begins at $n = 0$.

At zero. Every factor is assigned value 1 at zero, so both products equal 1 at the origin.

Convergence. The products converge locally uniformly on $\mathbb{C}$, hence define entire functions. A logarithm of the product is used only on a zero-free domain with an explicit branch.

Zeros. Multiplicities are computed by counting all pairs (n, k) for which $\xi/2^n = k \in \mathbb{Z} \setminus \{0\}$, not by treating the displayed factors as having disjoint zero sets.

Fourier series on the unit circle

Command. `\FourierSeriesCoefficient{h}{k}`. The printed subscript “FS” distinguishes a discrete Fourier-series coefficient from both continuous line-transform normalizations. For a period other than one, use `\FourierSeriesCoefficientAtPeriod{h}{k}{T}`; its third argument is mandatory and appears in the printed symbol.

Coefficient. For a 1-periodic integrable function h ,

$$\widehat{h}_{\text{FS}}[k] := \int_0^1 h(x) e^{-2\pi i k x} dx, \quad k \in \mathbb{Z}.$$

Series. The formal Fourier series is $\sum_{k \in \mathbb{Z}} \widehat{h}_{\text{FS}}[k] e^{2\pi i k x}$. Its mode of convergence—pointwise, L^2 , uniform, distributional, or a named summability method—is part of every theorem.

Separation from line transform. Square brackets and an integer index mark a Fourier-series coefficient. Parentheses and a real frequency mark $\widehat{h}_{2\pi}(\xi)$ or $\widehat{h}_{\text{ang}}(\omega)$ on the line.

Period T . For a T -periodic function, $T > 0$, $\widehat{h}_{\text{FS}; T}[k] = T^{-1} \int_0^T h(x) e^{-2\pi i k x/T} dx$. The period, the factor T^{-1} , and the kernel scale are never implicit. A finite measure or an unnormalized Fourier–Stieltjes coefficient is a different local object and does not use either function-coefficient command.

One-sided and bilateral Laplace transforms

Commands. `\LaplaceTransformSymbol` is the operator token and `\LaplaceTransformOf{f}` supplies a bracketed operand. An unsubscripted token is one-sided by this catalogue's convention.

One-sided. For $f : [0, \infty) \rightarrow \mathbb{C}$,

$$\mathcal{L}[f](s) := \int_0^{\infty} e^{-sx} f(x) dx, \quad s \in \mathbb{C}$$

where the integral converges.

Bilateral. For $f : \mathbb{R} \rightarrow \mathbb{C}$,

$$\mathcal{L}_{\text{bi}}[f](s) := \int_{-\infty}^{\infty} e^{-sx} f(x) dx.$$

The subscript bi is compulsory; it prevents a support assumption from being smuggled into an identity.

Abscissa. The abscissa of absolute convergence is $\sigma_a = \inf\{\sigma \in \mathbb{R} : \int_0^\infty e^{-\sigma x} |f(x)| dx < \infty\}$, with infimum in $\overline{\mathbb{R}}$.

Probability relation. For a nonnegative random variable X , $\mathbb{E}[e^{-sX}]$ is the Laplace–Stieltjes transform of its law. $M_X(t) = \mathbb{E}[e^{tX}]$ is obtained at $s = -t$ only where both sides are finite.

Moment-generating and exponential-generating functions

Moment-generating function. For a real random variable X ,

$$M_X(t) := \mathbb{E}[e^{tX}]$$

for real or complex t at which the expectation is absolutely finite. It is not declared on all of $\mathbb{C}$ merely because every formal moment exists.

Moment expansion. If the MGF is analytic at zero, $M_X(t) = \sum_{n=0}^\infty \mathbb{E}[X^n] t^n / n!$ locally.

Exponential-generating function. For an abstract sequence (a_n) , $A_{\text{egf}}(z) = \sum_{n \geq 0} a_n z^n / n!$. It is an MGF only after a random variable with $a_n = \mathbb{E}[X^n]$ has been specified.

Ordinary-generating function. $A_{\text{ogf}}(z) = \sum_{n \geq 0} a_n z^n$. The subscripts egf and ogf are retained wherever both occur.

Mellin transform

Commands. `\MellinTransformSymbol` is the operator token and `\MellinTransformOf{f}` supplies the operand. Neither command chooses a convergence strip or analytic continuation.

Definition. For $f : (0, \infty) \rightarrow \mathbb{C}$,

$$\mathcal{M}[f](s) := \int_0^\infty f(x) x^{s-1} dx, \quad s \in \mathbb{C},$$

on the maximal strip of absolute convergence.

Complex power. For $x > 0$, x^{s-1} means $\exp((s-1) \log x)$ with the real logarithm $\log x$; no complex branch is involved.

Scaling. For real $a > 0$, $\mathcal{M}[f(a \cdot)](s) = a^{-s} \mathcal{M}[f](s)$.

Continuation. A meromorphic continuation of the integral-defined transform is identified by the same symbol only after its initial strip and uniqueness of continuation have been stated. Poles, residues, and contour shifts refer to that continuation.

Cauchy and Stieltjes transforms

Cauchy transform. For a finite Borel measure μ on $\mathbb{R}$,

$$G_\mu(z) := \int_{\mathbb{R}} \frac{1}{z - x} d\mu(x), \quad z \in \mathbb{C} \setminus \text{supp}(\mu).$$

Alternative sign. Some sources define $\tilde{G}_\mu(z) = \int (x - z)^{-1} d\mu(x) = -G_\mu(z)$. The catalogue uses the $z - x$ denominator; a quoted alternative carries a tilde and the minus-sign conversion.

Stieltjes transform. For a measure on $[0, \infty)$, $S_\mu(s) = \int_0^\infty (s + x)^{-1} d\mu(x)$ for $s \in \mathbb{C} \setminus (-\infty, 0]$ when finite. Denominator signs must be accounted for before relating it to G_μ .

Boundary values. Symbols $G_\mu(x \pm i0)$ mean non-tangential limits from the upper or lower half-plane when those limits exist; they are not evaluation at a complex number with an infinitesimal component.

Transforms of distributions and generalized functions

Tempered distributions. For $T \in \mathcal{S}'(\mathbb{R})$, its 2π -Fourier transform is defined by duality $\langle \widehat{T}_{2\pi}, \phi \rangle = \langle T, \phi_{2\pi} \rangle$ after fixing the test-function transform above. The pairing is bilinear, not a Hermitian inner product.

Dirac identities. $\widehat{\delta}_a{}_{2\pi}(\xi) = e^{-2\pi i a \xi}$ and $\widehat{1}_{2\pi} = \delta_0$ in $\mathcal{S}'$.

Derivatives. $\widehat{D^m T}_{2\pi} = (2\pi i \xi)^m \widehat{T}_{2\pi}$, where D is distributional derivative.

Audit rule. A distributional identity is never presented as an absolutely convergent integral identity. The ambient test-function space and duality convention appear at the first use.

Nonlinear and linearized Fourier-decay objectives

Commands. For $n \in \mathbb{N}_0$ the exact optimization functional is $\text{Obj}_{\text{decay},n}$ and its first-order or otherwise explicitly specified linearization is $\text{Obj}_{\text{decay},n}^{\text{lin}}$. Their semantic names prevent either from being mistaken for a Fabius function, a Fourier operator, or a filtration.

Required declaration. At the first use a document supplies a variable space V_n , a feasible set $\mathcal{C}_n \subseteq V_n$, a codomain (normally $\mathbb{R} \cup \{+\infty\}$), and formulas

$$\text{Obj}_{\text{decay},n} : \mathcal{C}_n \rightarrow \overline{\mathbb{R}}, \quad \text{Obj}_{\text{decay},n}^{\text{lin}} : T_{v_0} \mathcal{C}_n \rightarrow \overline{\mathbb{R}}.$$

The base point v_0 , constraints, frequency normalization, logarithm base, weights, and all n -dependence are part of those formulas. The shared commands intentionally do not smuggle in one draft's choice.

Linearization contract. A label “linearized” identifies a new objective. An expansion such as $\text{Obj}_{\text{decay},n}(v_0 + h) = \text{Obj}_{\text{decay},n}(v_0) + \text{Obj}_{\text{decay},n}^{\text{lin}}(h) + r_n(h)$ states the norm and limit in which $r_n(h) = o_{h \rightarrow 0}(\|h\|)$. Dropping r_n produces an approximation, not an identity.

Optimization values. $\inf_{v \in \mathcal{C}_n} \text{Obj}_{\text{decay},n}(v)$ is an extended-real value. A symbol for a minimizer is introduced only after existence; uniqueness is an additional theorem. The optimum of the linearized problem is not substituted for the nonlinear optimum without a bound.

Status. The commands are **definitions of notation**. Particular decay bounds, saddle locations, convexity assertions, and asymptotic optima inherit the proof-status of the document that states them.

7 Binary arithmetic, Thue–Morse data, and exact combinatorics

Binary digit sum and bits

Binary expansion. Every $n \in \mathbb{N}_0$ has the unique finite expansion $n = \sum_{j=0}^{\infty} b_j(n)2^j$ with $b_j(n) \in \{0, 1\}$ and all but finitely many $b_j(n)$ zero. Leading zeros are allowed but contribute nothing.

Digit sum. The canonical binary weight is

$$\text{wt}_2(n) := \sum_{j=0}^{\infty} b_j(n), \quad n \in \mathbb{N}_0.$$

Thus $\text{wt}_2(0) = 0$.

Bit extraction. $b_j(n) = \lfloor n/2^j \rfloor - 2 \lfloor n/2^{j+1} \rfloor$ for $j \in \mathbb{N}_0$. Negative integers require an explicitly selected finite-word or 2-adic representation and are outside this global definition.

Retired aliases. The former digit-sum, population-count, and bare-weight spellings may occur in quotations. Normative prose uses `\BinaryDigitSum`; an unsubscripted weight is especially unsafe because it can denote an integration weight.

Thue–Morse bit and sign sequences

Bit sequence. $t_n := \text{wt}_2(n) \bmod 2 \in \{0, 1\}$ for $n \in \mathbb{N}_0$.

Sign sequence. The canonical sign is

$$\varepsilon_n := (-1)^{\text{wt}_2(n)} = (-1)^{t_n} \in \{-1, 1\}.$$

Consequently $\varepsilon_0 = 1$.

Command layer. `\ThueMorseSignSymbol` is a zero-argument reserved glyph token rendering ε ; it names the sign sequence inside operator labels, DFT displays, and prose where no index is being evaluated. `\ThueMorseSign{n}` takes exactly one argument $n \in \mathbb{N}_0$ and renders the indexed value ε_n . The former is not a constant value and the latter must not receive a second raw subscript.

Recurrences. $t_{2n} = t_n$, $t_{2n+1} = 1 - t_n$ and $\varepsilon_{2n} = \varepsilon_n$, $\varepsilon_{2n+1} = -\varepsilon_n$.

Generating product. For every $N \in \mathbb{N}_0$,

$$\sum_{n=0}^{2^N-1} \varepsilon_n z^n = \prod_{j=0}^{N-1} (1 - z^{2^j}).$$

At $N = 0$ both sides are 1: the sum contains the sole term $n = 0$ and the product is empty.

Lean crosswalk. The formal sign sequence uses declarations in `ThueMorse.lean` and related prefix modules. Exact theorem names are recorded in the crosswalk appendix; prose identities are not upgraded beyond their verified status.

p -adic and 2-adic valuations

Finite valuation. For a prime p and nonzero $n \in \mathbb{Z}$,

$$\nu_p(n) := \max\{k \in \mathbb{N}_0 : p^k \mid n\}.$$

The value depends only on $|n|$.

At zero. In analytic formulas using extended values, $\nu_p(0) = +\infty \in \overline{\mathbb{R}}$. In Lean or finite-natural APIs, the valuation of zero may instead be represented by a separate option or by a library-specific totalization; each theorem states which codomain it uses.

Two-adic abbreviation. $\nu_2(n) = \nu_2(n)$. For $n \neq 0$, write $n = 2^{\nu_2(n)} n_{\text{odd}}$ with n_{odd} odd.

Prime-power Pascal row. For a prime p , $m \in \mathbb{N}_0$, and $j \in \mathbb{N}_{>0}$ with $j \leq p^m$, the finite-natural valuation satisfies

$$\nu_p\binom{p^m}{j} + \nu_p(j) = m, \quad \nu_p\binom{p^m}{j} = m - \nu_p(j).$$

The right endpoint $j = p^m$ and the boundary $m = 0$ are included. The strict dyadic-comb specialization has $0 < j < 2^m$. These formulas do not cover the distinct endpoint-flat weights $\binom{2^m-2}{j-1}$.

Lean crosswalk. `primePowerChoose_padicValNat_add` and `primePowerChoose_padicValNat` at in `PrimePowerBinomialValuation.lean` prove the arbitrary-prime identities; `twoPowChoose_padicValNat` is the strict dyadic specialization. Lean’s `padicValNat` is totalized in $\mathbb{N}_0$, so its exact zero behavior is the declaration’s behavior rather than the extended-value convention above. Status: **Lean-proved**.

Order notation. For a zero or pole of a meromorphic function at z_0 , $\text{ord}_{z_0}(f)$ is a different valuation-like object; positive values denote zeros and negative values poles in this catalogue.

Dyadic blocks and prefix sums

Block. The level- N dyadic block of indices is $B_N = \{0, 1, \dots, 2^N - 1\}$ for $N \in \mathbb{N}_0$.

Signed prefix sum. For a function $P : \mathbb{N}_0 \rightarrow \mathbb{C}$,

$$\Sigma_N(P) := \sum_{n=0}^{2^N-1} \varepsilon_n P(n).$$

Prouhet cancellation. If P is a polynomial of degree $< N$, then $\Sigma_N(P) = 0$. At degree N the surviving value depends on the leading coefficient and the exact sign convention; it must not be quoted without that coefficient.

Arbitrary prefix. $S_M(P) = \sum_{n=0}^{M-1} \varepsilon_n P(n)$ for $M \in \mathbb{N}_0$. The half-open upper bound ensures $S_0(P) = 0$.

Shifted block. A sum over $a2^N \leq n < (a+1)2^N$ prints both $a \in \mathbb{N}_0$ and $N \in \mathbb{N}_0$; its sign is not inferred from the unshifted identity without applying the binary carry law.

Finite discrete Fourier transforms

Forward DFT. For a vector $a = (a_0, \dots, a_{N-1}) \in \mathbb{C}^N$, $N \in \mathbb{N}_{>0}$,

$$\hat{a}_k := \sum_{n=0}^{N-1} a_n e^{-2\pi i k n / N}, \quad k = 0, \dots, N-1.$$

Inverse. $a_n = N^{-1} \sum_{k=0}^{N-1} \hat{a}_k e^{2\pi i k n / N}$. The normalization is forward-unscaled, inverse- N^{-1} .

Root-of-unity form. With $\zeta_N = e^{2\pi i / N}$, $\hat{a}_k = \sum_n a_n \zeta_N^{-kn}$. The symbol ζ_N is a primitive root chosen with positive angle.

Thue–Morse specialization. For $N = 2^m$ and $a_n = \varepsilon_n$, the product formula evaluates the DFT at $z = \zeta_N^{-k}$. Multiplicities of zeros require counting vanishing factors, not only identifying a vanishing product.

Document-local command. In the Thue–Morse atlas, `\ThueMorseDFTCoefficient {m} {k}` has the two arguments $m \in \mathbb{N}_0$ and $k \in \mathbb{Z}$ in that order and means

$$\text{DFT}_{2^m}[\varepsilon][k] := \sum_{n=0}^{2^m-1} \varepsilon_n e^{-2\pi i k n / 2^m}.$$

The value depends only on k modulo 2^m ; the canonical representative range is $0 \leq k < 2^m$. The forward transform is unscaled, the exponential sign is negative, and inversion therefore carries the factor 2^{-m} and the positive exponential. This operator is a finite DFT, not either line-Fourier transform and not a non-Fourier accent.

Zero sets, multiplicities, and runs

Order of a zero. For analytic f not identically zero near z_0 , $\text{ord}_{z_0}(f) = m \in \mathbb{N}_0$ means $f(z) = (z - z_0)^m g(z)$ with $g(z_0) \neq 0$.

Run in a sequence. For a sequence (a_n) , a zero run of length ℓ from r means $a_r = \dots = a_{r+\ell-1} = 0$ and, when the adjacent indices exist, $a_{r-1} \neq 0$ and $a_{r+\ell} \neq 0$.

Boundary runs. At $r = 0$, only the right maximality condition is required. For a finite sequence ending at M , a terminal run has no right-neighbor condition.

Separation. A multiplicity of a zero of a generating polynomial is not a run of zero coefficients. The corpus contains both phenomena and keeps ord_{z_0} separate from ℓ_{run} .

Compositions, partitions, and multi-indices

Weak composition. A weak composition of n into k parts is $\alpha = (\alpha_1, \dots, \alpha_k) \in \mathbb{N}_0^k$ with $|\alpha| = \sum_{j=1}^k \alpha_j = n$.

Composition. A composition has $\alpha_j \in \mathbb{N}_{>0}$. For $n = 0$, the unique composition with zero parts is the empty tuple; there is no positive-part composition into $k > 0$ parts.

Partition. A partition $\lambda \vdash n$ is a finite nonincreasing sequence of positive integers summing to n . Its length is $\ell(\lambda)$ and its multiplicity of part j is $m_j(\lambda)$.

Multi-index. For $\alpha \in \mathbb{N}_0^d$, $|\alpha| = \sum_j \alpha_j$, $\alpha! = \prod_j \alpha_j!$, $x^\alpha = \prod_j x_j^{\alpha_j}$, and $\partial^\alpha = \prod_j \partial_{x_j}^{\alpha_j}$. Products over $d = 0$ are empty and equal 1.

Factorials. The shared $(x)^{\overline{n}} = x(x-1) \cdots (x-n+1)$ and $(x)^{\underline{n}} = x(x+1) \cdots (x+n-1)$ both equal 1 at $n = 0$.

8 q -series, products, and arithmetic functions

The q parameter and analytic regimes

Formal regime. q may be an indeterminate and all identities may live in $R[q]$ or $R[[q]]$. Convergence language is meaningless in this regime.

Analytic regime. For infinite products and series, the default explicit hypothesis is $q \in \mathbb{C}$ with $|q| < 1$. For probability models, it is strengthened to $q \in (0, 1)$.

Roots of unity. At $q = \zeta$ with $\zeta^N = 1$, denominators in rational q -expressions can vanish. Evaluation means polynomial evaluation after cancellation only when a polynomial representative has been proved; otherwise it is excluded.

Base-role rule. If a document also uses q for a denominator, quantile, frequency, or conjugate exponent, the q -series base is renamed locally q_{base} . Semantic subscripts take priority over traditional brevity.

Finite and infinite q -Pochhammer symbols

Finite definition. For $a, q \in \mathbb{C}$ and $n \in \mathbb{N}_0$,

$$(a; q)_n := \prod_{j=0}^{n-1} (1 - aq^j).$$

The product is empty at $n = 0$, so $(a; q)_0 = 1$ for all a, q , including values that make later factors vanish.

Infinite definition. For $|q| < 1$,

$$(a; q)_\infty := \prod_{j=0}^{\infty} (1 - aq^j),$$

with analytic meaning as a limit of finite products. The literal index ∞ denotes an infinite product, not an extended-natural argument to the finite definition.

Factorization. $(a; q)_{m+n} = (a; q)_m (aq^m; q)_n$ for $m, n \in \mathbb{N}_0$.

Retired signatures. Former two-argument macros hid the base and are retired. The canonical command always receives (a, q, n) in that order.

q -integers and q -factorials

Definition. For $n \in \mathbb{N}_0$,

$$[n]_q := \begin{cases} (1 - q^n)/(1 - q), & q \neq 1, \\ n, & q = 1. \end{cases}$$

Equivalently $[n]_q = 1 + q + \dots + q^{n-1}$ for $n > 0$, and $[0]_q = 0$.

q -factorial. $[n]_q! := \prod_{j=1}^n [j]_q$, with $[0]_q! = 1$. The exclamation mark belongs to the notation and does not apply an ordinary factorial to the polynomial $[n]_q$.

Limit. $\lim_{q \rightarrow 1} [n]_q = n$ and $\lim_{q \rightarrow 1} [n]_q! = n!$ for fixed n .

Alternative symmetric integer. The symmetric quantity $(q^n - q^{-n})/(q - q^{-1})$ is written $[n]_{q, \text{sym}}$ if needed; it is not the canonical `\QInteger`.

Gaussian binomial coefficients

Definition. For $n, k \in \mathbb{N}_0$,

$$\begin{bmatrix} n \\ k \end{bmatrix}_q := \begin{cases} \frac{(q; q)_n}{(q; q)_k (q; q)_{n-k}}, & 0 \leq k \leq n, \\ 0, & k > n, \end{cases}$$

interpreted as its polynomial in q , so removable singularities are filled.

Boundary values. $\begin{bmatrix} n \\ 0 \end{bmatrix}_q = \begin{bmatrix} n \\ n \end{bmatrix}_q = 1$, including $n = 0$.

Recurrence. For $1 \leq k \leq n$,

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \begin{bmatrix} n-1 \\ k \end{bmatrix}_q + q^{n-k} \begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_q.$$

Retired signatures. The corpus formerly used incompatible argument orders for its abbreviated Gaussian-binomial macro. The canonical `\GaussianBinomial` receives upper index, lower index, then base.

Limit. $\begin{bmatrix} n \\ k \end{bmatrix}_1 = \binom{n}{k}$, understood by polynomial evaluation, not substitution into the displayed rational quotient.

Basic hypergeometric series

Definition. For $|q| < 1$, the unilateral basic hypergeometric series is

$${}_r\phi_s \left(\begin{matrix} a_1, \dots, a_r \\ b_1, \dots, b_s \end{matrix}; q, z \right) := \sum_{n=0}^{\infty} \frac{\prod_{j=1}^r (a_j; q)_n}{(q; q)_n \prod_{j=1}^s (b_j; q)_n} \left((-1)^n q^{\binom{n}{2}} \right)^{1+s-r} z^n,$$

provided no denominator factor vanishes and the series converges.

Normalization warning. References differ on the $((-1)^n q^{\binom{n}{2}})^{1+s-r}$ factor. Every occurrence in this corpus either expands the definition above or cites this entry.

Termination. If an upper parameter is q^{-N} with $N \in \mathbb{N}_0$, the series terminates at $n = N$ unless an earlier denominator singularity makes the expression undefined.

Bilateral series. The bilateral basic hypergeometric symbol sums over $n \in \mathbb{Z}$ and is never identified with the unilateral symbol. Negative-index q -Pochhammers must be defined before use.

q -difference operators

Jackson derivative. For $q \neq 1$ and $x \neq 0$,

$$D_q f(x) := \frac{f(x) - f(qx)}{(1-q)x}.$$

At $x = 0$, $D_q f(0)$ is assigned $f'(0)$ only when that derivative exists and the local definition explicitly makes the continuous extension.

Shift operator. $T_q f(x) = f(qx)$. Thus $D_q = (1 - T_q)/((1-q)x)$ as an identity on functions where division by x is valid.

Base direction. The expression $(f(qx) - f(x))/((q-1)x)$ is algebraically identical. The genuinely different operator $(f(qx) - f(x))/((q-1)qx)$ is labelled separately.

Classical limit. If f is differentiable at x , then $D_q f(x) \rightarrow f'(x)$ as $q \rightarrow 1$ through real positive values. Uniformity in x is never inferred without a theorem.

Cyclotomic polynomials and roots of unity

Primitive root. $\zeta_N = e^{2\pi i/N}$ for $N \in \mathbb{N}_{>0}$ is the chosen positively oriented primitive N th root.

Cyclotomic polynomial. $\Phi_N(x) \in \mathbb{Z}[x]$ is the monic polynomial whose roots are ζ_N^k for $1 \leq k \leq N$ and $\gcd(k, N) = 1$.

Factorization. $x^N - 1 = \prod_{d|N} \Phi_d(x)$, where d ranges over the positive divisors of N .

Order of a root. For $\zeta^N = 1$, $\text{ord}(\zeta) = \min\{d \in \mathbb{N}_{>0} : \zeta^d = 1\}$. This group-theoretic order is distinct from the analytic $\text{ord}_{z_0}(f)$ of a zero.

Evaluation policy. A rational q -series expression is not evaluated at a root of unity until all denominator zeros and possible cancellations have been audited.

Divisibility, gcd, lcm, and standard arithmetic functions

Divisibility. $a \mid b$ for $a, b \in \mathbb{Z}$ means $b = ac$ for some $c \in \mathbb{Z}$. In a sum $d \mid n$, the divisor d is positive unless otherwise stated.

GCD and LCM. $\gcd(a, b) \in \mathbb{N}_0$ is the nonnegative greatest common divisor. The zero-argument command `\LeastCommonMultiple` supplies the operator token in $\text{lcm}(a_1, \dots, a_r)$; its parenthesized integer inputs are source arguments, not TeX macro arguments. Its value is the unique nonnegative generator of the ideal $a_1\mathbb{Z} \cap \dots \cap a_r\mathbb{Z}$ and is therefore insensitive to signs.

Zero and family conventions. For $r = 0$ the empty intersection is $\mathbb{Z}$. Explicitly, $\text{lcm}(a, b) = \text{lcm}(|a|, |b|)$, $\gcd(0, 0) = 0$, $\text{lcm}(0, b) = 0$ for every b , a singleton family has LCM $|a|$, and the empty family has LCM 1. The empty-family value is the unit for adjoining further inputs and is required in the dyadic-denominator formulas.

Möbius function. $\mu_M(n) = 0$ if a prime square divides n , and $(-1)^r$ if n is a product of r distinct primes; its domain is $\mathbb{N}_{>0}$ and $\mu_M(1) = 1$. The subscript prevents a collision with measures and means.

Euler totient. $\varphi_E(n) = |\{1 \leq k \leq n : \gcd(k, n) = 1\}|$, $n \in \mathbb{N}_{>0}$. The subscript separates it from characteristic functions.

Divisor sums. $\sigma_\alpha(n) = \sum_{d|n} d^\alpha$ for $n \in \mathbb{N}_{>0}$ and $\alpha \in \mathbb{C}$, with positive divisors and $d^\alpha = \exp(\alpha \log d)$.

9 Asymptotics, logarithms, Lambert W, and saddle analysis

Limits and named asymptotic regimes

Limit annotation. A limit variable and destination are part of every asymptotic assertion: $n \rightarrow \infty$, $x \rightarrow 0^+$, $z \rightarrow z_0$ in a sector, or $q \uparrow 1$ are different regimes. A subscript may abbreviate a regime only after it has been declared.

One-sided limits. $x \rightarrow a^+$ means $x > a$ and $x \rightarrow a$; $x \rightarrow a^-$ means $x < a$. For $q \uparrow 1$, q is real and approaches 1 from below.

Uniformity. An estimate uniform for $y \in Y$ means its threshold and implied constant may be chosen independently of y . Pointwise asymptotics never imply this.

Complex sectors. A sector is printed as $S(\theta_0, \alpha, r) = \{z : 0 < |z| < r, |\arg z - \theta_0| < \alpha\}$, with a selected continuous argument. Statements at a branch point include the slit and sector.

Parameter order. For iterated limits, notation such as $n \rightarrow \infty$ followed by $q \uparrow 1$ records the order. A joint limit instead states a relation such as $(1 - q)n \rightarrow \tau$.

Big O, little o, and comparison relations

Big O. $f = \mathcal{O}_{x \rightarrow a}(g)$ means that there are $C > 0$ and a punctured neighborhood in the declared domain such that $|f(x)| \leq C|g(x)|$. If g vanishes arbitrarily near a , the inequality definition remains authoritative.

Little o. $f = o_{x \rightarrow a}(g)$ means $f(x)/g(x) \rightarrow 0$ where the quotient is defined, or equivalently the epsilon inequality when that avoids isolated zeros.

Asymptotic equivalence. $f(x) \sim g(x)$ as $x \rightarrow a$ means $f(x)/g(x) \rightarrow 1$. It requires g to be nonzero in a punctured neighborhood.

Two-sided comparability. $f \asymp g$ in a regime means $c|g| \leq |f| \leq C|g|$ for constants $0 < c \leq C < \infty$. This is not asymptotic equivalence.

Vinogradov notation. $f \ll g$ means $f = \mathcal{O}(g)$ and $f \gg g$ means $g \ll f$. A dependency subscript, for example $\ll_\alpha$, lists every parameter on which the implied constant may depend.

Shared commands. Normative prose prefers `\BigOAt` and `\LittleOAt`; bare `\BigO` and `\LittleO` are reserved for formulas whose regime is printed immediately outside the expression.

Asymptotic sequences and expansions

Asymptotic sequence. Functions (ϕ_n) form an asymptotic sequence in a regime when $\phi_{n+1} = o(\phi_n)$ for every n .

Expansion. $f \sim \sum_{n \geq 0} a_n \phi_n$ means that for every $N \in \mathbb{N}_{>0}$,

$$f - \sum_{n=0}^{N-1} a_n \phi_n = o(\phi_{N-1})$$

in the stated regime. It does not assert convergence of the infinite series.

Poincaré scale. For powers at infinity, $f(x) \sim \sum_{n \geq 0} a_n x^{-\lambda_n}$ requires $\lambda_0 < \lambda_1 < \dots$ and x in a domain where the powers have been defined.

Exponentially small terms. A transseries term such as $e^{-A/\epsilon} \epsilon^\beta (\log \epsilon)^m$ lists the sign of $\operatorname{Re}(A/\epsilon)$, the logarithm branch, and the ordering used to compare exponential scales.

Truncation. An optimally truncated expansion names its truncation index $N(\epsilon)$ and the remainder norm; the infinite formal series is not assigned its optimal-truncation error.

Real logarithm, principal logarithm, and arguments

Real logarithm. $\log x$ for $x > 0$ is the real natural logarithm with base e.

Principal branch. The shared command denotes

$$\operatorname{Log} z = \log |z| + i \operatorname{Arg} z, \quad \operatorname{Arg} z \in (-\pi, \pi],$$

for $z \in \mathbb{C} \setminus (-\infty, 0]$, with the boundary convention $\operatorname{Arg}(-r) = \pi$ when a boundary value is explicitly taken from above. The analytic domain itself excludes the nonpositive real axis.

Other branches. On a simply connected zero-free domain D , a branch $\log_D z$ is an analytic function satisfying $\exp(\log_D z) = z$. Its normalization at one base point is stated.

Complex powers. $z^\alpha := \exp(\alpha \operatorname{Log} z)$ when the principal branch is intended. If $\alpha \in \mathbb{Z}$, the algebraic power has its global meaning and no branch is needed.

Boundary notation. $\operatorname{Log}(x \pm i0)$ denotes a specified limiting value. It is not silently substituted into a formula proved only inside the slit plane.

Lambert W and its branches

Defining equation. A branch of W satisfies

$$W_k(z) e^{W_k(z)} = z,$$

where $k \in \mathbb{Z}$ labels the branch. The inverse is multivalued before a branch is selected.

Principal branch. W_0 is real on $[-e^{-1}, \infty)$, analytic at 0, and satisfies $W_0(0) = 0$. Its standard complex cut is $(-\infty, -e^{-1}]$.

Lower real branch. W_{-1} is real on $[-e^{-1}, 0)$, takes values in $(-\infty, -1]$, and approaches $-\infty$ as $z \uparrow 0$.

Branch point. At $z = -e^{-1}$ the two real branches meet at -1 and have a square-root branch point. A Taylor series in $z + e^{-1}$ is therefore not asserted there.

Asymptotic branch. For large z away from cuts,

$$W_k(z) = L_1 - L_2 + \frac{L_2}{L_1} + \cdots, \quad L_1 = \log z + 2\pi i k, \quad L_2 = \log L_1,$$

with both logarithm branches and the sector specified.

Bare symbol policy. A branchless $W(z)$ is permitted only when the domain makes the real principal branch unambiguous; all complex and negative-real uses carry a subscript.

Polylogarithm, gamma, and zeta functions

Polylogarithm. For $|z| < 1$,

$$\operatorname{Li}_s(z) := \sum_{n=1}^{\infty} \frac{z^n}{n^s}, \quad s \in \mathbb{C}.$$

Elsewhere the symbol denotes its analytically continued branch, with the z -plane cut and logarithm convention stated.

Gamma function. For $\operatorname{Re} s > 0$, $\Gamma(s) = \int_0^{\infty} x^{s-1} e^{-x} dx$. Its meromorphic continuation has simple poles at $0, -1, -2, \dots$ and no zeros. For $n \in \mathbb{N}_0$, $\Gamma(n+1) = n!$.

Riemann zeta. For $\operatorname{Re} s > 1$, $\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$, extended meromorphically to $\mathbb{C}$ with a simple pole at $s = 1$.

Hurwitz zeta. $\zeta(s, a) = \sum_{n=0}^{\infty} (n+a)^{-s}$ initially for $\operatorname{Re} s > 1$ and $a \notin \{0, -1, -2, \dots\}$, with the branch of each power specified if a is complex.

Collision policy. The same Greek glyph may label a root of unity or a zeta function only with a semantic subscript or arguments that make its type immediate.

Coefficient extraction and formal series

Coefficient operator. For $A(z) = \sum_{n \in \mathbb{Z}} a_n z^n$,

$$[z^n]A(z) := a_n.$$

The bracket is an operator, not multiplication by z^n .

Cauchy formula. If A is analytic on and inside a positively oriented circle $|z| = r$,

$$[z^n]A(z) = \frac{1}{2\pi i} \int_{|z|=r} \frac{A(z)}{z^{n+1}} dz, \quad n \in \mathbb{N}_0.$$

Laurent coefficients. For an annular Laurent series the same operator permits $n \in \mathbb{Z}$; the annulus and contour winding number are stated.

Formal interpretation. In $R[[z]]$, coefficient extraction is algebraic and does not assert analytic convergence. Cauchy's formula belongs only to the analytic interpretation.

Residues and contour orientation

Residue. For meromorphic f near z_0 , $\text{Res}_{z=z_0} f(z)$ is the coefficient of $(z - z_0)^{-1}$ in the Laurent expansion.

Simple pole. If $f = g/h$, g, h analytic, $h(z_0) = 0$, and $h'(z_0) \neq 0$, then $\text{Res}_{z=z_0} f = g(z_0)/h'(z_0)$.

Contour. Unless an arrow says otherwise, a simple closed contour is positively oriented. Then $\int_{\gamma} f(z) dz = 2\pi i \sum \text{Ind}(\gamma, z_j) \text{Res}_{z=z_j} f$.

Indentations and cuts. A keyhole contour states the upper and lower boundary values of every multivalued factor and the direction of each segment. An omitted arc is discarded only with a displayed bound.

Saddles, phase functions, and steepest descent

Phase-amplitude form. A saddle integral is written

$$I(\lambda) = \int_{\Gamma} A(z, \lambda) e^{\lambda \Phi(z)} dz,$$

with λ 's limiting regime, the oriented contour Γ , and analytic domains of A and Φ stated.

Saddle. A simple saddle z_* satisfies $\Phi'(z_*) = 0$ and $\Phi''(z_*) \neq 0$. Degenerate saddles state the first nonzero derivative order.

Descent direction. A steepest-descent path has $\text{Im } \Phi$ constant and $\text{Re } \Phi$ decreasing away from the saddle. The square root in the Gaussian factor is fixed by the contour orientation.

Leading term. For a single nondegenerate contributing saddle, the schematic factor

$$A(z_*, \lambda) e^{\lambda \Phi(z_*)} \sqrt{\frac{2\pi}{-\lambda \Phi''(z_*)}}$$

is meaningful only after selecting the square-root branch and verifying the contour deformation. It is not a universal sign-free formula.

Multiple saddles. A set $\mathcal{S}(\lambda)$ lists all saddles; a contributing subset additionally depends on the original contour and Stokes region. Dominance is decided by $\operatorname{Re}(\lambda\Phi(z_*))$, not by modulus of z_* .

Lambert- W parametrization of saddle families

Typical reduction. When a saddle equation is rearranged to $ue^u = z(\lambda)$, its solutions are $u_k = W_k(z(\lambda))$, $k \in \mathbb{Z}$.

Back-substitution. If $u = \alpha z + \beta$, then $z_k = (W_k(z(\lambda)) - \beta)/\alpha$ with $\alpha \neq 0$. Every scaling and sign is retained; a Lambert solution is not the original saddle until this substitution is made.

Branch enumeration. The integer k labels Lambert branches, not necessarily the order in which saddle contributions dominate. A separate ordering permutation is used if needed.

Coalescence. At a branch point, two branch-labelled saddles may coalesce. Uniform asymptotics then use an Airy or higher canonical model rather than summing two singular Gaussian approximations.

Status. Lambert- W saddle catalogues in frontier documents are **frontier/conjectural** unless an exact contour theorem and error bound are cited.

Singularity analysis and transfer notation

Dominant singularity. For a power series with radius $\rho > 0$, a dominant singularity lies on $|z| = \rho$. There may be more than one; the symbol ρ alone does not choose one.

Delta domain. A dented domain at positive ρ is

$$\Delta(\rho, \eta, \theta) = \{z : |z| < \rho + \eta, z \neq \rho, |\arg(z - \rho)| > \theta\},$$

with $\eta > 0$ and $0 < \theta < \pi/2$. Alternate angular parametrizations must be expanded.

Basic transfer. For $\alpha \notin \{0, -1, -2, \dots\}$,

$$[z^n](1 - z/\rho)^{-\alpha} \sim \frac{n^{\alpha-1}}{\Gamma(\alpha)} \rho^{-n}.$$

The power uses the branch analytic at $z = 0$ in the dented domain.

Periodic singularities. Contributions from every dominant singularity are summed with their phases. Replacing them by the positive-real contribution can erase oscillation or exact zeros.

Remainder. A transfer theorem states the regularity and domain hypotheses that turn a local expansion into a coefficient error estimate; a formal local expansion alone does not supply it.

10 Fractional calculus, Volterra operators, and regularity classes

Riemann–Liouville fractional integrals

Left-sided operator. For $\alpha > 0$, lower terminal $a \in \mathbb{R}$, and suitable f ,

$$(I_{a+}^{\alpha}f)(x) := \frac{1}{\Gamma(\alpha)} \int_a^x (x-t)^{\alpha-1} f(t) \, dt, \quad x > a.$$

Right-sided operator. For upper terminal b ,

$$(I_{b-}^{\alpha}f)(x) := \frac{1}{\Gamma(\alpha)} \int_x^b (t-x)^{\alpha-1} f(t) \, dt, \quad x < b.$$

Order zero. $I_{a+}^0 = I_{b-}^0 = \text{id}$ by extension, not by substitution into the singular gamma-integral formula.

Semigroup. Under integrability hypotheses, $I_{a+}^{\alpha}I_{a+}^{\beta} = I_{a+}^{\alpha+\beta}$ for $\alpha, \beta > 0$. Terminals must match.

Complex order. For complex α with $\text{Re } \alpha > 0$, the kernel power uses the real logarithm because $x - t > 0$.

Riemann–Liouville and Caputo derivatives

Riemann–Liouville. Let $\alpha > 0$ and $m = \lceil \alpha \rceil$. The left derivative is

$$D_{a+}^{\alpha}f := \frac{d^m}{dx^m} I_{a+}^{m-\alpha}f.$$

Caputo. Under sufficient regularity,

$${}^C D_{a+}^{\alpha}f := I_{a+}^{m-\alpha}f^{(m)}.$$

These operators differ by explicit endpoint terms and are not interchangeable.

Integer order. If $\alpha = m \in \mathbb{N}_0$, both reduce to the ordinary m th derivative only under the appropriate endpoint/regularity interpretation.

Endpoint data. A fractional differential equation declares whether initial conditions are ordinary derivatives $f^{(j)}(a)$, fractional traces, or integral conditions. The operator symbol alone does not determine them.

Fourier derivative. On the whole line, a multiplier definition such as $\widehat{D}^{\alpha}f_{\text{ang}}(\omega) = (i\omega)^{\alpha}\widehat{f}_{\text{ang}}(\omega)$ requires a branch for the complex power and is a third, separately named construction.

Volterra integral operators and resolvents

Operator. For a kernel K on $a \leq t \leq x \leq b$,

$$(V_K f)(x) := \int_a^x K(x,t)f(t) \, dt.$$

Convolution kernel. If $K(x,t) = k(x-t)$ and $a = 0$, this is the one-sided convolution $(k *_+ f)(x) = \int_0^x k(x-t)f(t) \, dt$. It is not the whole-line \texttt{Convolution}.

Iterates. $V_K^0 = \text{id}$ and $V_K^{n+1} = V_K \circ V_K^n$. A superscript here is operator composition, not a pointwise power.

Resolvent. When the Neumann series converges, $(\text{id} - \lambda V_K)^{-1} = \sum_{n=0}^{\infty} \lambda^n V_K^n$. The topology and allowed λ are stated.

Kernel resolvent. A resolvent kernel $R_\lambda(x, t)$ represents the operator resolvent after its identity term is separated; conventions that include a Dirac identity kernel are labelled.

Lebesgue, Sobolev, and Hölder spaces

Lebesgue space. $L^p(E)$, $1 \leq p < \infty$, consists of almost-everywhere classes with norm $\|f\|_{L^p(E)} = (\int_E |f|^p)^{1/p}$; L^∞ uses essential supremum. The measure is Lebesgue measure unless printed otherwise.

Smoothness. $C^k(E)$ has continuous derivatives through order $k \in \mathbb{N}_0$; $C^\infty(E) = \bigcap_k C^k(E)$. On a closed interval, derivatives at endpoints are one-sided restrictions of derivatives on a neighborhood unless another convention is stated.

Hölder space. For $0 < \alpha \leq 1$,

$$[f]_{C^{0,\alpha}(E)} = \sup_{x \neq y \in E} \frac{|f(x) - f(y)|}{|x - y|^\alpha}.$$

$C^{k,\alpha}$ applies this seminorm to the k th derivative and includes lower sup norms in its norm.

Sobolev space. $W^{k,p}(E)$ uses weak derivatives through order k in L^p . H^s means the L^2 -based Sobolev space, with Fourier normalization specified when a Bessel-potential norm is displayed.

Local spaces. The suffix *loc* means membership after restriction to every compact subset of the open domain; it imposes no boundary control.

Gevrey classes, flatness, and nonanalyticity

Gevrey class. A smooth f is locally Gevrey of order $s \geq 1$ when every compact K admits $C, A > 0$ such that

$$\sup_{x \in K} |f^{(n)}(x)| \leq C A^n (n!)^s \quad (n \in \mathbb{N}_0).$$

For several variables, $n!$ is replaced by the multi-index factorial.

Analytic class. Order $s = 1$ corresponds locally to real analyticity under the standard derivative criterion. Membership for $s > 1$ does not imply analyticity.

Flatness. f is flat at a if $f^{(n)}(a) = 0$ for every $n \in \mathbb{N}_0$. A nonzero smooth function may be flat, so a zero Taylor series does not identify the function near a .

Fabi endpoints. up is flat at ± 1 and F_{bd} is flat at the transition points 0 and 1 relative to its constant tails. This is compatible with smoothness and with failure of analyticity there.

Quasianalyticity. A Denjoy–Carleman class is declared with its derivative weight sequence M_n ; whether it is quasianalytic is a theorem about that sequence, not synonymous with being Gevrey.

11 Exact Fabius moments and arithmetic normalizations

Even full moments of the up-function

Definition. For $n \in \mathbb{N}_0$,

$$c_n := \int_{\mathbb{R}} t^{2n} \text{up}(t) \, dt = \int_{-1}^1 t^{2n} \text{up}(t) \, dt.$$

The equality of the two integrals uses the support of up.

Initial value. $c_0 = 1$ because up is normalized to integral one. The full odd moments vanish by evenness.

Fourier expansion. With the 2π convention,

$$\widehat{\text{up}}_{2\pi}(z) = \sum_{n=0}^{\infty} \frac{(-1)^n c_n}{(2n)!} (2\pi z)^{2n}.$$

The series is entire; z may be complex.

Canonical integer numerator. Define

$$N_n^{(c)} := c_n (2n+1)!! \prod_{j=1}^n (2^{2j} - 1).$$

The empty product gives $N_0^{(c)} = 1$. Historical prose used a capital-letter sequence that collided with Fabius functions; new prose uses the semantic superscript.

Lean declarations. `momentn` and `momentIntegraln`.

Lean modules and status. Exact arithmetic results are in `Arithmetic.lean`; analytic results are in `AnalyticMoments.lean`. Status: **Lean-proved**.

Normalized half moments

Definition. Set $d_0 = 1$ and, for $n \in \mathbb{N}_{>0}$,

$$d_n := n \int_0^1 t^{n-1} \text{up}(t) \, dt.$$

The separate definition at zero avoids the meaningless factor $0 \int_0^1 t^{-1} \text{up}(t) \, dt$.

Binomial bridge. For $n \in \mathbb{N}_0$,

$$d_n = 2^{-n} \sum_{0 \leq k \leq n/2} \binom{n}{2k} c_k,$$

where the bound means $k = 0, \dots, \lfloor n/2 \rfloor$.

Dyadic-value bridge. For $n \in \mathbb{N}_0$,

$$F_{\text{bd}}(2^{-n}) = \frac{d_n}{n! 2^{\binom{n}{2}}}.$$

An inverse-dyadic value means evaluation at an inverse power of two, not a value of [the inverse function](#).

Canonical integer numerator. Define

$$N_n^{(d)} := d_n (n+1)! \prod_{j=1}^n (2^j - 1).$$

Again the empty product gives $N_0^{(d)} = 1$. A historical capital-letter numerator sequence is recorded as a retired alias rather than reused.

Two-adic value. For $n \geq 1$, the reduced numerator and denominator of $2d_n$ are odd, equivalently $\nu_2(d_n) = -1$ for the rational valuation extended additively to quotients.

Lean crosswalk. `halfMomentn` and `halfMomentIntegraln`. Status: **Lean-proved**.

Reshetnikov's integer normalization

Definition. For $n \in \mathbb{N}_{>0}$,

$$R_n^{\text{Res}} := 2^{\binom{n-1}{2}} (2n)! F_{\text{bd}}(2^{-n}) \prod_{j=1}^{\lfloor n/2 \rfloor} (2^{2j} - 1).$$

Type and theorem. $R_n^{\text{Res}} \in \mathbb{N}_{>0}$ and is odd. Its superscript is mandatory in cross-topic documents because R_n also denotes polynomial remainders, resolvents, and recurrence polynomials.

Factors. The power of two is triangular, the factorial is $(2n)!$, and the Mersenne-type product stops at $\lfloor n/2 \rfloor$. Removing or shifting any of these produces a different normalization.

Lean crosswalk. `reshetnikovn` and the corresponding integrality, positivity, and oddness theorems in `Arithmetic.lean` and `PaperStatements.lean`. Status: **Lean-proved**.

Dyadic common denominators

Grid. For $n \in \mathbb{N}_{>0}$, let $A_n = \{a \in \mathbb{N}_{>0} : a < 2^n, a \text{ odd}\}$.

Reduced denominator. For $r \in \mathbb{Q}$, write $r = p/q$ with $q \in \mathbb{N}_{>0}$ and $\gcd(p, q) = 1$; then $\text{den}(r) = q$, including $\text{den}(0) = 1$.

Definition. The semantic dyadic denominator is

$$D_n^{\text{dyad}} := \text{lcm}\{\text{den}(\text{up}(a/2^n)) : a \in A_n\}.$$

At $n = 0$ the index set is empty; the catalogue assigns $D_0^{\text{dyad}} = 1$, the LCM identity, when a zero-level value is needed.

Reflection. The same LCM is obtained from the corresponding bounded-Fabius values after applying the exact reflection/translation relation.

Collision policy. A bare D_n is never globally canonical: it also denotes determinants and asymptotic main terms. The semantic superscript is required outside a single-purpose local scope.

Lean declaration. `dyadicDenominatorn`.

Lean modules and status. The definition is in `Arithmetic.lean`.

Bounds are in `DenominatorBound.lean`. Status: **Lean-proved**.

12 Approximation, orthogonal polynomials, splines, and quadrature

Weights and polynomial inner products

Measure form. For a positive Borel measure μ with finite needed moments,

$$\langle f, g \rangle_\mu := \int f(x) \overline{g(x)} \, d\mu(x).$$

The inner product is linear in the first argument and conjugate-linear in the second in this catalogue.

Weight form. If $d\mu(x) = w(x) \, dx$ on an interval I , write $\langle f, g \rangle_w = \int_I f(x) \overline{g(x)} w(x) \, dx$, with $w \geq 0$ almost everywhere.

Norm. $\|f\|_\mu = \sqrt{\langle f, f \rangle_\mu}$. It is a norm only after quotienting functions equal μ -almost everywhere or restricting to a nondegenerate function space.

Fabius weights. When the measure has density up on $[-1, 1]$, write $\langle f, g \rangle_{\text{up}}$ after declaring that measure. An unlabeled upright-word subscript is retired because it bypasses the semantic command. A bounded-Fabius Stieltjes measure is a different but related object.

Orthogonal, monic, and orthonormal polynomials

Orthogonal family. Polynomials $(p_n)_{n \in \mathbb{N}_0}$ satisfy $\deg p_n = n$ and $\langle p_n, p_m \rangle_\mu = 0$ for $n \neq m$.

Monic normalization. P_n is monic when its coefficient of x^n is 1. This fixes an orthogonal polynomial uniquely when the moment functional is positive definite.

Orthonormal normalization. π_n is orthonormal when $\langle \pi_n, \pi_m \rangle_\mu = \delta_{nm}$ and its leading coefficient is positive. Thus $\pi_n = P_n / \|P_n\|_\mu$.

Three-term recurrence. For monic polynomials,

$$P_{n+1}(x) = (x - \alpha_n)P_n(x) - \beta_n P_{n-1}(x),$$

with $P_{-1} = 0$, $P_0 = 1$, $\alpha_n \in \mathbb{R}$, and $\beta_n > 0$ for $n \geq 1$. Orthonormal recurrence coefficients use a different normalization.

Classical names. Legendre, Chebyshev, Jacobi, and Gegenbauer polynomials carry their standard family parameters only after their normalization is displayed; several incompatible leading-value conventions exist.

Legendre and Chebyshev normalizations

Legendre. P_n^{Leg} is defined by $P_n^{\text{Leg}}(1) = 1$ and orthogonality for weight 1 on $[-1, 1]$. Its leading coefficient is $2^{-n} \binom{2n}{n}$, so it is not monic for $n > 1$.

First Chebyshev family. $T_n(\cos \theta) = \cos(n\theta)$, $T_n(1) = 1$, orthogonal for $(1 - x^2)^{-1/2}$ on $[-1, 1]$.

Second Chebyshev family. $U_n(\cos \theta) = \sin((n+1)\theta) / \sin \theta$, with the continuous endpoint values, orthogonal for $(1 - x^2)^{1/2}$.

Coefficient convention. A Legendre coefficient may be the raw inner product $\langle f, P_n^{\text{Leg}} \rangle$ or the expansion coefficient $(2n+1) \langle f, P_n^{\text{Leg}} \rangle / 2$. The corpus prints which one it means.

Collision. The letter P_n is local: it also denotes probability, prefix polynomials, and Padé denominators. A family superscript is required in a cross-topic statement.

Legendre Gaunt coefficients and the squared zero-row Wigner $3j$ datum

Indices and normalization. Throughout this entry $i, j, k \in \mathbb{N}_0$, and $P_n^{\text{Leg}}(1) = 1$. The Gaunt coefficient is

$$G_{ijk} := \int_{-1}^1 P_i^{\text{Leg}}(x) P_j^{\text{Leg}}(x) P_k^{\text{Leg}}(x) \, dx.$$

It is not a coefficient for monic or orthonormal Legendre polynomials.

Admissible support. Write $\text{Adm}(i, j, k)$ for the simultaneous conditions

$$i + j + k \equiv 0 \pmod{2}, \quad i \leq j + k, \quad j \leq i + k, \quad k \leq i + j.$$

Thus admissibility means even total degree and all three weak triangle inequalities; none is implicit. On this support, $s := (i + j + k)/2$ is a natural number and $s - i, s - j, s - k \in \mathbb{N}_0$.

Total rational square datum. Define

$$W_{ijk}^{2,0} := \begin{cases} \frac{\binom{2(s-i)}{s-i} \binom{2(s-j)}{s-j} \binom{2(s-k)}{s-k}}{(2s+1) \binom{2s}{s}}, & \text{Adm}(i, j, k), \\ 0, & \text{otherwise.} \end{cases}$$

This is a total element of $\mathbb{Q}$ and equals the square of the integer-index zero-row Wigner $3j$ datum. The superscript $2, 0$ records “square” and “zero magnetic row”; it does not select a signed Wigner symbol.

Gaunt and product-linearization identities. The exact relations are

$$G_{ijk} = 2W_{ijk}^{2,0}, \quad [P_k^{\text{Leg}}](P_i^{\text{Leg}} P_j^{\text{Leg}}) = (2k+1)W_{ijk}^{2,0},$$

where the bracket denotes the coefficient of P_k^{Leg} in the finite Legendre-basis expansion. Off the admissible support both sides of the first identity vanish.

Triangle coordinates and factorial form. Equivalently, there are $a, b, c \in \mathbb{N}_0$ with $i = b + c$, $j = a + c$, and $k = a + b$. Then

$$W_{ijk}^{2,0} = \frac{\binom{2a}{a} \binom{2b}{b} \binom{2c}{c}}{(2(a+b+c)+1) \binom{2(a+b+c)}{a+b+c}},$$

and expansion of each central binomial coefficient gives the source’s equivalent factorial quotient.

Rvachev Gram use family. Substitution into the finite Rvachev–Legendre Gaunt expansion gives rational Gram entries, rational Gram-matrix entries, and—under a bounded Fabius object plus its Fabius-specification proof—real up-law Gram entries as finite sums of canonical Rvachev–Legendre coefficients times $W_{ij,2r}^{2,0}$. This finite use does not authorize an infinite Legendre-series interchange.

Formal boundary. Only the total rational square of the integer-index, zero-magnetic-row datum is present. There is no sign or Condon–Shortley phase choice, no half-integer spin, no nonzero magnetic index, no general $3j$, $6j$, or $9j$ symbol, and no Wigner orthogonality or recoupling theory.

Lean crosswalk. `legendreGauntAdmissible` and `legendreWignerThreeJZeroSqRat` in `LegendreGauntClosedForm.lean` implement the support predicate and total square datum. The central identities are `legendreGauntRat_eq_two_mul_wignerThreeJZeroSqRat`, `legendreGaunt_eq_two_mul_wignerThreeJZeroSqRat`, and `legendreProductLinearizationCoefficientRat_eq_mul_wignerThreeJZeroSqRat`. `FabiusLegendreGauntClosedForm.lean` supplies the finite rational and real Rvachev Gram-entry sums, with `IsFabius` required for the analytic real formula. Status: **Lean-proved with the displayed hypotheses.**

Moment matrices and Hankel determinants

Moment sequence. For a finite measure μ , $m_n = \int x^n d\mu(x)$ for $n \in \mathbb{N}_0$ whenever finite.

Hankel matrix. The order- N moment matrix is

$$H_N(\mu) := (m_{i+j})_{0 \leq i, j < N} \in \mathbb{C}^{N \times N}, \quad N \in \mathbb{N}_0.$$

Thus H_0 is the unique 0×0 matrix.

Determinant. $\Delta_N(\mu) = \det H_N(\mu)$ and $\Delta_0(\mu) = 1$. A positive measure with infinite support has $\Delta_N > 0$ for every $N \geq 1$.

Shifted matrix. $H_N^{(r)} = (m_{i+j+r})_{0 \leq i, j < N}$ for $r \in \mathbb{N}_0$. The shift belongs in the superscript, not in an unstated change of the moment sequence.

Fabius parity. For the centered up measure, odd moments vanish and even moments are c_n . Reindexing the even block must show whether its entry is c_{i+j} or c_{i+j+r} .

Jacobi matrices and spectral measures

Matrix. For orthonormal polynomials with $x\pi_n = a_{n+1}\pi_{n+1} + b_n\pi_n + a_n\pi_{n-1}$, $a_n > 0$, the Jacobi operator is the tridiagonal matrix

$$J = \begin{pmatrix} b_0 & a_1 & 0 & \cdots \\ a_1 & b_1 & a_2 & \ddots \\ 0 & a_2 & b_2 & \ddots \\ \vdots & \ddots & \ddots & \ddots \end{pmatrix}.$$

Indexing. Rows and columns begin at 0; a_0 does not appear. A finite truncation J_N uses indices $0, \dots, N-1$.

Spectral measure. The measure μ satisfies $\langle e_0, p(J)e_0 \rangle = \int p(x) d\mu(x)$ for polynomials p , under the spectral theorem.

Resolvent. $G_\mu(z) = \langle e_0, (z \text{ id} - J)^{-1} e_0 \rangle$ matches the Cauchy-transform denominator $z - x$.

Frontier status. Any claimed closed recurrence coefficients for the Fabius measure remain **frontier/conjectural** unless linked to a proved Lean declaration.

Knots, B-splines, degree, and order

Knot vector. A knot vector is a nondecreasing finite sequence $\tau = (\tau_0, \dots, \tau_m)$. Repeated knots are allowed and their multiplicity affects smoothness.

Degree versus order. A B-spline of degree $p \in \mathbb{N}_0$ has order $p+1$. The corpus always says degree or order; a bare index is not assumed to be one or the other.

Cox-de Boor base case. $B_{i,0}(x) = \mathbf{1}_{[\tau_i, \tau_{i+1})}(x)$, except that a designated final endpoint may be closed to make a partition of unity on the full knot span.

Recursion. For $p \geq 1$,

$$B_{i,p}(x) = \frac{x - \tau_i}{\tau_{i+p} - \tau_i} B_{i,p-1}(x) + \frac{\tau_{i+p+1} - x}{\tau_{i+p+1} - \tau_{i+1}} B_{i+1,p-1}(x),$$

where a term with zero denominator is defined to be zero.

Support. $\text{supp}(B_{i,p}) \subseteq [\tau_i, \tau_{i+p+1}]$. Equality needs nondegenerate knot spans.

Cardinal splines. Uniform integer knots and a translated normalization are declared before using the shorter cardinal B-spline symbol.

Divided differences

Distinct nodes. For pairwise distinct $x_0, \dots, x_n$,

$$[x_0, \dots, x_n]f = \sum_{j=0}^n \frac{f(x_j)}{\prod_{0 \leq k \leq n, k \neq j} (x_j - x_k)}.$$

Recursive form. $[x_j]f = f(x_j)$ and $[x_0, \dots, x_n]f = ([x_1, \dots, x_n]f - [x_0, \dots, x_{n-1}]f)/(x_n - x_0)$.

Repeated nodes. If nodes coalesce and f is sufficiently differentiable, $\underbrace{[x, \dots, x]}_{n+1}f = f^{(n)}(x)/n!$. This

is a continuous extension, not substitution into a zero denominator.

Spline relation. B-splines expressed as divided differences state the truncated-power normalization and knot multiplicities.

Quadrature nodes, weights, and exactness

Rule. An N -node quadrature rule for μ is $Q_N(f) = \sum_{j=1}^N w_{j,N} f(x_{j,N})$, with nodes and weights indexed from 1.

Degree of exactness. It has degree of exactness d when $Q_N(p) = \int p \, d\mu$ for all polynomials of degree at most d , and this fails for at least one polynomial of degree $d + 1$.

Gaussian rule. For a positive measure, Gaussian nodes are the zeros of the degree- N orthogonal polynomial; the rule has positive weights and degree $2N - 1$.

Endpoint variants. Gauss–Radau fixes one endpoint and Gauss–Lobatto fixes both. Their exactness degrees and endpoint weights differ from the Gaussian rule and are printed separately.

Error. An expression $E_N(f) = \int f \, d\mu - Q_N(f)$ fixes the sign. The derivative form of an error requires the exact smoothness interval and normalization constant.

Shifted dyadic Rvachev self-sampling. Let F be a bounded Fabius object satisfying the Fabius specification, $N \in \mathbb{N}_0$, $P \in \mathbb{R}[X]$ with $\deg P \leq N$, and $\theta \in \mathbb{R}$. Then

$$\int_{\mathbb{R}} P(x) \operatorname{up}_F(x) \, dx = 2^{-N} \sum_{k \in \mathbb{Z}} P(2^{-N}(\theta + k)) \operatorname{up}_F(2^{-N}(\theta + k)).$$

The right side is formally a bilateral infinite sum, but compact support makes it finite. The theorem proves exactness for every polynomial of degree at most N ; it does not assert failure in degree $N + 1$, so it does not by itself establish that the degree of exactness is exactly N .

Lean crosswalk. The displayed rule is `integral_polynomial_mul_rvachevUp_eq_dyadic_tsum` in `PolynomialCombExactness.lean`; its normalized source identity is `tsum_shifted_polynomial_eq_integral`. Both retain the bounded-Fabius and Fabius-specification hypotheses. Status: **Lean-proved**.

Padé approximation

Definition. For a formal series $f(z) = \sum_{n \geq 0} a_n z^n$, a type $[L/M]$ Padé approximant is a rational function $P_L(z)/Q_M(z)$ with $\deg P_L \leq L$, $\deg Q_M \leq M$, $Q_M(0) = 1$, and

$$Q_M(z)f(z) - P_L(z) = \mathcal{O}_{z \rightarrow 0}(z^{L+M+1}).$$

Existence and uniqueness. A normalized pair may fail to exist with exact degrees or may have a common factor. The rational function is unique under the standard normality hypothesis, which is stated when used.

Poles. Zeros of Q_M after cancellation are poles of the approximant. Uncancelled numerator-denominator zero pairs are not counted as poles.

Collision policy. P, Q are local polynomial names; they do not denote probability or the Fabius quantile inside this entry. Outside it, the type- $[L/M]$ subscript or semantic label is retained.

13 Linear algebra, operators, kernels, and spectral notation

Vectors, matrices, and indices

Vectors. A column vector is $x = (x_1, \dots, x_n)^\top \in \mathbb{K}^n$, where $\mathbb{K}$ is explicitly $\mathbb{R}$ or $\mathbb{C}$. Algorithmic zero-based coordinates are marked separately.

Matrices. $A = (a_{ij})_{1 \leq i \leq m, 1 \leq j \leq n} \in \mathbb{K}^{m \times n}$. The first dimension is the number of rows.

Transpose and adjoint. $A^\top$ is transpose, $\overline{A}$ entrywise conjugation, and $A^* = \overline{A}^\top$ conjugate transpose. For real matrices, $A^* = A^\top$.

Identity and zeros. I_n is the $n \times n$ identity and $0_{m \times n}$ the rectangular zero matrix. A bare 0 inherits type only inside a displayed typed equation.

Empty size. The unique 0×0 matrix has determinant 1, trace 0, and rank 0. Matrix products with a zero inner dimension are zero matrices by empty sums.

Trace, rank, determinant, and diagonal

Trace. For $A \in \mathbb{K}^{n \times n}$, $\text{tr } A = \sum_{j=1}^n a_{jj}$, with $\text{tr } A = 0$ at $n = 0$.

Trace-class extension. For a trace-class operator T on a real or complex Hilbert space and any orthonormal basis (e_j) , $\text{tr } T := \sum_j \langle T e_j, e_j \rangle$. Trace class makes this series absolutely convergent and its value independent of the orthonormal basis; in finite dimension it agrees with the matrix trace above. No trace is inferred for an arbitrary bounded infinite-dimensional operator.

Rank. $\text{rank } A$ is the dimension of the image of the associated linear map. Row rank and column rank agree over a field.

Determinant. $\det A$ is defined only for a square matrix or an endomorphism of a finite free module with a chosen basis-free determinant construction.

Diagonal. $\text{diag}(x_1, \dots, x_n)$ is the square diagonal matrix with those diagonal entries. Applying `\DiagonalOperator` to a matrix in order to extract a vector is forbidden; extraction is written $\text{diagvec}(A)$ to prevent an input-type overload.

Span. For vectors v_j , $\text{span}_{\mathbb{K}}\{v_j : j \in J\}$ is their linear span over $\mathbb{K}$. The scalar field is printed when not obvious.

Tensor, Kronecker, and Hadamard products

Tensor product. $V \otimes_{\mathbb{K}} W$ is the algebraic tensor product of vector spaces; completed topological tensor products carry a norm or topology subscript.

Kronecker product. For matrices $A = (a_{ij})$ and B , $A \otimes B$ is the block matrix $(a_{ij}B)$. Its dimensions are the pairwise products of the input dimensions.

Hadamard product. $A \circ B = (a_{ij}b_{ij})$ for matrices of identical shape. It is not composition.

Composition. $S \circ T$ means first T , then S . When Hadamard products occur in the same section, they use $\odot$ instead of $\circ$.

Powers. $A^0 = I_n$ for square $A \in \mathbb{K}^{n \times n}$, including the zero matrix. Entrywise powers are written $A^{\circ k}$.

Vector, matrix, and operator norms

Euclidean norm. For $x \in \mathbb{C}^n$, $\|x\|_2 = (\sum_j |x_j|^2)^{1/2}$.

Induced norm. For $A : \mathbb{K}^n \rightarrow \mathbb{K}^m$, $\|A\|_{p \rightarrow q} = \sup_{x \neq 0} \|Ax\|_q / \|x\|_p$. The spectral norm is $\|A\|_{2 \rightarrow 2}$.

Frobenius norm. $\|A\|_F = (\sum_{i,j} |a_{ij}|^2)^{1/2} = (\text{tr } A^* A)^{1/2}$.

Unsubscripted norm. The shared `\Norm` supplies delimiters only. Each local scope declares the norm behind them; it is not globally the Euclidean or supremum norm.

Condition number. For invertible A , $\kappa_p(A) = \|A\|_{p \rightarrow p} \|A^{-1}\|_{p \rightarrow p}$; for singular A it is $+\infty$.

Spectrum, eigenvalues, and resolvents

Finite spectrum. For $A \in \mathbb{C}^{n \times n}$, $\sigma(A) = \{\lambda \in \mathbb{C} : \det(\lambda I_n - A) = 0\}$.

Multiplicity. Algebraic multiplicity is root multiplicity in the characteristic polynomial; geometric multiplicity is $\dim \ker(A - \lambda I_n)$. They are not conflated.

Spectral radius. $\rho(A) = \max_{\lambda \in \sigma(A)} |\lambda|$ for $n > 0$; for the 0×0 matrix the spectrum is empty and the catalogue sets $\rho(A) = 0$ when a total convention is required.

Operator spectrum. For a bounded operator T on a complex Banach space, $\sigma(T) = \{\lambda : \lambda \text{ id} - T \text{ is not boundedly invertible}\}$.

Resolvent. $R_T(z) = (z \text{ id} - T)^{-1}$ for $z \notin \sigma(T)$. This operator-valued R_T is distinct from R_n^{Res} .

Hankel and Toeplitz matrices

Hankel. A finite Hankel matrix has entries $H_{ij} = a_{i+j}$, with declared index origins. In this catalogue $0 \leq i, j < N$ is the default.

Toeplitz. A Toeplitz matrix has entries $T_{ij} = a_{i-j}$ and therefore requires a two-sided sequence for zero-based finite indices.

Moment interpretation. A Hankel matrix (m_{i+j}) is positive semidefinite when (m_n) is a moment sequence of a positive real measure.

Operator interpretation. Infinite Hankel or Toeplitz matrices define operators only after a sequence space and boundedness/domain conditions are supplied.

Collision. A transform, matrix, and determinant may all be called H in source literature. Cross-topic prose uses $H_N(\mu)$, $T_N(a)$, or a semantic operator subscript.

Minors and total positivity

Minor. For row index set I and column index set J of equal finite size, $A[I, J]$ is the corresponding submatrix and $\Delta_{I,J}(A) = \det A[I, J]$. Both index sets are listed in increasing order.

Totally nonnegative. A real matrix is totally nonnegative if every minor is nonnegative.

Totally positive. It is totally positive if every minor that is not forced to vanish by the matrix shape is positive. For a rectangular matrix, all square orders up to $\min(m, n)$ are included.

Strict versus order r . The suffix TP_r restricts to minors of order at most r ; TN_r is the nonnegative analogue. Neither implies unqualified total positivity when r is finite below matrix size.

Kernel version. A kernel $K(x, y)$ is totally positive when determinants $\det(K(x_i, y_j))$ have the specified sign for strictly ordered point tuples. Weak orders and repeated nodes require a confluent formulation.

Status. Total-positivity claims for Fabius moment or collocation matrices are **frontier/conjectural** unless their exact minor family is linked to a proof.

Positive-definite kernels and RKHS notation

Positive definite. A Hermitian kernel $K : X \times X \rightarrow \mathbb{C}$ is positive semidefinite if $\sum_{i,j} \bar{c}_i c_j K(x_i, x_j) \geq 0$ for all finite choices.

Strictness. Strict positive definiteness requires positivity for nonzero coefficient vectors when the points are pairwise distinct.

RKHS. The reproducing-kernel Hilbert space $\mathcal{H}_K$ satisfies $f(x) = \langle f, K(\cdot, x) \rangle_{\mathcal{H}_K}$ under the catalogue's first-slot-linear convention. References using second-slot linearity conjugate this identity.

Gram matrix. $G_X = (K(x_i, x_j))_{i,j=1}^N$. Its rank may be smaller than N even for a positive-semidefinite kernel.

Collision. Kernel K , compact set K , and field $\mathbb{K}$ may coexist only with type-bearing subscripts or a local declaration.

14 Research-frontier namespaces

The entries in this section define the symbols needed to read the semi-formalized frontier documents. They do not assert that the proposed Fabius specializations are theorems. Every such specialization carries one of the proof-status labels in [section 1.3](#).

Appell and Sheffer polynomial families

Appell family. A polynomial sequence $(A_n)_{n \in \mathbb{N}_0}$ is Appell when A_0 is a nonzero constant and $A'_n(x) = nA_{n-1}(x)$ for $n \geq 1$.

Generating function. With the normalization $A_0 = 1$,

$$\sum_{n=0}^{\infty} A_n(x) \frac{z^n}{n!} = g(z)e^{xz}, \quad g(0) = 1,$$

as a formal identity or analytic identity on a printed disk.

Associated law. If $g(z) = 1/M_X(z)$, the family is associated with a law only where the reciprocal series exists; this does not imply that all A_n are orthogonal.

Sheffer family. A Sheffer sequence has EGF $g(z) \exp(xf(z))$ with $g(0) \neq 0$, $f(0) = 0$, and $f'(0) \neq 0$. Appell is the case $f(z) = z$.

Collision. The local A_n is neither a matrix nor an event. A Fabius-derived family carries the semantic superscript Fab outside its defining section.

Lagrange and cardinal interpolation

Nodes. Let $x_0, \dots, x_N$ be pairwise distinct scalars. The cardinal polynomial at node j is

$$\ell_j(x) = \prod_{\substack{0 \leq k \leq N \\ k \neq j}} \frac{x - x_k}{x_j - x_k}, \quad \ell_j(x_k) = \delta_{jk}.$$

Interpolant. $I_N f(x) = \sum_{j=0}^N f(x_j) \ell_j(x)$ is the unique polynomial of degree at most N matching f at all nodes.

Empty case. For $N = 0$, the product is empty and $\ell_0 = 1$. There is no $N = -1$ interpolant unless a separate empty-data convention is introduced.

Barycentric weights. $w_j = (\prod_{k \neq j} (x_j - x_k))^{-1}$ and $I_N f(x) = (\sum_j w_j f(x_j) / (x - x_j)) / (\sum_j w_j / (x - x_j))$ away from nodes, with the value at a node assigned by continuity. Scaling every w_j by the same nonzero constant changes neither interpolant.

Cardinal infinite grids. An infinite cardinal function L satisfying $L(k) = \delta_{0k}$ for $k \in \mathbb{Z}$ is a different object and requires a function space and convergence mode for $\sum_k f(k) L(x - k)$.

Finite Rvachev decoder. Let s be a finite index set, $v : s \rightarrow \mathbb{R}$ a node map, and ℓ_i the Lagrange basis polynomial for $i \in s$. With $\mathcal{D}_{\text{up}}$ the Rvachev deconvolution operator, $M \in \mathbb{N}_{>0}$, and $k \in \mathbb{Z}$, define

$$D_{ki}^{s,v,M} := (\mathcal{D}_{\text{up}} \ell_i)(k/M).$$

For nodal data $y : s \rightarrow \mathbb{R}$, the corresponding atom coefficient is

$$a_k^{s,v,y,M} := \frac{1}{M} \sum_{i \in s} D_{ki}^{s,v,M} y_i.$$

The superscripts display the finite node family, node locations, data, and mesh; a source that suppresses them must declare those objects in the immediately governing scope.

Exact finite synthesis. For a bounded Fabius object F satisfying the Fabius specification, $i \in s$, $M \neq 0$, $|s| - 1 \leq \nu_2(M)$, and $x \in [-1, 1]$,

$$\frac{1}{M} \sum_{-2M < k < 2M} D_{ki}^{s,v,M} \text{up}_F(x - k/M) = \ell_i(x).$$

The index set is the open integer interval $-2M < k < 2M$, including neither endpoint. Summing against y_i reconstructs the complete Lagrange interpolant from the coefficients $a_k^{s,v,y,M}$. Node injectivity is unnecessary for the polynomial reconstruction identity but is required for the Kronecker-delta cardinal interpretation; the unit decoder-row-sum theorem additionally assumes $s \neq \emptyset$.

Lean crosswalk. `lagrangeRvachevDecoder` and `lagrangeRvachevAtomCoefficient` in `LagrangeRvachevSynthesis.lean` implement the two displayed definitions. The exact reconstruction, node-evaluation, coefficient/deconvolution, full-interpolation, and row-sum results are its named `normalized_sum_Ioo_lagrangeRvachevDecoder_mul_shifted_rvachevUp`, `normalized_sum_Ioo_lagrangeRvachevDecoder_eval_node`, `lagrangeRvachevAtomCoefficient_eq_deconvolved_interpolate`, `sum_Ioo_lagrangeRvachevAtomCoefficient_mul_shifted_rvachevUp`, and `sum_lagrangeRvachevDecoder_eq_one` theorems. Lean's definitions remain total at $M = 0$ through field division; the reconstruction theorem requires $M \neq 0$. No optimized decoder, bundled matrix inverse, Gaussian q -binomial closed form, or minimum-variation solution is supplied. Status: **Lean-proved with the displayed hypotheses**.

Sampling, reconstruction, and band limitation

Band limitation. Under the 2π Fourier convention, f is bandlimited to $[-B, B]$ when $\text{supp}(\widehat{f}_{2\pi}) \subseteq [-B, B]$.

Critical lattice. The Shannon lattice spacing is $(2B)^{-1}$ for $B > 0$. With suitable L^2 or regularity hypotheses,

$$f(x) = \sum_{k \in \mathbb{Z}} f\left(\frac{k}{2B}\right) \text{sinc}_\pi(2Bx - k).$$

The theorem states whether convergence is in L^2 , uniform, or pointwise.

Oversampling. A denser lattice adds frame redundancy; it does not change the definition of `\SincPi`.

Fabius caveat. Compact support in physical space makes the Fourier transform entire, not bandlimited. Sampling formulas in the inverse-Fabius frontier therefore require an independently proved interpolation or frame theorem.

Alias policy. A sample index k , a frequency index, and a Lambert branch index may all be integers; each sum declares which role its index has.

Refinement equations and dilation operators

Dilation. For nonzero real a , $(D_a f)(x) = f(ax)$. A unitary L^2 dilation is instead $(U_a f)(x) = |a|^{1/2} f(ax)$; the two are not conflated.

Translation. $T_b f(x) = f(x - b)$. Thus $D_a T_b = T_{b/a} D_a$ for $a \neq 0$ under this convention.

Refinement equation. A refinable function satisfies

$$\phi(x) = \sum_{k \in \mathbb{Z}} h_k \phi(ax - k)$$

with dilation $|a| > 1$, a declared mask (h_k) , and a stated convergence/pointwise interpretation. A factor such as $\sqrt{|a|}$ or $|a|$ is part of the mask normalization and is never implicit.

Mask symbol. $H(z) = \sum_k h_k z^k$ is a Laurent polynomial or series according to the mask support. Its Fourier-domain refinement law depends on the selected Fourier normalization.

Fabius scaling. The equation $F'_{\text{gl}}(x) = 2F_{\text{gl}}(2x)$ is a differential scaling law, not by itself a standard refinability equation for F_{gl} .

Function iterates, orbits, and conjugacy

Iterate. For a self-map $f : X \rightarrow X$, $f^{\circ 0} = \text{id}_X$ and $f^{\circ(n+1)} = f \circ f^{\circ n}$. The circle distinguishes iteration from pointwise power $f(x)^n$.

Orbit. The forward orbit of x is $\mathcal{O}_f^+(x) = \{f^{\circ n}(x) : n \in \mathbb{N}_0\}$.

Periodic point. x has least period $p \in \mathbb{N}_{>0}$ when $f^{\circ p}(x) = x$ and no smaller positive exponent does. A fixed point has period one.

Conjugacy. Maps $f : X \rightarrow X$ and $g : Y \rightarrow Y$ are conjugate through a bijection $h : X \rightarrow Y$ when $h \circ f = g \circ h$. If h is only a surjection, the relation is a semiconjugacy.

Fabius inverse. Iterations of `\FabiusQuantile` stay in $[0, 1]$. Iterations of the clamped inverse on all reals are a different dynamical system and use `\FabiusClampedQuantile` explicitly.

Koopman and transfer operators

Koopman. For a measurable map $T : X \rightarrow X$, $(\mathcal{K}_T f)(x) = f(Tx)$. Its function space and measure are part of the operator type.

Perron–Frobenius. A transfer operator $\mathcal{P}_T$ is defined by the duality

$$\int (\mathcal{P}_T f) g \, d\mu = \int f(g \circ T) \, d\mu$$

when such a density operator exists. It is not automatically pointwise inverse composition.

Invariant measure. μ is T -invariant when $T_{\#}\mu = \mu$, equivalently $\int f \circ T \, d\mu = \int f \, d\mu$ for bounded measurable f .

Spectrum. A Koopman eigenfunction satisfies $\mathcal{K}_T f = \lambda f$. Spectral claims state the Banach or Hilbert space because the spectrum changes with it.

Status. Fabius Koopman models in the frontier corpus are **frontier/conjectural** unless an operator, invariant measure, and proved spectral statement are all linked.

Stein operators and zero-bias transforms

Stein operator. An operator $\mathcal{A}$ characterizes a law μ on a test class $\mathcal{T}$ when $\int \mathcal{A}f \, d\mu = 0$ for every $f \in \mathcal{T}$, together with a uniqueness theorem. The operator alone is not a characterization without the class.

Stein equation. For a test h , $\mathcal{A}f_h = h - \int h \, d\mu$. A bound on f_h specifies its norm and the class of h .

Zero bias. For mean-zero X with variance $\sigma^2 > 0$, a zero-biased variable X^* satisfies

$$\mathbb{E}[X f(X)] = \sigma^2 \mathbb{E}[f'(X^*)]$$

for an appropriate class of absolutely continuous f .

Iteration. A zero-bias tower declares how centering and variance renormalization are applied at every stage; the star is a distributional transform, not complex conjugation.

Status. Proposed Fabius Stein characterizations and zero-bias towers retain **frontier/conjectural** status until uniqueness and domain conditions are proved.

Carleman linearization and Carleman matrices

Analytic map. Let $f(z) = \sum_{n \geq 0} a_n z^n$ near a fixed expansion point. The Carleman matrix is

$$C(f)_{mn} := [z^n]f(z)^m, \quad m, n \in \mathbb{N}_0.$$

Zeroth row. Since $f(z)^0 = 1$, $C(f)_{0n} = \delta_{0n}$, including the convention $0^0 = 1$ inside the polynomial identity.

Composition. Formally, $C(f \circ g) = C(f)C(g)$ when both sides are defined with compatible expansion points and the infinite matrix product is coefficientwise finite or convergent.

Truncation. $C_N(f) = (C(f)_{mn})_{0 \leq m, n < N}$ is a finite truncation, not the exact Carleman operator. Spectral convergence of truncations requires a theorem.

Collision. This Carleman matrix is distinct from the Carleman condition $\sum_n m_{2n}^{-1/(2n)} = \infty$ for moment determinacy.

Holonomic and D -finite functions

D -finite. A formal power series $f(z)$ over a characteristic-zero field is D -finite when it satisfies

$$p_r(z)f^{(r)}(z) + \cdots + p_0(z)f(z) = 0$$

for polynomials p_j , not all zero.

P -recursive. A sequence (a_n) is P -recursive when $\sum_{j=0}^r p_j(n)a_{n+j} = 0$ for all sufficiently large n , with polynomial coefficients $p_j(n)$.

Equivalence. Over characteristic zero, an ordinary generating function is D -finite exactly when its coefficient sequence is P -recursive, subject to the standard formal-series interpretation.

Nonholonomicity. Failure of one guessed differential equation does not prove nonholonomicity. A nonholonomicity claim identifies a rigorous obstruction such as infinitely many singularities or incompatible coefficient asymptotics.

Status. Fabius holonomicity questions in the active frontier reports are **open** unless a theorem is explicitly cited.

Chaos expansions, cumulants, and Edgeworth terms

Orthogonal chaos. A chaos decomposition $L^2(\Omega) = \bigoplus_{n \geq 0} \mathcal{H}_n$ is relative to a specified probability structure. J_n denotes the orthogonal projection onto $\mathcal{H}_n$ only within that scope.

Cumulant generating function. For an MGF nonzero near zero,

$$K_X(t) = \log M_X(t) = \sum_{r \geq 1} \kappa_r(X) \frac{t^r}{r!}, \quad K_X(0) = 0,$$

where the analytic logarithm is the branch normalized by the displayed value at zero.

Standardization. For variance $\sigma^2 > 0$, $Z = (X - \mathbb{E}X)/\sigma$. Standardized cumulants are $\lambda_r = \kappa_r(X)/\sigma^r$ for $r \geq 3$.

Edgeworth expansion. An order- s Edgeworth approximation identifies the normal density or CDF, Hermite-polynomial normalization, scaling parameter, and a uniform remainder regime. The formal polynomial correction alone is not a probability density theorem.

Hermite choice. Probabilists' Hermite polynomials satisfy $\text{He}_n(x) = (-1)^n e^{x^2/2} \frac{d^n}{dx^n} e^{-x^2/2}$. Physicists' polynomials are separately labelled H_n^{phys} .

Large deviations and rate functions

Log-MGF. $\Lambda(t) = \log \mathbb{E}[e^{tX}]$ on its effective domain $\mathcal{D}_\Lambda = \{t : M_X(t) < \infty\}$.

Legendre–Fenchel transform. $\Lambda^*(x) = \sup_{t \in \mathbb{R}} \{tx - \Lambda(t)\} \in [0, \infty]$.

LDP. A sequence X_n obeys a large-deviation principle with speed $a_n \rightarrow \infty$ and rate I when closed-set upper and open-set lower logarithmic bounds are stated explicitly.

Good rate function. I is good when every sublevel set $\{x : I(x) \leq c\}$ is compact. Convexity is not part of the definition of a general rate function.

Endpoint scaling. An endpoint or small-ball asymptotic for $F_{\text{bd}}(x)$ as $x \downarrow 0$ is not called an LDP until a family, speed, and normalization have been identified.

Entropy, relative entropy, and Fisher information

Differential entropy. For a density f , $h(f) = - \int f(x) \log f(x) \, dx$ when the positive and negative parts are well defined, with $0 \log 0 = 0$ by continuity. It can be negative.

Relative entropy. For probability measures $\mu \ll \nu$,

$$D_{\text{KL}}(\mu \parallel \nu) = \int \log \left(\frac{d\mu}{d\nu} \right) d\mu,$$

and it is $+\infty$ when absolute continuity fails or the integral diverges. The order of the arguments matters.

Fisher information. For a differentiable positive density, $J(f) = \int |(\log f)'|^2 f \, dx = \int |f'|^2 / f \, dx$ when finite. Zeros require a weak or extended-value interpretation.

Mutual information. $I(X; Y) = D_{\text{KL}}(\text{Law}((X, Y)) \parallel \text{Law}(X) \otimes \text{Law}(Y))$.

Status. Numerical entropy or Fisher-information tables are **computational evidence** unless accompanied by verified error bounds.

Minkowski sums and zonoids

Minkowski sum. For $A, B \subseteq \mathbb{R}^d$, $A + B = \{a + b : a \in A, b \in B\}$ and $\lambda A = \{\lambda a : a \in A\}$.

Zonotope. A finite zonotope generated by vectors $v_1, \dots, v_N$ is

$$Z_N = \sum_{j=1}^N [0, v_j] = \left\{ \sum_{j=1}^N t_j v_j : 0 \leq t_j \leq 1 \right\}.$$

Zonoid. A zonoid is a Hausdorff limit of zonotopes. A random-vector representation such as $\mathbb{E}[0, X]$ requires integrability and a declared selection/Aumann-expectation convention.

Support function. $h_K(u) = \sup_{x \in K} \langle u, x \rangle$ for a nonempty compact convex set K . It satisfies $h_{A+B} = h_A + h_B$.

Common-digit construction. When coordinates share binary or uniform digits, the joint random series and ambient dimension are declared. Marginal Fabius laws alone do not determine the zonoid or dependence structure.

Radon transforms and projections

Hyperplane Radon transform. For suitable $f : \mathbb{R}^d \rightarrow \mathbb{C}$,

$$(\mathcal{R}f)(\theta, s) = \int_{\{x : \langle x, \theta \rangle = s\}} f(x) \, d\mathcal{H}^{d-1}(x), \quad \theta \in \mathbb{S}^{d-1}, s \in \mathbb{R}.$$

Redundancy. $\mathcal{R}f(-\theta, -s) = \mathcal{R}f(\theta, s)$. A parameter domain may quotient this redundancy or retain it explicitly.

Measure projection. For a measure μ , the one-dimensional projection in direction θ is $(x \mapsto \langle x, \theta \rangle)_{\#} \mu$. Its density, when it exists, is related to the Radon transform but is not assumed.

Fourier slice. With compatible 2π conventions, $(\widehat{\mathcal{R}f})(\widehat{\theta}, \cdot)_{2\pi}(\xi) = \widehat{f}_{2\pi}(\xi\theta)$, where the transform on the right is d -dimensional and uses exponent $-2\pi i \langle x, \eta \rangle$.

Status. Dyadic Radon-profile identifications are **frontier/conjectural** unless their measure and slice normalizations are proved.

Heat traces, spectral zeta functions, and determinants

Eigenvalue list. For a nonnegative self-adjoint operator L with discrete spectrum, eigenvalues are $0 \leq \lambda_0 \leq \lambda_1 \leq \dots$, repeated by multiplicity.

Heat trace. $\Theta_L(t) = \text{tr}(e^{-tL}) = \sum_j e^{-t\lambda_j}$ for $t > 0$ when trace class.

Spectral zeta. $\zeta_L(s) = \sum_{\lambda_j > 0} \lambda_j^{-s}$ initially in its convergence half-plane. Zero eigenvalues are excluded and their multiplicity is recorded separately.

Zeta determinant. If ζ_L is regular at zero, $\det' L = \exp(-\zeta'_L(0))$. The prime means zero modes are omitted; it is not an ordinary finite determinant.

Heat coefficients. An expansion $\Theta_L(t) \sim \sum_j a_j t^{\alpha_j}$ as $t \downarrow 0$ specifies the exponent order, possible logarithmic terms, and whether the expansion is proved or numerically fitted.

Digital spectrum. A digital or dyadic spectrum defined from a sequence or matrix is not automatically the spectrum of a self-adjoint operator; the construction must supply the operator and Hilbert space.

Noncommutative probability and free cumulants

Noncommutative probability space. A pair $(\mathcal{A}, \tau)$ consists of a unital algebra and a unital linear functional τ ; positivity and traciality are additional hypotheses, not automatic.

Moments. For $X \in \mathcal{A}$, $m_n = \tau(X^n)$. Products preserve order; XY need not equal YX .

Free cumulants. Free cumulants (r_n) are defined by

$$m_n = \sum_{\pi \in NC(n)} \prod_{B \in \pi} r_{|B|},$$

where $NC(n)$ is the lattice of noncrossing partitions of $\{1, \dots, n\}$ and the scalar formula assumes a single variable.

Freeness. Subalgebras are free when alternating centered products from different subalgebras have zero τ . It is not classical independence.

Status. Noncommutative Fabius analogues in the frontier corpus are **proposed definitions** or **conjectural**; they are never crosswalked to the commutative Lean object without an explicit construction.

Matrix dilations and multivariate scaling

Expansive matrix. A real $d \times d$ matrix A is expansive when every eigenvalue has modulus greater than one.

Dilation operator. For $f : \mathbb{R}^d \rightarrow \mathbb{C}$, $(D_A f)(x) = f(Ax)$. The unitary L^2 version is $(U_A f)(x) = |\det A|^{1/2} f(Ax)$.

Lattice mask. A matrix refinement equation has the form $\phi(x) = \sum_{k \in \mathbb{Z}^d} h_k \phi(Ax - k)$ with a declared mask and convergence mode.

Fourier scaling. Under the d -dimensional 2π convention, $\widehat{f(A \cdot)}_{2\pi}(\xi) = |\det A|^{-1} \widehat{f}_{2\pi}(A^{-T}\xi)$.

Scalar specialization. For $d = 1$ and $A = (2)$ this reduces to the familiar dyadic scaling. A matrix theorem does not follow from the scalar theorem without controlling anisotropy and lattice geometry.

15 Namespace governance and collision register

The shared commands above are global semantic names. Every other symbol is local and is governed by the rules in this section. A local declaration is not optional merely because a letter is conventional: the active corpus combines probability, analysis, arithmetic, special functions, approximation theory, spectral theory, and proposed frontier constructions, so nearly every conventional letter has several legitimate meanings.

15.1 Scope hierarchy and declaration schema

The narrowest enclosing declaration wins among local names. Scopes, from narrowest to widest, are: a displayed definition, a theorem/lemma/conjecture environment, an algorithm or table, a subsection explicitly headed “Notation”, a section, and finally one standalone document. A fragment included with `\input` does not create an independent preamble or global namespace; it inherits the standalone root’s shared input and may introduce only locally scoped names. A local meaning ends at the closing environment or section boundary unless a later sentence explicitly renews it.

Every local notation declaration supplies the following data. “Not applicable” is a permitted answer; silence is not.

Field	Required content
Kind	Scalar, set, proposition, sequence, function, measure, matrix, operator, formal series, equivalence class, or other fully named mathematical kind.
Type	Domain and codomain, including scalar field, measurable/topological structure, and whether equality is pointwise or modulo an equivalence relation.
Indices	Complete index set, origin, inclusivity of endpoints, and empty-index behavior.
Parameters	Which quantities are fixed and varying; admissible domains, inequalities, integrality, primality, nonvanishing, and branch indices.
Formula	Exact definition with every sum, product, integral, supremum, and infimum bound.
Normalization	Fourier frequency/sign, measure mass, polynomial leading coefficient, orthogonality scale, logarithm/power branch, and any factorial or transform prefactor.
Exceptional cases	Values at zero, endpoints, removable singularities, empty sets, empty matrices, divergent parameters, and nonexistence cases.
Status	Definition, Lean-proved, classical, frontier-derived, conjectural, refuted, historical alias, or numerical evidence.
Scope end	The environment, section, or document boundary after which the letter is available for reassignment.

15.2 High-frequency letter collisions

This register does not ban ordinary local letters. It records the qualification needed when two meanings can meet in one document or in cross-document synthesis.

Glyph	Meanings found in the active corpus	Normative resolution
F	Bounded Fabius function, signed global extension, CDF, Fourier transform, polynomial family, matrix, and optimization functional.	Use F_{bd} and F_{gl} for the two Fabius functions, normalization-labelled Fourier commands for transforms, and $\text{Obj}_{\text{decay},n}$ for the decay objective. A local F_a , F_m , or matrix F_N retains its typed subscript and declaration.
Q	Rational numbers, Fabius quantile, quadrature rule, polynomial, matrix, and a q -dependent quantity.	Use $\mathbb{Q}$ for the number field and Q_F or $Q_{F,\text{cl}}$ for inverses. Quadrature is Q_N only inside its declared numerical-analysis scope; the base of a q -series is lower-case q and always explicit.

Glyph	Meanings found in the active corpus	Normative resolution
N	Natural-number type, positive-integer type, truncation level, matrix size, node count, and asymptotic parameter.	Use $\mathbb{N}_0$ or $\mathbb{N}_{>0}$ for sets. A local N carries a declaration such as $N \in \mathbb{N}_{>0}$ and one role; zero-based and one-based counts are never inferred.
A	Event, matrix, Appell family, algebra, feasible set, and dyadic filtration.	Use $\mathcal{A}_n$ for the canonical dyadic filtration. An event is typed $A \in \mathcal{A}$, a matrix has dimensions, and an Appell polynomial has a degree subscript and preferably a semantic superscript.
D	Derivative, differential operator, dilation, denominator, determinant-derived sequence, and domain.	The upright differential is d ; derivatives are f' or $D^n f$ after declaring D . Dilations use D_a or D_A , and dyadic denominators retain the superscript D_n^{dyad} .
E	Expectation, event, energy, error, and exponential generating function.	Use $\mathbb{E}$ for expectation. Events are roman-labelled or typed in the event σ -field; energies and errors retain descriptive subscripts; an EGF is labelled “egf”.
G	Cauchy transform, generating function, Gram matrix, graph, group, and inverse alias in historical drafts.	A transform is G_μ , a Gram matrix is G_N or G_X , and a group is typed. The Fabius inverse is never bare G ; use the quantile commands.
H	Hilbert space, Hankel matrix, Hermite polynomial, Hessian, entropy, and transfer function.	Use $\mathcal{H}$ for a Hilbert/RKHS space, $H_N(\mu)$ for a finite Hankel matrix, He_n or H_n^{phys} for the two Hermite normalizations, and $\nabla^2 f$ for a Hessian.
I	Identity, indicator, interpolant, interval, information, and inverse.	Use id for an identity map and $\mathbf{1}_A$ for an indicator. Interpolants carry a degree such as I_N , intervals print endpoints, and neither Fabius inverse is denoted by bare I .
K	Kernel, compact set, number field, Koopman operator, and cumulant generating function.	A kernel is $K(x, y)$, a compact set remains $K \subseteq X$, the scalar field is $\mathbb{K}$, the Koopman operator is $\mathcal{K}_T$, and the cumulant generator is $K_X(t)$ only inside its probability scope.
L	Laplace transform, Lagrange/cardinal function, linear operator, likelihood, and slowly varying factor.	Use $\mathcal{L}$ for Laplace. Cardinal functions carry an index, operators declare domain/codomain, and slowly varying functions are labelled L_{sv} when transforms are nearby.
M	Mellin transform, moment-generating function, matrix, moment cutoff, mask, and manifold.	Use $\mathcal{M}$ and M_X for the two transforms. Matrices carry dimensions; cutoffs and masks are locally typed.
P	Probability, polynomial, projection, transition kernel, Padé numerator, and partition.	Use $\mathbb{P}$ for probability. Polynomials carry degree indices, projections are P_V with their range, Markov kernels are $P(x, dy)$, and Padé numerators are P_L only in a declared $[L/M]$ scope.
R	Real-number type, radius, residue, resolvent, recurrence polynomial, remainder, and Reshetnikov integer.	Use $\mathbb{R}$ and Res for the first two semantic roles. Resolvents are $R_T(z)$, remainders carry an error superscript, and the arithmetic integer is R_n^{Res} .
S	Sum, support, Stieltjes transform, saddle action, spline space, and random series.	Use $\text{supp}(f)$ for support and S_μ for Stieltjes transforms. Saddle actions, spline spaces, and random series carry semantic subscripts and declared domains.
T	Translation, transformation, Toeplitz matrix, random time, Thue–Morse object, and truncation.	Translations are T_b , Toeplitz matrices are $T_N(a)$, and the sign sequence uses ε_n . A dynamical map $T : X \rightarrow X$ is typed at first use.
W	Lambert function, Wasserstein distance, Walsh function, and weight.	Use W_k with branch k and $W_p(\mu, \nu)$ with transport exponent p . Walsh functions and weights carry their own indices and domains.
ε	Thue–Morse sign, small positive tolerance, perturbation parameter, and error sequence.	Use ε_n for the sign. A tolerance is quantified $\varepsilon > 0$; perturbations and errors receive semantic subscripts when signs occur in the same scope.

Finite-spline, Fourier-product, and endpoint-kernel notation in the inverse-synthesis volume

Scope. The commands `\FiniteSpline`, `\FiniteSplineInverse`, `\Fn`, `\Gn`, `\PhiAng`, `\PhiCyc`, and `\EndpointKernel` are local to the standalone inverse-analyticity, asymptotics, and computability volume and its included chapters. They are not shared-API commands, and their meanings end with that standalone document.

Finite-prefix probability data. For $k \in \mathbb{N}_{>0}$, let $U_1, U_2, \dots$ be independent random variables uniformly distributed on $[-1, 1]$, set

$$S_k := \sum_{j=1}^k 2^{-j} U_j,$$

and let $p_k : \mathbb{R} \rightarrow [0, \infty)$ be the Lebesgue density of S_k .

Finite spline family. `\FiniteSpline{k}` has TeX arity one and prints F_k^{spl} . Its mathematical argument is $k \in \mathbb{N}_{>0}$, and

$$F_k^{\text{spl}} : [0, 1] \longrightarrow [0, \infty), \quad F_k^{\text{spl}}(x) := p_k(x - 1).$$

The superscript `spl` marks the finite convolution prefix and is never an iteration exponent.

Exact box-spline form. For $x \in [0, 1]$, the same finite spline is the explicit Thue–Morse box spline

$$F_k^{\text{spl}}(x) = \frac{2^{\binom{k}{2}}}{(k-1)!} \sum_{j=0}^{2^k-1} (-1)^{s_2(j)} \left(x - 2^{-k} - j2^{1-k} \right)_+^{k-1}, \quad s_2(j) := \text{wt}_2(j).$$

Here $\text{wt}_2(j)$ is the number of one-bits in the nonnegative integer j , and $(t)_+ := \max\{t, 0\}$. Thus neither the Thue–Morse sign nor the truncation at zero is implicit.

Local inverse family. `\FiniteSplineInverse{k}` has TeX arity one and prints G_k^{spl} . On a stated interval $J \subset (0, 1)$ on which F_k^{spl} is strictly increasing, it denotes the branch

$$G_k^{\text{spl}} : F_k^{\text{spl}}(J) \longrightarrow J, \quad G_k^{\text{spl}} := \left(F_k^{\text{spl}}|_J \right)^{-1}.$$

No inverse value is implied unless the branch interval and monotonicity hypothesis have been supplied. In the asymptotic uses, J is a fixed compact subinterval of $(0, 1)$, and the branch exists for all sufficiently large k .

Fixed-index aliases. `\Fn` and `\Gn` each have TeX arity zero. With an ambient $n \in \mathbb{N}_{>0}$, they expand respectively to F_n^{spl} and G_n^{spl} . They do not denote the forward composition iterate $F^{\circ n}$, the inverse composition iterate $(F^{-1})^{\circ n}$, or either globally catalogued Fabius inverse.

Finite-prefix use family. The spline and local-inverse symbols are used for finite-prefix approximation, uniform inverse-transfer asymptotics, and Richardson extrapolation. The prefix depth, evaluation variable, and—for the inverse—the selected monotone branch interval are ambient mathematical data, not additional TeX arguments.

Angular-frequency product. `\PhiAng` has TeX arity zero; its evaluation variable is external. It prints Φ_{ang} and denotes the entire extension of

$$\Phi_{\text{ang}}(\xi) := \widehat{\text{up}}_{\text{ang}}(\xi) = \prod_{j=1}^{\infty} \text{sinc}_{\text{rad}}(2^{-j}\xi), \quad \xi \in \mathbb{C},$$

where $\text{sinc}_{\text{rad}} z = \sin z / z$ for $z \neq 0$ and $\text{sinc}_{\text{rad}} 0 = 1$.

Cycles-per-unit product. `\PhiCyc` has TeX arity zero and an external variable. It prints $\Phi_{2\pi}$ and denotes the same entire transform in the cycles-per-unit coordinate:

$$\Phi_{2\pi}(z) := \widehat{\text{up}}_{2\pi}(z) = \Phi_{\text{ang}}(2\pi z) = \prod_{j=0}^{\infty} \text{sinc}_{\pi}(z/2^j), \quad z \in \mathbb{C}.$$

Thus neither the factor 2π nor the transform normalization is suppressed.

Fourier-product use families. Φ_{ang} is used for reciprocal-sinc coefficients, moment-generating/Fourier rotation, and polynomial deconvolution. $\Phi_{2\pi}$ is used for integer zero multiplicities, differentiated transform identities, moments, and dyadic self-sampling. Both symbols suppress the fixed normalized Rvachev up-function, but their printed subscripts make the coordinate normalization explicit; they are never silently interchanged.

Endpoint variables. For the endpoint regime $y \downarrow 0$, set $L = \log 2$, $T = \log(1/y)$, $\rho = \sqrt{2T/L}$, $z = \rho^{-1}$, $\ell = \log \rho$, $\beta = L^{-1} + 1/2$, and $\theta = \rho + \beta \pmod{1}$. The phase functions Ψ , A_1 , $A_2, \dots$ are the fixed one-periodic coefficient functions in the forward expansion

$$\log F(x) = -\frac{L}{2}\lambda^2 + L\beta\lambda - \frac{1}{2}\log \lambda + C_{\sharp} + \Psi(\lambda) + \sum_{j=1}^{N-1} \frac{A_j(\lambda)}{\lambda^j} + \mathcal{O}_N(\lambda^{-N}),$$

where $\lambda = -W_{-1}(-Lx)/L$ and the remainder is uniform in the real phase.

Endpoint kernel. `\EndpointKernel` has TeX arity zero and prints $\mathcal{K}$; both displayed arguments are mathematical arguments external to the macro. In the formal two-scale coefficient algebra generated by ℓ , the one-periodic functions above, and their derivatives, define

$$\begin{aligned} \mathcal{R}(z, w) &= -\frac{L}{2}\beta^2 + \frac{L}{2}w^2 + \frac{1}{2}\ell + \frac{1}{2}\log(1 + \beta z + zw) - C_{\sharp} - \Psi(\theta + w) \\ &\quad - \sum_{j \geq 1} \frac{A_j(\theta + w)z^j}{(1 + \beta z + zw)^j}, \\ \mathcal{K}(z, w) &:= -\frac{1}{L}\mathcal{R}(z, w). \end{aligned}$$

The inverse endpoint correction $D \in z\mathcal{A}[\ell][[z]]$ is then characterized by $D = z\mathcal{K}(z, D)$. Here w is the second formal argument; it is not a hidden function of z until the fixed-point substitution $w = D$ is made. The same source later writes $\mathcal{K}_W = -\mathcal{R}_W/L$ for a distinct Wright–omega-carrier kernel. The form `\EndpointKernel_w` prints that local symbol; it does not denote $\mathcal{K}(z, W)$, a derivative, or an automatically shared subscripted family.

Endpoint-kernel use family. The kernel is used in the Lagrange diagonal formulas for the fixed point, the all-orders inverse-endpoint recursion, and comparison with the distinct Wright–omega carrier. Its suppressed ambient data are $L, \beta, \ell, \theta, C_{\sharp}, \Psi$, the full coefficient sequence $(A_j)_{j \geq 1}$, and the chosen two-scale coefficient algebra.

Script composition. `\FiniteSpline`, `\FiniteSplineInverse`, `\Fn`, and `\Gn` already own a subscript and the superscript `spl`; a derivative or further order must group the whole object, as in $(F_n^{\text{spl}})^{(m)}$. `\PhiAng` and `\PhiCyc` already own their normalization subscripts; powers and derivatives may follow the grouped object, but no second raw subscript may be appended. `\EndpointKernel` owns no script, yet any local suffix such as $_W$ must be declared as a new semantic object rather than inferred as evaluation or differentiation.

Status. The finite-prefix objects and the two Fourier products are exact definitions. The endpoint kernel is an exact formal definition inside the volume’s human-proved all-orders coefficient construction; convergence or analytic equality is asserted only where a separate theorem supplies it.

15.3 Accents, scripts, and overloaded punctuation

Hats, tildes, bars, stars, primes, and scripts

Hats. A normalization-labelled hat produced by `\FourierTwoPi` or `\FourierAngular` is a Fourier transform. A unit-period coefficient uses `\FourierSeriesCoefficient`; any other period uses `\FourierSeriesCoefficientAtPeriod` and prints that period. A hat on a polynomial, matrix, estimator, accelerated approximant, or lifted object is local: all use sites call a descriptive document-local macro, and the raw accent occurs only in that macro's one-line definition. The immediately preceding source comment has the form `LOCAL-NON-FOURIER-HAT: \MacroName meaning`, names the same macro, and explains its mathematical role. A generic local alias for `\widehat` or `\hat` is forbidden. Thus no unlabelled hat is silently interpreted as a Fourier transform in a document that also uses monic or normalized polynomials.

Tildes. A tilde may mark centering, normalization, continuation, lifting, or an alternative sign convention. The transformed object's type and the defining map are given at first use. It is never a substitute for the signed global Fabius function.

Bars. For a scalar, bars mean $|x|$; for a finite set they mean $|A|$; a restriction is $f|_S$; divisibility is $a \mid b$; conditional probability is $\mathbb{P}(A \mid B)$ with $\mathbb{P}(B) > 0$. These roles are distinguished by type and spacing, not guessed from visual similarity.

Stars. For a scalar, $\bar{z}$ is conjugation; A^* is an adjoint; X^* in the zero-bias entry is a distributional transform; X^* may also denote a dual space only after a type declaration. None of these implies optimality; an optimizer is labelled x_{opt} .

Primes. f' is a derivative, ζ'_L a derivative of the spectral zeta function, and $\det' L$ omits zero modes. A primed variable such as x' is a new local variable only after declaration.

Superscripts. Parenthesized $f^{(n)}$ means the n th derivative; $f^{\circ n}$ means iteration; f^{*n} means convolution power; A^T means transpose; A^* means adjoint. Ordinary f^n remains pointwise or algebraic power.

Subscripts. A shared symbol that already owns a semantic subscript is not followed by a second raw subscript. Use the command's arguments or introduce a wrapper whose expansion has one syntactically valid script. This rule covers number systems, quantiles, transform hats, and any future decorated semantic command.

15.4 Exhaustive register of document-local non-Fourier hats

This register inventories every distinct command named by an active `LOCAL-NON-FOURIER-HAT` declaration. There are 35 marked declarations and 33 distinct command names; the two Bailey-pair commands are each declared in two document scopes. The phrase *prints a wide hat over X* describes literal output without reintroducing a generic accent command. TeX arguments, mathematical arguments, and scripts appended by a caller are distinguished explicitly. These commands are document-local: their definitions and ambient data do not survive the document or included-fragment scope stated below. Each heading contains one machine-readable `\localaccentname` token, so the strict audit rejects a missing, extra, or duplicate register entry.

15.4.1 Extensions, certified data, and formal series

`\ClampedExtensionOf`

TeX arity is one; argument 1 is a continuous function f on $I = [0, 1]$. The command prints a wide hat over f . Its mathematical value is the continuous map

$$E_I f : \mathbb{R} \longrightarrow \mathbb{R}, \quad (E_I f)(x) := f(\text{proj}_I x) = \begin{cases} f(0), & x \leq 0, \\ f(x), & 0 \leq x \leq 1, \\ f(1), & 1 \leq x, \end{cases}$$

where $f(x)$ in the middle branch uses the canonical membership $x \in I$. In the Lean construction this is precomposition with $\text{Set.projIcc}(0, 1)$; admissibility of f supplies the bounds and reflection identity used later but is not required to define $E_I f$. The only use family is the fixed-point primitive $A_f(y) = \int_0^y (E_I f)(t) dt$. The accent means endpoint-clamped extension, not Fourier transformation. Status is an exact definition; the later admissibility and fixed-point theorems are separate assertions. The clamping interval, endpoint values, and projection map are suppressed by the printed symbol. The function argument must include any script belonging to f inside the braces; a derivative or other decoration of the resulting map must group the complete expansion before adding the decoration.

`\FullRvachevLegendreCoefficientRat`

TeX arity is one; argument 1 is the full Legendre degree $k \in \mathbb{N}_0$. The command prints a wide hat over c , with subscript k and superscript $\mathbb{Q}$. It denotes the function $c^{\text{full}, \mathbb{Q}} : \mathbb{N}_0 \rightarrow \mathbb{Q}$ defined by

$$c_k^{\text{full}, \mathbb{Q}} := \begin{cases} c_{k/2}^{\mathbb{Q}}, & 2 \mid k, \\ 0, & 2 \nmid k, \end{cases} \quad c_{2r}^{\text{full}, \mathbb{Q}} = c_r^{\mathbb{Q}}, \quad c_{2r+1}^{\text{full}, \mathbb{Q}} = 0.$$

Here $c_r^{\mathbb{Q}}$ is the canonical even Rvachev–Legendre coefficient. Equivalently,

$$c_k^{\text{full}, \mathbb{Q}} = \frac{2k+1}{2} \mathcal{M}_{\text{Rv}}(P_k^{\mathbb{Q}}),$$

where $P_k^{\mathbb{Q}}$ is the rational Legendre polynomial and $\mathcal{M}_{\text{Rv}}$ is the moment functional with moment sequence `rvachevRawMomentRat`. The use family consists of the rational Legendre–Gaunt formulas and their real-cast comparison. The accent means parity completion of an even-indexed sequence. The canonical moment sequence, Legendre normalization, and coefficient family are suppressed ambient data. Status is an exact rational sequence definition; comparison with analytic coefficients retains the source’s explicit Fabius-function hypotheses. The command already owns both its subscript and its $\mathbb{Q}$ -superscript; appending any further raw script would be a script-composition error.

`\CertifiedEnclosureCenter`

TeX arity is one; argument 1 is the exact quantity whose validated numerical center is being named. The command prints a wide hat over that argument. Its sole active use has $Q, Q_{\text{num}} \in \mathbb{C}$ and means only the center component of the certificate

$$\varepsilon_Q \geq 0, \quad |Q - Q_{\text{num}}| \leq \varepsilon_Q.$$

In the validated argument-principle scope, let Ω be a bounded Jordan domain, let f be holomorphic on a neighborhood of $\overline{\Omega}$, assume $0 \notin f(\partial\Omega)$, and decompose the positively oriented boundary into finitely many C^3 panels $\gamma_k : [0, 1] \rightarrow \partial\Omega$. Define

$$g_k(s) = \frac{f'(\gamma_k(s))\gamma_k'(s)}{f(\gamma_k(s))}, \quad Q = \sum_k \frac{g_k(0) + g_k(1)}{2},$$

$$M_k \geq \sup_{0 \leq s \leq 1} |g_k''(s)|, \quad E = \frac{1}{12} \sum_k M_k.$$

The panelwise trapezoid estimate gives

$$\left| \oint_{\partial\Omega} \frac{f'(z)}{f(z)} dz - Q \right| \leq E.$$

Outward-rounded arithmetic supplies Q_{num} and ε_Q . The resulting certified zero-count disk has center $Q_{\text{num}}/(2\pi i)$ and radius $(E + \varepsilon_Q)/(2\pi)$. Status is therefore a certificate schema: the accented center

alone is not a certificate. The ambient normed field, domain Ω , function f , boundary orientation and panelization (γ_k) , derivative bounds (M_k) , trapezoid rule, rounding method, error radius, and exact target are all suppressed data. The accent means validated numerical center, not Fourier transformation. The complete exact-quantity name belongs inside the single argument; callers must group the expansion before attaching any further script.

`\FormalGevreySeriesSymbol`

TeX arity is one; argument 1 is the name R of a formal series, not its formal variable. The command prints a wide hat over R . In its cyclotomic resurgence scope it denotes

$$R_{\text{for}}(t) := \sum_{n \geq 0} c_n t^n \in \mathbb{C}[[t]], \quad |c_n| \leq C A^n n! \quad (n \in \mathbb{N}_0)$$

for some constants $C, A > 0$. The document's exact formal Borel convention is

$$\mathcal{B}R_{\text{for}}(\xi) := \sum_{n \geq 0} \frac{c_{n+1}}{n!} \xi^n;$$

the constant c_0 is carried separately by Borel–Laplace summation. Status is a formal-series definition used immediately before a conjectural resurgence statement; the Gevrey estimate itself does not assert Borel summability or analytic continuation. The coefficient field, formal variable, coefficients, Gevrey order, sector, and Borel convention are suppressed by the printed symbol. The accent means formal asymptotic series prior to Borel transformation. The series name, including any semantic script, must be placed inside the one argument; any subsequent decoration must group the expansion.

15.4.2 Bailey-pair transforms

`\BaileyTransformedAlpha`

TeX arity is zero. The command prints a wide hat over α ; the index n is an external source-level subscript rather than a TeX argument. Let $a, q \in \mathbb{C}$ and let $\alpha, \beta : \mathbb{N}_0 \rightarrow \mathbb{C}$ be a Bailey pair relative to a , so

$$\beta_n = \sum_{r=0}^n \frac{\alpha_r}{(q; q)_{n-r} (aq; q)_{n+r}}.$$

Both active document scopes define the transformed alpha sequence by

$$\alpha_n^{\text{B}} := a^n q^{n^2} \alpha_n.$$

The q -series monograph calls this the alpha component of a Bailey-chain step and also uses it for the transformed unit pair in the Rogers–Ramanujan proof. The proof-oriented article calls the same formula the alpha component of the weak Bailey-lemma transformed pair and abbreviates $(z; q)_n$ by $(z)_n$ locally. Status is an exact finite transformation; in analytic limiting statements the standing convention is $q \in \mathbb{C}$, $0 < |q| < 1$, with all displayed denominators nonzero. The source pair, a , q , coefficient field, denominator hypotheses, and index are suppressed. The accent means Bailey transformation, not Fourier transformation. Appending the single external subscript is currently valid because the zero-argument expansion itself owns no script, but it is fragile: no second subscript or simultaneous replacement superscript may be appended, and any future scripted macro body would require an indexed wrapper.

`\BaileyTransformedBeta`

TeX arity is zero. The command prints a wide hat over β ; the index n is appended externally and is not a TeX argument. For a Bailey pair (α, β) relative to a , with $a, q \in \mathbb{C}$, its exact transformed beta component is

$$\beta_n^B := \sum_{j=0}^n \frac{a^j q^{j^2}}{(q; q)_{n-j}} \beta_j.$$

Together with $\alpha_n^B = a^n q^{n^2} \alpha_n$, this is again a Bailey pair relative to a . In the monograph it appears in the Bailey-chain step, in the limiting weak Bailey lemma through β_N^B , and after transforming the unit pair, where $\beta_n^B = 1/(q; q)_n$. In the proof-oriented article the same formula is the beta component of its weak Bailey-lemma corollary, with $(z)_n = (z; q)_n$ in that local section. Status is an exact finite transformation, with convergence assumptions added only when a subsequent infinite limit is taken. The input pair, a, q , coefficient field, nonvanishing denominators, and index are suppressed. The accent means Bailey transformation. The external `_n` or `_N` convention is safe only while the zero-argument expansion remains unscripted; callers must not compose it with a second raw subscript or conflicting superscript.

15.4.3 Inverse, sampling, and prediction constructions`\FactorialScaledInverseTransferCoefficient`

TeX arity is one; argument 1 is the coefficient index $m \in \{1, \dots, J-1\}$. The command prints a wide hat over c with subscript m . The evaluation variable y is external to the macro and is suppressed at the bare-symbol use sites.

Fix $0 < \eta < 1/2$, $J \in \mathbb{N}_{>0}$, $q = 1/4$, and $\varepsilon_n = q^n$. In the uniform inverse-transfer expansion

$$G_n^{\text{spl}}(y) = Q_F(y) + \sum_{j=1}^{J-1} c_j(y) \varepsilon_n^j + \mathcal{O}_{\eta, J}(\varepsilon_n^J), \quad y \in [\eta, 1 - \eta],$$

the theorem constructs

$$c_m \in C^\infty([\eta, 1 - \eta]; \mathbb{R}) \quad (1 \leq m \leq J-1).$$

With $x = Q_F(y)$, the accented coefficient is exactly

$$\hat{c}_m(y) := m! c_m(y).$$

The source normally suppresses the displayed argument y and writes only $\hat{c}_m$ inside partial exponential Bell polynomials.

More explicitly, put

$$a_j(x) := (-1)^j b_j F^{(2j)}(x),$$

and let $\mathcal{B}_{r,k}$ be the partial exponential Bell polynomial with $\mathcal{B}_{0,0} = 1$. Then

$$c_m(y) = -\frac{1}{F'(x)} \left[\frac{1}{m!} \sum_{k=2}^m F^{(k)}(x) \mathcal{B}_{m,k}(\hat{c}_1, \dots, \hat{c}_{m-k+1}) + \sum_{j=1}^m \frac{1}{(m-j)!} \sum_{k=0}^{m-j} a_j^{(k)}(x) \mathcal{B}_{m-j,k}(\hat{c}_1, \dots, \hat{c}_{m-j-k+1}) \right],$$

where the term with $m = j$ is interpreted as $a_m(x)$.

Status is an exact factorial rescaling inside the source's proved uniform inverse-transfer theorem. The source supplies a proof but makes no Lean formalization claim. The printed symbol suppresses y ,

η , J , q , the spline inverse G_n^{spl} , the base function F , the perturbation coefficients (a_j) , and the Bell-polynomial convention. The accent means conversion from ordinary inverse-expansion coefficients to the factorial scaling used by exponential Bell polynomials. It is not a Fourier transform and not the tomographic Fourier-coefficient estimator elsewhere printed as a hatted c .

The macro already owns the subscript m . An explicit evaluation may be written after the complete expansion as $\widehat{c}_m(y)$; callers must not encode y as a second raw subscript. Any derivative, superscript, or other decoration must group the complete expanded symbol.

`\FormalTaylorGerm`

TeX arity is one; argument 1 is a function or germ label, not the center or local variable. The command prints a wide hat over that label. For a smooth local map $h : \mathbb{R} \rightarrow \mathbb{R}$ at x_0 , its full formal Taylor series in the increment z is

$$\mathcal{T}_{x_0} h(z) := \sum_{m \geq 0} \frac{h^{(m)}(x_0)}{m!} z^m \in \mathbb{R}[[z]].$$

When both source and target centers are translated, the document instead uses the zero-constant germ

$$\mathcal{T}_{x_0}^0 h(z) := \mathcal{T}_{x_0} h(z) - h(x_0) = \sum_{m \geq 1} \frac{h^{(m)}(x_0)}{m!} z^m.$$

The active use families are formal compositional inversion of smooth local inverses, the affine center jets of G_n and K_n , numerical coefficient-reversion residuals, and the exact quarter-point germ of F . Status is formal algebra: positive radius and equality with the original smooth map are separate assertions. The center, increment variable, whether the output center was subtracted, coefficient field, and convergence status are all suppressed and must be stated at each use. The accent means Taylor germ, not Fourier transformation. A scripted function label such as $F_{1/4}$ belongs wholly inside the one argument; any further decoration must group the complete expansion.

`\GeometricPrefixReproduction`

TeX arity is one; argument 1 is the prefix depth $m \in \mathbb{N}_{>0}$. The command prints a wide hat over X with subscript m . For

$$X_q = (1 - q) \sum_{j \geq 0} q^j U_j, \quad U_j \stackrel{\text{iid}}{\sim} \text{Unif}[-1, 1], \quad 0 < q < 1,$$

it denotes the real square-integrable random variable

$$S_{q,m} := (1 - q) \sum_{j=0}^{m-1} q^j U_j \in L^2(\Omega; \mathbb{R}).$$

This is the deterministic first- m -digit reproduction used in the squared-error test channel, with

$$I(X_q; S_{q,m}) = m \log(1/q), \quad \mathbb{E}[(X_q - S_{q,m})^2] = q^{2m} \text{Var}(X_q).$$

Status is an exact definition and exact rate/distortion calculation. The parameter q , probability space, digit sequence, digit law, and distortion measure are suppressed. The accent means reproduction or estimation, not Fourier transformation. The argument already becomes the subscript of X ; no additional raw subscript or superscript may follow the expanded command.

`\LeastSquaresProjectionEstimate`

TeX arity is one; argument 1 is the complete target random-variable expression. The command prints a wide hat over that target. In its sole scope,

$$X_q = (1 - q) \sum_{n \geq 0} q^n U_n, \quad U_n \stackrel{\text{iid}}{\sim} \text{Unif}[0, 1], \quad Y_q = \frac{X_q - \frac{1}{2}}{\sqrt{(1 - q)/(12(1 + q))}}.$$

For $0 < q, q_1, \dots, q_m < 1$, with the comparison nodes pairwise distinct and $q \notin \{q_1, \dots, q_m\}$, the accented target is the unique real L^2 -orthogonal projection

$$\Pi_q Y_q := \operatorname{argmin}_{Z \in \text{span}(Y_{q_1}, \dots, Y_{q_m})} \mathbb{E}[(Y_q - Z)^2],$$

and

$$\mathbb{E}[(Y_q - \Pi_q Y_q)^2] = \prod_{j=1}^m \left(\frac{q - q_j}{1 - qq_j} \right)^2.$$

Status is an exact Hilbert-space projection and prediction-error identity. The common probability space and digits, standardization, comparison nodes, coefficient field, and projection subspace are suppressed. The accent means least-squares predictor. The full target, including its own subscript, must be enclosed in the one argument; any subsequent decoration must group the complete expansion.

`\NestedLambertLogApproximation`

TeX arity is one; argument 1 is the iterate index $r \in \{0, \dots, n\}$. The command prints a wide hat over T with subscript r . The endpoint input, recursion depth, truncation order, and evaluation phases are not TeX arguments.

In its sole active scope, $I = Q_F$ on $(0, 1)$. Put

$$y_0 = y, \quad y_{r+1} = I(y_r), \quad T_r := \log(1/y_r),$$

in the endpoint regime where every attained y_r is positive and belongs to the domain of I . The source does not assert a larger maximal domain for this construction. For a fixed depth $n \geq 1$, put

$$q_n := T_0^{-2^{-n}}, \quad \ell := \log T_0,$$

and set $L = \log 2$ and $\beta = L^{-1} + 1/2$. For a chosen truncation order $N \in \mathbb{N}_{>0}$ and the one-step coefficient functions h_j supplied by the preceding all-orders theorem, define

$$\mathcal{I}_N(T) := L\rho + L\beta - \log \rho - \sum_{j=1}^N \frac{h_j(\log \rho, \rho + \beta)}{\rho^j}, \quad \rho := \sqrt{2T/L}.$$

The accented sequence is the procedural recursion

$$\widehat{T}_0 := T_0,$$

followed, for each r , by expansion of $\mathcal{I}_N(\widehat{T}_r)$ in powers of q_n and deletion of every term whose q_n -exponent is at least N ; the retained expression is called $\widehat{T}_{r+1}$.

The source presents this construction inside the open *Nested Lambert phase closure* problem. It asks for, but does not prove, the uniform estimate

$$T_r - \widehat{T}_r = \mathcal{O}_{n,N} \left((1 + \ell)^{D'_{n,N}} q_n^N \right) \quad (0 \leq r \leq n),$$

the phase-substitution bounds, nonconstant transport of each periodic phase, and closure or minimal enlargement of the exponent lattice $q_n^{\mathbb{Z}}$. Because that closure is itself part of the problem, the source

does not yet supply a proved ambient formal-series ring or a proved global function domain and codomain for every $\hat{T}_r$. It intends real asymptotic expressions along the attained endpoint regime. No Lean formalization is claimed.

The printed symbol suppresses $y, I, n, N, T_0, q_n, \ell, L, \beta$, the coefficient family (h_j) , the phase vector, and the expansion/truncation convention. The accent means recursive asymptotic truncation, not Fourier transformation and not the lacunary-product tail extrapolation elsewhere printed with a hatted T . The macro already owns the subscript r ; callers must not append a second raw subscript. Any further decoration must group the complete expanded symbol and receive a separate semantic declaration.

15.4.4 Omission marks, endpoint matrices, and lifted operators

`\OmittedSymmetricArgument`

TeX arity is one; argument 1 is the complete variable to delete from a symmetric-polynomial argument list. The command prints a wide hat over that variable. It is omission metanotation, not a map on the scalar q_j . Its exact active meaning is

$$T_k := \sum_{j \geq 1} q_j^2 e_{k-1}(q_1, \dots, q_{j-1}, q_{j+1}, \dots), \quad q_j = Q^j, \quad Q = \frac{1}{4},$$

so the j -th entry is absent from the arguments of e_{k-1} . The identity

$$e_1 e_k = T_k + (k+1)e_{k+1}$$

supplies the first correction in the small-field q -binomial chaos asymptotic. Status is exact combinatorial deletion notation. The ambient infinite sequence, deleted position, convergence of the symmetric sums, and value $Q = 1/4$ are suppressed. The accent means omission, not Fourier transformation. A scripted item such as q_j belongs inside the one argument, and the resulting omission marker receives no further script.

`\GeneratedEndpointCoefficientMatrix`

TeX arity is zero. The command prints a wide hat over B ; the degree subscript 2 in the generated fragment is appended externally rather than passed as an argument. The fixed evidence object is

$$B_2^{\text{end}} = \begin{bmatrix} \frac{392}{9} & -\frac{496}{9} & \frac{176}{9} \\ -\frac{752}{9} & \frac{1072}{9} & -\frac{392}{9} \\ \frac{680}{9} & -\frac{1072}{9} & \frac{464}{9} \\ -\frac{464}{9} & \frac{496}{9} & -\frac{104}{9} \end{bmatrix} \in \mathbb{Q}^{4 \times 3}.$$

It is the endpoint-basis coefficient matrix for degree $d = 2$, scale $M = 2^d = 4$, nodes $x = (-1, 0, 1)$, transported nodes $t = (0, \frac{1}{2}, 1)$, and retained integer centers $k = (1, 2, 3, 4)$. Its columns are the exact endpoint-atom coefficient vectors of the three Lagrange cardinal polynomials. The generator constructs those cardinals, applies the reciprocal-moment differential operator at scale 4, samples the active shifts, removes the annihilator-recurrence null component, and retains the final $d + 2$ coefficients. Status is an exact generated companion-evidence fragment, not an independent globally parameterized matrix family. All degree, node, scale, basis, and row/column data are suppressed. The accent means endpoint augmentation/compression. The external degree subscript is presently valid because the zero-argument expansion owns no script; regeneration must preserve that invariant or replace the command with an indexed wrapper.

`\AugmentedEvaluationMatrix`

TeX arity is zero. The command prints a wide hat over U ; entry indices such as i, r are appended externally. Fix $d \in \mathbb{N}_0$, distinct nodes $x_0, \dots, x_d \in [-1, 1]$, and the $d + 2$ normalized endpoint atoms

$$c_{d,r} = 2^{d+1} - 2d - 3 + 2r, \quad \Phi_{d,r}(x) := 2^{-d} \operatorname{up}\left(\frac{x - c_{d,r}}{2^{d+1}}\right), \quad 0 \leq r \leq d + 1.$$

The matrix is

$$U_X^{\operatorname{end}} := [\Phi_{d,r}(x_i)]_{0 \leq i \leq d, 0 \leq r \leq d+1} \in \mathbb{R}^{(d+1) \times (d+2)}.$$

It maps an endpoint-basis coefficient vector to its $d + 1$ nodal values. Status is an exact finite evaluation matrix. The degree, ordered node set, endpoint basis, index ranges, and normalization of the atoms are suppressed. The accent means augmentation of the polynomial evaluation system by one endpoint atom. A single external entry subscript is valid because the zero-argument expansion has no internal script; callers must not create a second subscript, and a future scripted body would require an entry wrapper.

`\AugmentedCoefficientMatrix`

TeX arity is zero. The command prints a wide hat over B ; transpose or entry scripts are external. With the degree, nodes, atoms, and evaluation matrix fixed as above, let $\ell_j \in \Pi_d$ be the Lagrange cardinal satisfying $\ell_j(x_i) = \delta_{ij}$. The coefficient matrix

$$B_X^{\operatorname{end}} = [b_{rj}]_{0 \leq r \leq d+1, 0 \leq j \leq d} \in \mathbb{R}^{(d+2) \times (d+1)}$$

is defined by the unique endpoint-basis expansions

$$\ell_j = \sum_{r=0}^{d+1} b_{rj} \Phi_{d,r}, \quad U_X^{\operatorname{end}} B_X^{\operatorname{end}} = I_{d+1}.$$

Define the canonical cofactor vectors by

$$v_r = (-1)^r \det U_X^{(r)}, \quad \zeta_r = (-1)^r \det B_X^{(r)},$$

where $U_X^{(r)}$ deletes column r from U_X^{end} and $B_X^{(r)}$ deletes row r from B_X^{end} . Then $U_X^{\operatorname{end}} v = 0$, $\zeta^T B_X^{\operatorname{end}} = 0$, and Cauchy–Binet gives the normalization $\zeta^T v = 1$. With

$$\mathcal{M}_X = \begin{pmatrix} U_X^{\operatorname{end}} \\ \zeta^T \end{pmatrix},$$

then the defect-completed inverse formula is

$$B_X^{\operatorname{end}} = \mathcal{M}_X^{-1} \begin{pmatrix} I_{d+1} \\ 0 \end{pmatrix}, \quad \zeta^T B_X^{\operatorname{end}} = 0.$$

Status is an exact finite right inverse selected by the endpoint-basis polynomiality constraint. The degree, nodes, cardinals, endpoint atoms, polynomiality row, and matrix dimensions are suppressed. The accent means endpoint-atom augmentation. An external transpose is valid only because the zero-argument expansion has no script; do not combine it with an internal or second external superscript.

`\OmittedMatrixIndex`

TeX arity is one; argument 1 is the index $r \in \{0, \dots, d+1\}$ to omit. The command prints a wide hat over r . It is deletion metanotation, not a map on indices. In the evaluation-matrix context,

$$U_X^{(r)} \in \mathbb{R}^{(d+1) \times (d+1)}$$

means U_X^{end} with column r deleted; in the coefficient-matrix context,

$$B_X^{(r)} \in \mathbb{R}^{(d+1) \times (d+1)}$$

means B_X^{end} with row r deleted. The cofactor vectors are

$$v_r = (-1)^r \det U_X^{(r)}, \quad \zeta_r = (-1)^r \det B_X^{(r)},$$

and Cauchy–Binet yields $\zeta^T v = 1$. Status is exact maximal-minor notation. Which matrix is present, whether a row or column is deleted, the matrix dimensions, and the permitted index set are suppressed. The accent means omission. The argument is used as a nested marker inside the matrix's one subscript; it must not itself acquire any further script.

`\LiftedCoefficientProjector`

TeX arity is two. Argument 1 is the interpolation level q , and argument 2 is the terminal coefficient-space level s , in that order. The command prints a wide hat over $\mathcal{P}$, with subscript q and parenthesized superscript s . Fix nested node sets $X^{(0)} \subset \dots \subset X^{(s)}$, interpolation projectors $I_q : C[-1, 1] \rightarrow \Pi_{d_q}$, a terminal coefficient space C_s , a synthesis map $T_s : C_s \rightarrow C[-1, 1]$, and a canonical coefficient map $E_s : \Pi_{d_s} \rightarrow C_s$ satisfying $T_s E_s = I$ on Π_{d_s} . Then

$$\mathcal{P}_q^{(s)} := E_s I_q T_s : C_s \longrightarrow C_s, \quad 0 \leq q \leq s.$$

These operators satisfy

$$\mathcal{P}_q^{(s)} \mathcal{P}_r^{(s)} = \mathcal{P}_{\min(q,r)}^{(s)}, \quad \text{rank } \mathcal{P}_q^{(s)} = d_q + 1.$$

Status is an exact coefficient-space lift of an interpolation projector. The nested grids, degrees, coefficient dictionary, synthesis and analysis maps, and terminal level are suppressed. The accent means coefficient lift. Both available scripts are already built from the two arguments; no raw subscript, superscript, or prime may be appended.

`\LiftedCoefficientDetail`

TeX arity is two. Argument 1 is the detail level q , and argument 2 is the terminal coefficient-space level s , in that order. The command prints a wide hat over Δ , with subscript q and parenthesized superscript s . Using the lifted projectors just defined,

$$\Delta_0^{(s)} := \mathcal{P}_0^{(s)}, \quad \Delta_q^{(s)} := \mathcal{P}_q^{(s)} - \mathcal{P}_{q-1}^{(s)} \quad (1 \leq q \leq s).$$

Each $\Delta_q^{(s)} : C_s \rightarrow C_s$ is idempotent, distinct details annihilate one another, and

$$\text{rank } \Delta_0^{(s)} = d_0 + 1, \quad \text{rank } \Delta_q^{(s)} = d_q - d_{q-1} \quad (q \geq 1).$$

Status is an exact finite multiresolution-detail definition. The nested node sets, interpolation projectors, terminal dictionary, degree sequence, and coefficient-space maps are suppressed. The accent means coefficient-space lift, not frequency transformation. Both scripts are owned by the two arguments; appending any further raw script is invalid.

15.4.5 Monic and hypothetical polynomial families

\MonicDeconvolvedLegendre

TeX arity is one; argument 1 is the degree $n \in \mathbb{N}_0$. The command prints a wide hat over Q , with subscript n and superscript minus. Let

$$Z := \sum_{j \geq 1} 2^{-j} U_j, \quad U_j \stackrel{\text{iid}}{\sim} \text{Unif}[-1, 1],$$

so Z has the normalized Rvachev up-density on $[-1, 1]$. Let P_n be the classical Legendre polynomial and set

$$M_u(z) := \mathbb{E}[e^{zZ}] = \prod_{j \geq 1} \frac{\sinh(z/2^j)}{z/2^j}, \quad \frac{1}{M_u(z)} = \sum_{r \geq 0} \gamma_r \frac{z^r}{r!}, \quad \mathcal{D}_u := \sum_{r \geq 0} \gamma_r \frac{D^r}{r!}.$$

The deconvolved polynomial is $Q_n^- = \mathcal{D}_u P_n$. Since $[x^n]P_n = 2^{-n} \binom{2n}{n}$ and $\mathcal{D}_u$ preserves leading coefficients, its exact monic normalization is

$$Q_n^{-,\text{mon}} := \frac{2^n}{\binom{2n}{n}} \mathcal{D}_u P_n \in \mathbb{Q}[x].$$

For example,

$$\begin{aligned} Q_2^{-,\text{mon}} &= x^2 - \frac{4}{9}, & Q_3^{-,\text{mon}} &= x^3 - \frac{14}{15}x, \\ Q_4^{-,\text{mon}} &= x^4 - \frac{32}{21}x^2 + \frac{1072}{4725} = xQ_3^{-,\text{mon}} - \frac{62}{105}Q_2^{-,\text{mon}} - \frac{8}{225}Q_0^{-,\text{mon}}. \end{aligned}$$

Status is an exact rational polynomial definition; the final equality is the explicit degree-four Favard obstruction. The up-law, moment-generating function, reciprocal moments, differential operator, Legendre normalization, and coefficient field are suppressed. The accent means monic normalization. The command already owns its degree subscript and minus superscript; no additional raw script may follow it.

\GenericMonicOrthogonalPolynomial

TeX arity is one; argument 1 is the degree $n \in \mathbb{N}_0$. The command prints a wide hat over Q with subscript n . It is a schema for a hypothetical monic, parity-alternating scalar orthogonal-polynomial family

$$Q_n^{\text{mon}} \in \mathbb{R}[x], \quad \deg Q_n^{\text{mon}} = n, \quad [x^n]Q_n^{\text{mon}} = 1.$$

For a scalar quasi-definite moment functional, Favard's theorem would force

$$Q_0^{\text{mon}} = 1, \quad Q_1^{\text{mon}} = x, \quad \lambda_n \neq 0 \quad (n \geq 1),$$

and the parity-reduced three-term recurrence

$$Q_{n+1}^{\text{mon}} = xQ_n^{\text{mon}} - \lambda_n Q_{n-1}^{\text{mon}} \quad (n \geq 1).$$

The symbol does not name a separately constructed global family; it is used only to contrast this three-term schema with the nonzero degree-zero defect in the actual monic deconvolved Legendre family. Status is therefore a generic proof schema, not a definition of new canonical polynomials. The scalar functional, orthogonality normalization, recurrence coefficients, coefficient field, and parity assumptions are suppressed. The accent means monic normalization, not Fourier transformation. The degree is already placed by the one argument, so no second subscript or conflicting superscript may be appended.

15.4.6 Lacunary-product extrapolants

Common lacunary-product data.

For the product extrapolants below, let $b > 1$, $N \in \mathbb{N}_0$, $q = b^{-2} \in (0, 1)$, and

$$P_b(t) := \prod_{k=1}^{\infty} \text{sinc}_{\text{rad}}(t/b^k), \quad P_{b,N}(t) := \prod_{k=1}^N \text{sinc}_{\text{rad}}(t/b^k),$$

where $P_{b,0} = 1$. Put

$$T_N(t) := \prod_{k=N+1}^{\infty} \text{sinc}_{\text{rad}}(t/b^k), \quad T_{N,j}(t) := \prod_{k=N+1}^{N+j} \text{sinc}_{\text{rad}}(t/b^k), \quad T_{N,0} = 1,$$

and

$$D_N := \{t \in \mathbb{C} : |t| < \pi b^{N+1}\}.$$

Every $T_{N,j}$, including T_N , is zero-free on D_N . Write $L_{N,j}$ for the unique holomorphic logarithm of $T_{N,j}$ on D_N satisfying $L_{N,j}(0) = 0$. For $r \in \mathbb{N}_0$ and $0 \leq j \leq r$, define

$$(q; q)_m := \prod_{\ell=1}^m (1 - q^\ell), \quad (q; q)_0 := 1, \quad \lambda_{r,j}(q) := \frac{(-1)^{r-j} q^{(r-j)(r-j+1)/2}}{(q; q)_j (q; q)_{r-j}}.$$

\FactorizedTailExtrapolant: factorized tail extrapolant

TeX interface. Arity 2, with arguments N, r in that order; it prints $\widehat{T}_{N,r}$. The evaluation variable t is written after the macro and is not a TeX argument.

Definition and type. For $N, r \in \mathbb{N}_0$,

$$\widehat{T}_{N,r}(t) := \exp \left(\sum_{j=0}^r \lambda_{r,j}(q) L_{N,j}(t) \right), \quad \widehat{T}_{N,r} : D_N \longrightarrow \mathbb{C}^\times.$$

It extrapolates the exact omitted tail T_N , is holomorphic and zero-free on D_N , and is positive on $D_N \cap \mathbb{R}$.

Scope and status. The printed notation suppresses the fixed base b , hence also $q = b^{-2}$, and leaves t external. The construction, its exact error, and its zero-free properties are proved in the factorized frontier report; that report claims no Lean counterpart at its snapshot.

Accent and scripts. The hat means logarithmic q -Richardson extrapolation, not Fourier transformation. The symbol owns only the subscript N, r , so a later grouped or direct superscript is syntactically available.

\FactorizedProductExtrapolant: zero-preserving product extrapolant

TeX interface. Arity 3, with arguments b, N, r in that order; it prints $\widehat{P}_{b,N,r}^{\text{fac}}$. The evaluation variable t is external.

Definition and type.

$$\widehat{P}_{b,N,r}^{\text{fac}}(t) := P_{b,N}(t) \exp \left(\sum_{j=0}^r \lambda_{r,j}(b^{-2}) L_{N,j}(t) \right), \quad \widehat{P}_{b,N,r}^{\text{fac}} : D_N \longrightarrow \mathbb{C}.$$

It has exactly the zero divisor of P_b in D_N , including multiplicities, and agrees with the unfactorized logarithmic product extrapolant wherever $P_{b,N} \neq 0$.

Scope and status. The derived parameter $q = b^{-2}$ and the argument t are not printed as indices. Holomorphic continuation, zero-divisor preservation, exact zero-normalized quotients, and local jet identities are proved in the frontier report but are not claimed there as Lean theorems.

Accent and scripts. The hat means a factorized approximation to P_b , not a Fourier transform. The macro already owns the superscript fac; any derivative, power, or further semantic superscript must be attached to the grouped expression $(\widehat{P}_{b,N,r}^{\text{fac}})$, never appended directly.

\RelativeTailExtrapolant: dyadic relative-tail extrapolant

TeX interface. Arity 2, with arguments N, r in that order; it prints $\widehat{T}_{N,r}$. The evaluation variable x is external, and the dyadic base is fixed rather than printed.

Definition and type. Put

$$\mathcal{T}_{N,j}(x) := \prod_{h=N+1}^{N+j} \text{sinc}_{\text{rad}}(\pi x/2^h), \quad \mathcal{T}_{N,0} = 1,$$

and let $L_{N,j}^{\text{dy}}$ be its holomorphic logarithm on $|x| < 2^{N+1}$, normalized by $L_{N,j}^{\text{dy}}(0) = 0$. Then

$$\widehat{T}_{N,r}(x) := \exp\left(\sum_{j=0}^r \lambda_{r,j}(1/4) L_{N,j}^{\text{dy}}(x)\right), \quad \widehat{T}_{N,r} : \{x \in \mathbb{C} : |x| < 2^{N+1}\} \longrightarrow \mathbb{C}^\times.$$

Its target is $\mathcal{T}_N(x) = \prod_{h=N+1}^{\infty} \text{sinc}_{\text{rad}}(\pi x/2^h)$.

Scope and status. The fixed data $b = 2$, $q = 1/4$, and the argument x are suppressed. The source introduces this object by cross-reference rather than by a standalone equality; the formula above is the explicit specialization used in its proved dyadic bracket theorem.

Accent and scripts. The hat means extrapolation of a product tail that is zero-free on the complex disk and positive on its real slice, not Fourier transformation. The symbol owns only the subscript N, r .

\DyadicFactorizedProductExtrapolant: zero-preserving dyadic Fourier-product approximant

TeX interface. Arity 2, with arguments N, r in that order; it prints $\widehat{\Phi}_{N,r}$. The evaluation variable x is external.

Definition and type. Let

$$\Phi(x) := \widehat{\text{up}}_{2\pi}(x) = \prod_{h=0}^{\infty} \text{sinc}_{\text{rad}}(\pi x/2^h), \quad \Phi_N(x) := \prod_{h=0}^N \text{sinc}_{\text{rad}}(\pi x/2^h).$$

Then

$$\widehat{\Phi}_{N,r}(x) := \Phi_N(x) \widehat{T}_{N,r}(x), \quad \widehat{\Phi}_{N,r} : \{x \in \mathbb{C} : |x| < 2^{N+1}\} \longrightarrow \mathbb{C}.$$

It preserves every included integer zero of Φ , its multiplicity, and the real Thue–Morse lobe sign.

Scope and status. The fixed dyadic data $b = 2$, $q = 1/4$, the relative-tail logarithms, and x are suppressed. The source defines the object by reference to factorization; its zero, jet, bracket, and sign properties are proved in the manuscript but are not claimed as Lean results.

Accent and scripts. Although Φ is itself the 2π -normalized Fourier transform of up, the hat on $\widehat{\Phi}_{N,r}$ means numerical/analytic extrapolation of Φ , not a second Fourier transform. The macro owns only a subscript, so the current derivative notation $\widehat{\Phi}_{N,r}^{(\mu)}$ is syntactically safe.

15.4.7 Radial-angular and unfactorized product estimators

\RadialJetEstimator: radial endpoint-coefficient estimator

TeX interface. Arity 1, whose sole argument is the coefficient index j ; it prints $\hat{A}_j$. The phase θ is an external function argument.

Definition and type. Let $a, d \in \mathbb{Z}$ satisfy $d \geq 1$ and $0 \leq a < d$, put $\theta = a/d$, and let the integer $\rho > 1$ satisfy $\rho \equiv 1 \pmod{d}$. Choose $n \in \mathbb{N}_{>0}$ with $n + \theta \geq 1/\log 2$. For fixed $p \in \mathbb{N}_0$, put

$$q = \rho^{-1}, \quad H = (n + \theta)^{-1}, \quad y_s = \mathcal{E}(\rho^s(n + \theta)) = \sum_{m=0}^p A_m(\theta) H^m q^{sm} + r_s,$$

and

$$\ell_s(z) := \prod_{\substack{0 \leq u \leq p \\ u \neq s}} \frac{z - q^u}{q^s - q^u}.$$

For $0 \leq j \leq p$,

$$\hat{A}_j(\theta) := H^{-j} [z^j] \sum_{s=0}^p y_s \ell_s(z).$$

For the real Fabius endpoint expansion this is real-valued; the same finite interpolation formula is valid over $\mathbb{C}$.

Scope and status. The printed symbol suppresses p, n, ρ, q, H , the sample vector $(y_s)_{s=0}^p$, and the normalized endpoint residual $\mathcal{E}$. Exact recovery in the polynomial model and the remainder bound are conditional on this externally supplied normalization and sample vector: the source intentionally leaves the precise nonperiodic subtraction inside $\mathcal{E}$ schematic. The resulting conditional theorem is proved in the tomography manuscript, not formalized there in Lean.

Accent and scripts. The hat means recovery of $A_j(\theta)$ by geometric radial interpolation. It is not a Fourier transform. The symbol owns only the subscript j .

\TomographicCoefficientEstimator: radial-angular Fourier-coefficient estimator

TeX interface. Arity 1, whose sole macro argument is the Fourier mode k ; the base form is $\hat{c}_k$. The source appends the external configuration superscript (d, p, n) , printing $\hat{c}_k^{(d,p,n)}$.

Definition and type. If

$$A_0(u) = \sum_{m \in \mathbb{Z}} c_m e^{2\pi i m u}, \quad \mathcal{R}_{0;p,n,\rho}(\theta) := [z^0] \sum_{s=0}^p y_s \ell_s(z),$$

where y_s, ℓ_s are the radial data in the preceding entry, then

$$\hat{c}_k^{(d,p,n;\rho)} := \frac{1}{d} \sum_{a=0}^{d-1} \mathcal{R}_{0;p,n,\rho}(a/d) e^{-2\pi i k a/d} \in \mathbb{C}.$$

The semicolon- ρ augmentation is catalogue-only disambiguation: the active source prints $\hat{c}_k^{(d,p,n)}$ and suppresses ρ . The theorem uses $d \geq 2$, $p \in \mathbb{N}_0$, $n \in \mathbb{N}_{>0}$, $k \in \mathbb{Z}$, $|k| \leq (d-1)/2$, and an integer $\rho > 1$ satisfying $\rho \equiv 1 \pmod{d}$.

Scope and status. The printed superscript records d, p, n but suppresses ρ , the radial samples, and the endpoint normalization. Under the stated uniform endpoint expansion and Wiener-strip hypothesis, the report proves

$$\hat{c}_k^{(d,p,n)} - c_k = \mathcal{O}(n^{-p-1}) + \mathcal{O}(\exp[-2\pi\sigma(d - |k|)])$$

as $n \rightarrow \infty$, uniformly over the grid phases a/d , with d, p, ρ , and the strip width σ fixed. It does not claim a Lean formalization.

Accent and scripts. The hat means an estimate of the already defined Fourier coefficient c_k . The finite Fourier transform occurs inside the construction, but the hat is not the Fourier transform of c_k . Because the macro owns only k as a subscript, the externally appended configuration superscript is currently safe; changing the macro to own a superscript would require a dedicated wrapper.

\GeometricProductExtrapolant: multiplicative q -Richardson product extrapolant

TeX interface. Arity 3, with arguments b, N, r in that order; it prints $\widehat{P}_{b;N,r}$. The semicolon separates the base from the two discrete indices. The evaluation variable t is external.

Definition and type. For real t satisfying

$$\frac{|t|}{b^{N+1}} < \pi, \quad P_{b,N}(t) \neq 0,$$

define

$$\widehat{P}_{b;N,r}(t) := \operatorname{sgn}(P_{b,N}(t)) \exp \left(\sum_{j=0}^r \lambda_{r,j}(b^{-2}) \log |P_{b,N+j}(t)| \right) \in \mathbb{R}^\times.$$

Equivalently,

$$\left| \widehat{P}_{b;N,r}(t) \right| = \prod_{j=0}^r |P_{b,N+j}(t)|^{\lambda_{r,j}(b^{-2})}.$$

Scope and status. The derived value $q = b^{-2}$ and t are not printed as indices. This is the real logarithmic construction away from prefix zeros, unlike the holomorphic factorized continuation. Its exact error series, sign, brackets, and fixed-parameter convergence are proved in the report; the associated analytic claims remain separate Lean obligations.

Accent and scripts. The hat means multiplicative extrapolation of $P_b(t)$, not Fourier transformation. The macro owns only a subscript, so the hybrid-correction superscript $[s]$ used in the source is syntactically safe and denotes correction depth, not differentiation.

\DyadicProductExtrapolant: dyadic multiplicative product extrapolant

TeX interface. Arity 2, with arguments N, r in that order; it prints $\widehat{P}_{N,r}$. The base $b = 2$ is fixed, and t is an external function argument.

Definition and type. With

$$P_N(t) := \prod_{k=1}^N \operatorname{sinc}_{\operatorname{rad}}(t/2^k),$$

define, whenever $|t|/2^{N+1} < \pi$ and $P_N(t) \neq 0$,

$$\widehat{P}_{N,r}(t) := \operatorname{sgn}(P_N(t)) \exp \left(\sum_{j=0}^r \lambda_{r,j}(1/4) \log |P_{N+j}(t)| \right) \in \mathbb{R}^\times.$$

Thus it is exactly the $b = 2, q = 1/4$ specialization of the preceding geometric estimator.

Scope and status. The base, $q = 1/4$, and t are suppressed. The target is the tail product $P_2(t)$, not the full Fourier product $\Phi(x) = \operatorname{sinc}_{\operatorname{rad}}(\pi x) P_2(\pi x)$. Explicit low-order formulas, error bounds, and the finite Thue–Morse realization are proved in the report. The raw finite complex-exponential identity and the sine/sinc Thue–Morse normalization are Lean-formalized in the complex-product bridge modules; the analytic Richardson estimator, error, bracket, and convergence claims are not.

Accent and scripts. The hat means dyadic q -Richardson extrapolation, not Fourier transformation. The symbol owns only the subscript N, r .

15.4.8 Phase-locked lower-Lambert estimators

Common phase-locked data.

Let $L = \log 2$. For $\lambda > 1/L$, set

$$x(\lambda) := \lambda 2^{-\lambda}, \quad \lambda(x) := -\frac{1}{L} W_{-1}(-Lx) \quad \left(0 < x < \frac{1}{eL}\right),$$

so that $\lambda(x(\lambda)) = \lambda$. The phase-locked nodes and weights are

$$x_j(\lambda) := (\lambda + j) 2^{-(\lambda+j)}, \quad \omega_{r,j}(\lambda) := \frac{(-1)^{r-j}}{r!} \binom{r}{j} (\lambda + j)^r \quad (0 \leq j \leq r).$$

They satisfy

$$\sum_{j=0}^r \omega_{r,j}(\lambda) = 1, \quad \sum_{j=0}^r \omega_{r,j}(\lambda) (\lambda + j)^{-m} = 0 \quad (1 \leq m \leq r).$$

With the endpoint theorem's fixed constant C_F , define

$$\mathcal{W}_{\text{MSE}}(x) := -\frac{L}{2} \lambda(x)^2 + \left(1 + \frac{L}{2}\right) \lambda(x) - \frac{1}{2} \log \lambda(x) + C_F, \quad Y(x) := \log F_{\text{bd}}(x) - \mathcal{W}_{\text{MSE}}(x).$$

For every fixed $M \in \mathbb{N}_0$, as $x \downarrow 0$ (equivalently $\lambda(x) \rightarrow \infty$), uniformly along a fixed compact phase set, the all-orders input has

$$Y(x) = \Psi(\lambda(x)) + \sum_{m=1}^M A_m(\lambda(x)) \lambda(x)^{-m} + \mathcal{O}(\lambda(x)^{-M-1}),$$

where Ψ and every A_m are one-periodic in λ .

`\PhaseLockedLogEstimator`: phase-locked logarithmic endpoint estimator

TeX interface. Arity 1, whose argument is the extrapolation order r ; it prints $\hat{\Psi}_r$. The lower-Lambert coordinate λ is an external function argument.

Definition and type. For $r \in \mathbb{N}_0$ and $\lambda > 1/\log 2$,

$$\hat{\Psi}_r(\lambda) := \sum_{j=0}^r \omega_{r,j}(\lambda) Y(x_j(\lambda)) \in \mathbb{R}.$$

The weights preserve the periodic constant term and cancel the first r inverse- λ corrections exactly, so the target is $\Psi(\lambda)$.

Scope and status. The printed form suppresses λ , the nodes x_j , the reduced observable Y , the main-saddle normalization C_F , and the weights. At each fixed r and finite residual depth, the logarithmic extraction and its first-omitted-term estimate are Lean-backed under the repository's all-orders endpoint input. No convergence of the infinite residual series and no uniformity in growing r are asserted.

Accent and scripts. The hat means a phase-locked reciprocal-scale estimate of the one-periodic correction $\Psi(\lambda)$, not a Fourier transform. The symbol owns only the subscript r .

\PhaseLockedExponentialEstimator: exponentiated phase-locked endpoint estimator

TeX interface. Arity 1, whose argument is the extrapolation order r ; it prints $\widehat{\mathcal{A}}_r$. The coordinate λ is external.

Definition and type. For $r \in \mathbb{N}_0$, $\lambda > 1/\log 2$, and the common nodes, weights, and reduced observable,

$$\widehat{\mathcal{A}}_r(\lambda) := \exp\left(\sum_{j=0}^r \omega_{r,j}(\lambda) Y(x_j(\lambda))\right) = \prod_{j=0}^r \left(\frac{F_{\text{bd}}(x_j(\lambda))}{\exp(\mathcal{W}_{\text{MSE}}(x_j(\lambda)))}\right)^{\omega_{r,j}(\lambda)} \in \mathbb{R}_{>0}.$$

Its exact target is the periodic endpoint amplitude $\mathcal{A}(\lambda) := \exp(\Psi(\lambda))$.

Scope and status. The printed form suppresses λ , x_j , Y , $\mathcal{W}_{\text{MSE}}$, and $\omega_{r,j}$. The displayed equality is exact exponentiation of the logarithmic estimator; the multiplicative estimator's Bell relative-error hierarchy is not part of the closed Lean phase-extraction tranche.

Accent and scripts. The hat means an estimated positive periodic amplitude, not Fourier transformation. The symbol owns only the subscript r .

\PhaseLockedCoefficientEstimator: phase-locked endpoint coefficient extractor

TeX interface. Arity 2, with arguments s, r in that exact order: s is the coefficient index and r is the interpolation order. It prints $\widehat{A}_s^{(r)}$. The coordinate λ is external.

Definition and type. For $\lambda > 1/\log 2$, set

$$h_j := (\lambda + j)^{-1}, \quad \ell_j(h) := \prod_{\substack{0 \leq k \leq r \\ k \neq j}} \frac{h - h_k}{h_j - h_k}.$$

For $r \in \mathbb{N}_0$ and $0 \leq s \leq r$,

$$\widehat{A}_s^{(r)}(\lambda) := \frac{1}{s!} \sum_{j=0}^r \ell_j^{(s)}(0) Y(x_j(\lambda)) \in \mathbb{R}.$$

The factor $1/s!$ makes this the coefficient of h^s in the interpolating polynomial, rather than its unnormalized s th derivative. Locally, $\widehat{A}_0^{(r)} = \widehat{\Psi}_r$; for $s \geq 1$, the target is the periodic endpoint coefficient $A_s(\lambda)$.

Scope and status. The printed symbol suppresses λ , the nodes, the observable Y , and the Lagrange basis. The extraction formula is exact finite interpolation, but the higher coefficient extractors remain a formalization target; only the logarithmic $s = 0$ tranche has the compiled fixed-order status described above.

Accent and scripts. The hat means coefficient reconstruction, not Fourier transformation. The built-in superscript (r) records interpolation order and is not a derivative. Any additional derivative, power, or semantic superscript must be attached to the grouped expression $(\widehat{A}_s^{(r)})$.

15.4.9 Tail and inverse-root approximations

\SimpleTailEstimate: first-order negative-power tail estimate

TeX interface. Arity 1, whose argument is the tail-start index N ; it prints $\widehat{R}_N$. The exponent parameter a is an external function argument.

Definition and type. Let X be an $[0, 1]$ -valued random variable with distribution function F_{bd} , put $d_n := \mathbb{E}X^n$, and, for $a > 0$, define

$$(a)^{\overline{k}} := \begin{cases} 1, & k = 0, \\ a(a+1) \cdots (a+k-1), & k \geq 1, \end{cases} \quad \binom{a+k-1}{k} = \frac{(a)^{\overline{k}}}{k!},$$

$$t_k(a) := \binom{a+k-1}{k} \frac{d_{k+1}}{k+1}, \quad R_N(a) := \sum_{k \geq N} t_k(a).$$

With $L = \log 2$, the simple hazard estimate is

$$\widehat{R}_N(a) := \frac{N t_N(a)}{\log N/L - (a - \frac{1}{2})}.$$

It is a real number for $N \in \mathbb{N}_{>0}$ whenever the denominator is nonzero, and has the intended positive-tail interpretation for N sufficiently large that the denominator is positive.

Scope and status. The printed form suppresses a , the moment sequence d_n , and the fact that the more precise conjectural hazard also contains $-\Psi'(\log_2 N)/L$. This is a heuristic first-order estimator: as $N \rightarrow \infty$ with fixed $a > 0$, the displayed numerical ratios are structural evidence, not certification, and the corresponding uniform tail asymptotic remains conjectural.

Accent and scripts. The hat means approximation of the positive series remainder $R_N(a)$, not Fourier transformation. The symbol owns only the subscript N .

\LambertRootApproximation: numerical Lambert-root approximation

TeX interface. Arity 0; it prints the ambient scalar $\widehat{w}$. The input x , branch k , numerical method, and iteration count are not TeX arguments.

Definition and domain. The exact target is

$$r = W_k(x), \quad re^r = x,$$

where

$$W_0 : [-1/e, \infty) \longrightarrow [-1, \infty), \quad W_{-1} : [-1/e, 0) \longrightarrow (-\infty, -1].$$

The value $\widehat{w} \in \mathbb{R}$ is any numerical approximation chosen on the same monotone real inverse interval as r ; the symbol itself fixes no iteration. If I is the closed interval joining r and $\widehat{w}$, I avoids -1 , and

$$m_I := \min_{t \in I} |e^t(1+t)| > 0,$$

then the ordinary residual gives

$$|\widehat{w} - r| \leq \frac{|\widehat{w}e^{\widehat{w}} - x|}{m_I}.$$

For $x \neq 0$, if I also avoids 0, define

$$G_x(w) := w + \log |w| - \log |x|, \quad \mu_I := \min_{t \in I} |1 + 1/t| > 0.$$

The logarithmic residual then gives the alternative certificate

$$|\widehat{w} - r| \leq \frac{|G_x(\widehat{w})|}{\mu_I}.$$

Scope and status. The branch and input are mandatory ambient data. Both residual inequalities are proved a posteriori theorems. Near the branch point their denominators correctly record the conditioning loss. At $x = -1/e$, where $r = -1$, the interval necessarily meets -1 , so neither certificate applies; the guide returns -1 exactly or changes to the signed square-root branch coordinate. No particular Newton, Halley, series, or rational approximation is privileged.

Accent and scripts. The hat means an approximate inverse root, not Fourier transformation. The symbol owns no subscript or superscript, but any later decoration must state whether it records branch, iteration, precision, or another numerical parameter.

15.5 Local alphabet register

The following roles are intentionally local because assigning them global commands would create more collisions than it removes. Each occurrence must nevertheless instantiate the declaration schema above.

Family	Mandatory local data
Polynomials p_n, P_n, A_n	Coefficient field, variable, degree, leading-coefficient normalization, orthogonality measure if any, and index origin.
Matrices A_N, H_N, T_N, G_N	Row and column index types, dimensions, scalar field, entry formula, and zero-size convention.
Kernels $K(x, y)$	Two domain spaces, scalar codomain, symmetry/Hermitian convention, regularity, and positive-definiteness status.
Measures μ, ν	Underlying measurable space, positivity/signedness, total mass, and density only if one exists.
Random variables X, Y, U_j	Probability space or a statement that only their joint law is relevant, codomain, independence/dependence, and distribution.
Generating functions $A(z), G(z), M(z)$	Ordinary, exponential, Dirichlet, or other series type; coefficient indexing; formal versus analytic interpretation; convergence domain.
Recurrences	Initial values, index range on which the recurrence applies, coefficient ring, and treatment of singular leading coefficients.
Optimization variables	Feasible set, topology/norm, objective codomain, existence and uniqueness status, and whether infima are attained.
Asymptotic phases and amplitudes	Limit variable and direction, uniform parameters, branch choices, order of truncation, and quantified remainder.
Frontier constructions	Complete proposed definition plus one of the status labels in section 1.3 ; a suggestive name does not confer a theorem.

16 Retired-alias index

This index is a reading aid for historical sources and a migration contract, not a second namespace. Names in the first column are printed through `\commandname` so that the retired control sequences are not executed. Migration is semantic: determine the old occurrence's type and normalization first, then choose the target. A textual replacement performed before that classification is invalid whenever the target column contains “or”.

Retired source name(s)	Canonical target	Required disambiguation
<code>\R</code>	<code>\RealNumbers</code>	Number field only; a radius remains a typed local variable.
<code>\C</code>	<code>\ComplexNumbers</code>	Number field only; constants and contours remain local.
<code>\Q</code>	<code>\RationalNumbers</code>	Number field only; do not rewrite a quantile, quadrature rule, or polynomial.
<code>\N</code>	<code>\NaturalNumbers</code> <code>\PositiveIntegers</code>	or Decide from whether zero belongs to the set; never append another raw subscript to the chosen command.
<code>\Z</code>	<code>\IntegerNumbers</code>	Number field only.
<code>\Up</code> , <code>\up</code> , <code>\upf</code>	<code>\RvachevUp</code>	The even compactly supported function, not the bounded or global Fabius function.
<code>\sinc</code>	<code>\SincRad</code> or <code>\SincPi</code>	Inspect the defining formula or zero set. Radian zeros are at nonzero multiples of π ; normalized zeros are at nonzero integers.
<code>\sincpi</code> , <code>\sincpiPartc</code>	<code>\sincp</code> , <code>\SincPi</code>	These source-local names were explicitly normalized by π .
<code>\sincPartx</code> , <code>\sincPartii</code> , <code>\sincPartxx</code> , <code>\sincPartll</code> , <code>\sincPartcc</code> , <code>\sincPartvvv</code>	<code>\sincPartl</code> , <code>\SincRad</code>	These generated-fragment aliases were classified from their defining formulas as radian sinc.
<code>\wt</code>	<code>\BinaryDigitSum</code>	Only for binary Hamming weight; a general graph or code weight remains local.
<code>\tm</code> , <code>\TM</code>	<code>\ThueMorseSignSymbol</code> <code>\ThueMorseSign</code>	or Use the symbol-only command in declarations and the argument-taking command in formulas.
<code>\e</code> , <code>\ee</code>	<code>\EulerE</code>	Only the exponential base; an error variable remains local.
<code>\ii</code>	<code>\ImaginaryUnit</code>	The complex unit, not an index.
<code>\dd</code>	<code>\Differential</code>	Upright integral differential; a derivative operator is declared separately.
<code>\Li</code>	<code>\Polylogarithm</code>	The special function, with its order printed as a subscript.
<code>\Var</code>	<code>\Variance</code>	Probability variance only.
<code>\Cov</code>	<code>\Covariance</code>	Probability covariance only.
<code>\E</code>	<code>\Expectation</code>	Expectation only; events and energies are local and semantically labelled.
<code>\Prob</code> , <code>\Pp</code>	<code>\Probability</code>	Probability measure, not polynomial or projection.
<code>\bigO</code> , <code>\Oh</code> , <code>\OO</code> , <code>\bigOPartl</code>	<code>\BigO</code> or <code>\BigOAt</code>	Prefer the explicit-regime command; retain the token-only form only when the same sentence quantifies the limit.

Retired source name(s)	Canonical target	Required disambiguation
<code>\smallo</code> , <code>\smalloPart1</code> , <code>\littleo</code> , <code>\littleoh</code> <code>\supp</code>	<code>\LittleO</code> or <code>\LittleOAt</code> <code>\SupportOperator</code> <code>\SupportOf</code>	Prefer the explicit-regime command and state uniformity parameters. Use the argument-delimited form in prose; distinguish pointwise, topological, measure, and essential support.
<code>\sgn</code> <code>\TV</code>	<code>\Sign</code> <code>\TotalVariationOf</code> <code>\TotalVariationDistance</code>	Real sign with value zero at zero. Count the mathematical arguments: one typed object means total variation; two probability measures mean the fixed distance.
<code>\dist</code> <code>\ord</code>	<code>\MetricDistance</code> <code>\EqualInLaw</code> <code>\OrderOperator</code>	Inspect the relation: a numeric metric distance is not equality in distribution. Multiplicative order, vanishing order, or another order operator only after the local arguments define which notion is meant.
<code>\lcm</code>	<code>\LeastCommonMultiple</code>	State the finite family and empty-family convention.
<code>\vtwo</code>	<code>\TwoAdicValuation</code>	State the value at zero or restrict to nonzero arguments.
<code>\qbinom</code>	<code>\GaussianBinomial</code>	Supply all three arguments: upper index, lower index, and base.
<code>\qpoch</code> , <code>\poch</code>	<code>\QPochhammer</code>	Supply element, base, and finite length or ∞ explicitly.
<code>\Law</code>	<code>\LawOperator</code> or <code>\LawOf</code>	Use the argument-delimited form for a random variable.
<code>\Lop</code>	<code>\LaplaceTransformSymbol</code> or <code>\LaplaceTransformOf</code>	Decide between an operator token and application; mark bilateral transforms separately.
<code>\abs</code>	<code>\AbsoluteValue</code>	Scalar absolute value or complex modulus; do not use for set cardinality.
<code>\norm</code>	<code>\Norm</code>	Declare the normed space and norm species.
<code>\floor</code> , <code>\ceil</code> <code>\pospart</code>	<code>\Floor</code> , <code>\Ceiling</code> <code>\PositivePart</code>	Preserve exact endpoint inequalities. Ordered-scalar positive part with value zero on nonpositive inputs.
<code>\set</code>	<code>\SetOf</code>	Supply the ambient type and predicate/list grammar.
<code>\angles</code>	<code>\InnerProduct</code>	Declare inner product versus dual pairing and the linear slot.
<code>\diag</code>	<code>\DiagonalOperator</code>	Matrix constructor only; diagonal extraction uses a separately named local operator.
<code>\Span</code> , <code>\spanop</code> <code>\Tr</code>	<code>\SpanOperator</code> <code>\TraceOperator</code>	State the scalar field. Matrix or trace-class operator trace; state the space.
<code>\rank</code> <code>\Res</code>	<code>\RankOperator</code> <code>\ResidueOperator</code>	Rank over the stated field. Complex residue; a named arithmetic sequence retains its semantic superscript.
<code>\Id</code> <code>\Log</code>	<code>\IdentityOperator</code> <code>\PrincipalLogarithm</code>	Identity map on a stated domain. Only the principal complex branch; a formal logarithm or another branch is separately labelled.
<code>\defeq</code>	<code>\DefinitionEquals</code>	Introduction of a definition, not a proved equality.

Retired source name(s)	Canonical target	Required disambiguation
<code>\Fglobal</code> , <code>\Fs</code> , <code>\extF</code> , <code>\Fext</code> <code>\InvF</code>	<code>\FabiusGlobal</code> <code>\FabiusQuantile</code> <code>\FabiusClampedQuantile</code>	Signed global Fabius extension only. or Decide from the input domain: $[0, 1]$ for the ordinary inverse, all reals for the clamped totalization.

The alias index deliberately does not assign global replacements to bare local letters, generic hats, primes, stars, or calligraphic alphabets. Those collisions require a type- and-scope decision, recorded in [section 15](#), rather than a lexical substitution.

17 Symbol and shared-command indexes

17.1 Rendered-symbol index

The rendered column is intentionally redundant with the semantic command name. The redundancy makes a PDF search by glyph useful while preserving a stable source-level identifier.

Rendered symbol family	Shared command(s)	Distinguishing data
2026 – 08 – 31	<code>\FabiusNotationVersion</code>	Date-valued snapshot stamp for the shared notation contract; it is meta-data, not a mathematical variable.
$\mathbb{N}_0, \mathbb{N}_{>0}$	<code>\NaturalNumbers, \PositiveIntegers</code>	Zero included versus excluded.
$\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}, \overline{\mathbb{R}}$	<code>\IntegerNumbers, \RationalNumbers, \RealNumbers, \ComplexNumbers, \ExtendedRealNumbers</code>	Algebraic number system versus ordered extension by two infinities.
$F_{\text{bd}}, F_{\text{gl}}, \text{up}$	<code>\FabiusBounded, \FabiusGlobal, \RvachevUp</code>	Bounded CDF, signed global extension, and even compactly supported density-shaped function.
$Q_F, Q_{F,\text{cl}}$	<code>\FabiusQuantile, \FabiusClampedQuantile</code>	Inverse on $[0, 1]$ versus clamped totalization on $\mathbb{R}$.
e, i, dx	<code>\EulerE, \ImaginaryUnit, \Differential</code>	Exponential base, complex unit, and integration differential.
$ x , \ x\ , A $	<code>\AbsoluteValue, \Norm, \CardinalityOf</code>	Scalar modulus, norm, and set cardinality; the types disambiguate identical bars.
$\lfloor x \rfloor, \lceil x \rceil, \{x\}, (x)_+$	<code>\Floor, \Ceiling, \FractionalPart, \PositivePart</code>	Integer rounding, fractional residue, and ordered positive part.
$\mathbf{1}_A, \{x : P(x)\}, \langle x, y \rangle$	<code>\IndicatorOf, \SetOf, \InnerProduct</code>	Function, set-builder delimiters, and inner-product/declared duality delimiters.
$:=, \text{sgn}, \text{id}$	<code>\DefinitionEquals, \Sign, \IdentityOperator</code>	Definition relation, real sign, and identity map.
$\text{span}, \text{tr}, \text{rank}, \text{diag}$	<code>\SpanOperator, \TraceOperator, \RankOperator, \DiagonalOperator</code>	Linear-algebra operators with an explicit field and space.
$\text{Res}, \text{lcm}, \text{ord}, \text{Log}$	<code>\ResidueOperator, \LeastCommonMultiple, \OrderOperator, \PrincipalLogarithm</code>	Complex residue, arithmetic LCM, qualified order species, and principal complex branch.
$(x)^{\underline{n}}, (x)^{\overline{n}}$	<code>\FallingFactorial, \RisingFactorial</code>	Unit-step falling versus rising products; both equal one at length zero.
$\text{supp}, \text{supp}(f), \text{tsupp}(f)$	<code>\SupportOperator, \SupportOf, \TopologicalSupportOf</code>	Operator token, pointwise support, and its topological closure.
$*, \text{dist}, \text{TV}(a), d_{\text{TV}}$	<code>\Convolution, \MetricDistance, \TotalVariationOf, \TotalVariationDistance</code>	Algebraic/measure convolution, generic metric, a one-object variation functional, and the fixed two-measure probability TV convention.
$\mathbb{P}, \mathbb{E}, \text{Var}, \text{Cov}$	<code>\Probability, \Expectation, \Variance, \Covariance</code>	Measure, integral operator, and second-order statistics.
$\text{Law}, \text{Law}(X), \stackrel{\text{law}}{=}, \stackrel{\text{law}}{\longrightarrow}$	<code>\LawOperator, \LawOf, \EqualInLaw, \ConvergesInLaw</code>	Law operator/application, law equality, and weak convergence of laws.
$\mathcal{A}_n$	<code>\DyadicSigmaField</code>	Level- n filtration σ -field with explicitly stated generators.
$\text{Obj}_{\text{decay}, n}, \text{Obj}_{\text{decay}, n}^{\text{lin}}$	<code>\DecayOptimizationObjective, \LinearizedDecayObjective</code>	Exact versus linearized Fourier-decay objectives.
$\mathcal{F}_{2\pi}, \mathcal{F}_{2\pi}^{-1}$	<code>\FourierTwoPiOperator, \InverseFourierTwoPiOperator</code>	Forward and inverse cycles-per-unit operators.

Rendered symbol family	Shared command(s)	Distinguishing data
$\mathcal{F}_{\text{ang}}, \mathcal{F}_{\text{ang}}^{-1}$	<code>\FourierAngularOperator,</code> <code>\InverseFourierAngularOperator</code>	Forward and inverse angular-frequency operators.
$\mathcal{F}_{2\pi}[f], \widehat{f}_{2\pi}$	<code>\FourierTwoPiTransformOf,</code> <code>\FourierTwoPi</code>	Bracketed operator application and hat rendering of the same 2π transform.
$\mathcal{F}_{\text{ang}}[f], \widehat{f}_{\text{ang}}$	<code>\FourierAngularTransformOf,</code> <code>\FourierAngular</code>	Bracketed operator application and hat rendering of the same angular transform.
$\widehat{h}_{\text{FS}}[k], \widehat{h}_{\text{FS}; T}[k]$	<code>\FourierSeriesCoefficient,</code> <code>\FourierSeriesCoefficientAtPeriod</code>	Unit-period and explicitly T -periodic normalized coefficients; neither is a continuous line transform.
$\mathcal{L}, \mathcal{L}[f]$	<code>\LaplaceTransformSymbol,</code> <code>\LaplaceTransformOf</code>	One-sided operator token and application; bilateral use is subscripted.
$\mathcal{M}, \mathcal{M}[f]$	<code>\MellinTransformSymbol,</code> <code>\MellinTransformOf</code>	Operator token and application on a stated convergence strip.
M_X, φ_X	<code>\MomentGeneratingFunction,</code> <code>\CharacteristicFunction</code>	Exponential moment transform and positive-sign probabilistic Fourier transform.
$\text{sinc}_{\text{rad}}, \text{sinc}_{\pi}$	<code>\SincRad,</code> <code>\SincPi</code>	Radian versus π -normalized entire sinc.
$\text{wt}_2, \varepsilon, \varepsilon_n$	<code>\BinaryDigitSum,</code> <code>\ThueMorseSignSymbol,</code> <code>\ThueMorseSign</code>	Bit count, reserved sequence glyph, and indexed sign.
ν_p, ν_2	<code>\PadicValuation,</code> <code>\TwoAdicValuation</code>	Prime-parametric and specialized two-adic valuations; zero convention local.
$(a; q)_n, \begin{bmatrix} n \\ k \end{bmatrix}_q, [n]_q$	<code>\QPochhammer,</code> <code>\GaussianBinomial,</code> <code>\QInteger</code>	Every base and finite/infinite length explicit.
$\text{Li}, W, \Gamma, \zeta$	<code>\Polylogarithm,</code> <code>\LambertW,</code> <code>\GammaFunction,</code> <code>\RiemannZeta</code>	Named special functions; order, branch, and continuation domains printed as applicable.
$\mathcal{O}, o, \mathcal{O}_a(g), o_a(g)$	<code>\BigO,</code> <code>\LittleO,</code> <code>\BigOAt,</code> <code>\LittleOAt</code>	Token-only and explicit-limit asymptotic operators.

17.2 Alphabetical shared-command index

The following is the complete source-level index for version 2026-08-31. Argument counts are part of the API; braces displayed in a command name are mathematical arguments, not optional typography.

Command	Arity	Normative entry
<code>\AbsoluteValue</code>	1	Absolute value, norm, distance.
<code>\BigO</code>	0	Asymptotic operators.
<code>\BigOAt</code>	2	Asymptotic operators.
<code>\BinaryDigitSum</code>	0	Binary digit sum.
<code>\CardinalityOf</code>	1	Semantic delimiters.
<code>\Ceiling</code>	1	Rounding and positive part.
<code>\CharacteristicFunction</code>	1	Characteristic functions.
<code>\ComplexNumbers</code>	0	Number systems.
<code>\ConvergesInLaw</code>	0	Law and weak convergence.
<code>\Convolution</code>	0	Convolution.
<code>\Covariance</code>	0	Expectation and moments.
<code>\DecayOptimizationObjective</code>	1	Decay objectives.
<code>\DefinitionEquals</code>	0	Semantic delimiters.

Command	Arity	Normative entry
\DiagonalOperator	0	Matrix operators.
\Differential	0	Calculus.
\DyadicSigmaField	0	Dyadic filtration.
\EqualInLaw	0	Law and weak convergence.
\EulerE	0	Elementary constants.
\Expectation	0	Expectation and moments.
\ExtendedRealNumbers	0	Number systems.
\FabiBounded	0	Bounded Fabius function.
\FabiClampedQuantile	0	Fabius inverses.
\FabiGlobal	0	Global extension.
\FabiNotationVersion	0	Contract version.
\FabiQuantile	0	Fabius inverses.
\FallingFactorial	2	Powers and factorials.
\Floor	1	Rounding and positive part.
\FourierAngular	1	Angular Fourier transform.
\FourierAngularOperator	0	Angular Fourier transform.
\FourierAngularTransformOf	1	Angular Fourier transform.
\FourierSeriesCoefficient	2	Fourier series.
\FourierSeriesCoefficientAtPeriod	3	Fourier series.
\FourierTwoPi	1	2π Fourier transform.
\FourierTwoPiOperator	0	2π Fourier transform.
\FourierTwoPiTransformOf	1	2π Fourier transform.
\FractionalPart	1	Rounding and positive part.
\GammaFunction	0	Gamma function.
\GaussianBinomial	3	Gaussian binomial.
\IdentityOperator	0	Maps and the identity map.
\ImaginaryUnit	0	Elementary constants.
\IndicatorOf	1	Indicators.
\InnerProduct	1	Semantic delimiters.
\IntegerNumbers	0	Number systems.
\InverseFourierAngularOperator	0	Angular Fourier transform.
\InverseFourierTwoPiOperator	0	2π Fourier transform.
\LambertW	0	Lambert (W).
\LaplaceTransformOf	1	Laplace transforms.
\LaplaceTransformSymbol	0	Laplace transforms.
\LawOf	1	Law and weak convergence.
\LawOperator	0	Law and weak convergence.
\LeastCommonMultiple	0	Arithmetic functions.
\LinearizedDecayObjective	1	Decay objectives.
\LittleO	0	Asymptotic operators.
\LittleOAt	2	Asymptotic operators.
\MellinTransformOf	1	Mellin transform.
\MellinTransformSymbol	0	Mellin transform.
\MetricDistance	0	Absolute value, norm, distance.
\MomentGeneratingFunction	1	Moment-generating functions.
\NaturalNumbers	0	Number systems.
\Norm	1	Absolute value, norm, distance.
\OrderOperator	0	Logarithm and order.
\PadicValuation	1	Valuations.
\Polylogarithm	0	Polylogarithm.
\PositiveIntegers	0	Number systems.
\PositivePart	1	Rounding and positive part.
\PrincipalLogarithm	0	Logarithm and order.
\Probability	0	Probability spaces.
\QInteger	2	(q)-integers.
\QPochhammer	3	(q)-Pochhammer symbols.

Command	Arity	Normative entry
\RankOperator	0	Matrix operators.
\RationalNumbers	0	Number systems.
\RealNumbers	0	Number systems.
\ResidueOperator	0	Residues.
\RiemannZeta	0	Zeta functions.
\RisingFactorial	2	Powers and factorials.
\RvachevUp	0	Rvachev up-function.
\SetOf	1	Semantic delimiters.
\Sign	0	Rounding and positive part.
\SincPi	0	Sinc normalizations.
\SincRad	0	Sinc normalizations.
\SpanOperator	0	Matrix operators.
\SupportOf	1	Support.
\SupportOperator	0	Support.
\ThueMorseSign	1	Thue–Morse sign.
\ThueMorseSignSymbol	0	Thue–Morse sign.
\TopologicalSupportOf	1	Support.
\TotalVariationDistance	0	Probability metrics.
\TotalVariationOf	1	Probability metrics.
\TraceOperator	0	Matrix operators.
\TwoAdicValuation	0	Valuations.
\Variance	0	Expectation and moments.

18 Lean crosswalk and formalization boundary

The crosswalk maps semantic mathematical objects to actual current Lean declarations. It does not imply that every identity printed in a frontier document has been formalized. A row marked “notation only” records precisely that no single global Lean object is asserted by the shared command. Lean names are case-sensitive and module paths are relative to Analysis/FabiusFunction/Lean/FabiusFunction/.

Catalogue object	Lean declaration and module	Exact boundary
F_{bd}	<code>fabiusReal</code> in <code>Basic.lean</code> ; canonical witness <code>fabius</code> and proof <code>fabius_spec</code> in <code>PaperStatements.lean</code>	<code>fabiusReal</code> is the real-valued view of a <code>BoundedFabius</code> ; the witness and <code>IsFabius</code> proof select the canonical object.
Abstract Fabius specification	<code>IsFabius</code> in <code>Basic.lean</code>	The declaration’s actual type is authoritative for hypotheses; the catalogue summary does not manufacture a weaker premise-free theorem.
up	<code>rvachevUp</code> in <code>Basic.lean</code>	Takes a <code>BoundedFabius</code> argument. Evenness, bounds, and other properties are separate theorems, not fields silently attached to an arbitrary function.
F_{gl}	<code>extendedFabius</code> in <code>Basic.lean</code>	Takes a <code>BoundedFabius</code> ; translate-cell and calculus results occur in later dedicated modules.
Q_F and $Q_{F,\text{cl}}$	<code>fabiusOrderIso</code> and <code>fabiusInv</code> in <code>FabiusInverse.lean</code>	The formal definitions take both a bounded object and its <code>IsFabius</code> proof. Domain behavior must be checked against the exact declaration before claiming ordinary versus clamped inverse laws.
F_n^{spl} and its local inverse G_n^{spl}	<code>reportFiniteFabiusApproximant</code> in <code>QuarterSplineLocalPolynomial.lean</code> ; <code>fabiusUniformSpline</code> in <code>FabiusUniformSpline.lean</code>	The exact index bridge identifies the report approximant at n with the uniform spline at $n - 1$. Strict monotonicity is proved only for $n \geq 3$ on $[1/4, 1/4 + 2^{-n}]$. No global G_n^{spl} is defined in Lean; the quantile theorem assumes a supplied local left inverse and is not <code>fabiusInv</code> .
Φ_{ang} and $\Phi_{2\pi}$	<code>centeredSincProduct</code> in <code>FourierLaplaceRotation.lean</code> ; <code>rvachevFourierProduct</code> in <code>Basic.lean</code> ; bridge <code>centeredSincProduct_eq_rvachevFourierProduct</code>	The bridge is $\text{centeredSincProduct}(z) = \text{rvachevFourierProduct}(z/(2\pi))$. Identifying the product with the integral transform uses <code>rvachevFourier_eq_product</code> and retains $F : \text{BoundedFabius}$ and $h_F : \text{IsFabius } F$.
Endpoint kernel $\mathcal{K}(z, w)$	Notation only	Formal frontier construction in the two-scale endpoint coefficient algebra. No exact current Lean definition of $\mathcal{K}$, $\mathcal{R}$, or its fixed-point algebra is claimed.

Catalogue object	Lean declaration and module	Exact boundary
ε_n	thueMorseSign in Arithmetic.lean	Integer-valued sign sequence on Nat; digit, recurrence, and product identities live in their named Thue–Morse modules.
Prime-power Pascal-row valuations	primePowerChoose_padicValNat_add, primePowerChoose_padicValNat, and twoPowChoose_padicValNat in PrimePowerBinomialValuation.lean	Arbitrary-prime formulas include $j = p^m$ and $m = 0$; the dyadic theorem uses $0 < j < 2^m$. No theorem for the distinct endpoint-flat weights $\binom{2^m-2}{j-1}$ is inferred.
Exact full moments	moment in Arithmetic.lean; momentIntegral in Basic.lean	Rational recurrence and real integral are distinct objects; AnalyticMoments.lean supplies bridge theorems with hypotheses.
Exact half moments	halfMoment in Arithmetic.lean; halfMomentIntegral in Basic.lean	Rational recurrence and real integral are not identified by notation alone.
R_n^{Res}	reshetnikov in Arithmetic.lean	The Lean value is rational at the definition boundary; integrality, positivity, and oddness require their named theorems.
D_n^{dyad}	dyadicDenominator in Arithmetic.lean	A natural number definition; bounds and divisibility results occur in DenominatorBound.lean and related modules.
Number systems and elementary algebra	Mathlib types Nat, Int, Rat, Real, Complex and their standard APIs	Printed coercions are mathematical exposition. Lean coercions and side conditions remain those required by the elaborated theorem.
Intervals	Set.Icc, Set.Ioo, Set.Ico, Set.Ioc	Endpoint order in the Lean constructor matches the four catalogue interval conventions.
Probability laws and pushforwards	Mathlib Measure.map and probability-measure APIs	A prose random-series representation is not Lean-proved unless a project theorem states the corresponding measure equality.
Fourier commands	Mathlib Fourier-transform APIs plus project-specific theorems in dedicated Fourier modules	The shared commands choose print normalization. They are not names of one new Lean definition and do not erase Mathlib’s hypotheses.
Finite-node Lagrange–Rvachev decoder	lagrangeRvachevDecoder and lagrangeRvachevAtomCoefficient in LagrangeRvachevSynthesis.lean	Exact synthesis uses $M \neq 0$, $ s - 1 \leq \nu_2(M)$, and $x \in [-1, 1]$. Node injectivity is required for cardinal evaluation, not polynomial reconstruction; row-sum normalization also requires a nonempty node family.
Shifted dyadic polynomial self-sampling	integral_polynomial_mul_rvachevUp_eq_dyadic_tsum in PolynomialCombExactness.lean	Exact for every polynomial of degree at most N , every real shift, and a bounded Fabius object satisfying IsFabius. Compact support makes the displayed bilateral sum finite; exactness failure at degree $N + 1$ is not claimed.

Catalogue object	Lean declaration and module	Exact boundary
Legendre Gaunt and zero-row Wigner square	<code>legendreGauntAdmissible</code> and <code>legendreWignerThreeJZeroSqRat</code> in <code>LegendreGauntClosedForm.lean</code> ; finite Rvachev Gram sums in <code>FabiusLegendreGauntClosedForm.lean</code>	Only the total rational square for natural-number indices and zero magnetic row is defined. No signed symbol, phase, half-integer/nonzero-magnetic theory, or general $3j/6j/9j$ API is claimed; the real Gram formula retains <code>IsFabius</code> .
$\mathcal{A}_n$	Notation only	Each document declares its generators and probability space; no single project-wide Lean filtration is asserted by the command.
$\text{Obj}_{\text{decay},n}$ and $\text{Obj}_{\text{decay},n}^{\text{lin}}$	Notation only	They name document-local typed functionals in the Fourier-decay frontier. No general formalization or equality between exact and linearized objectives is claimed.
Generic q -series, fractional, matrix, spectral, and frontier entries	Definition or classical notation unless a row or document cites an exact declaration	Their presence in this encyclopedic catalogue is not evidence that the general theory or a proposed Fabius specialization is formalized.

The safe direction of use is Lean-to-prose: inspect a declaration and then state exactly that theorem with all hypotheses. Similar names, compiled object files, an earlier audit, or a catalogue entry are not substitutes for inspecting the current declaration.

19 Active-source manifest and reproducibility

19.1 Snapshot boundary

The inventory below is the recursive active tree observed on 2026-08-31 at the source-validation checkpoint. It excludes every path below docs/archive/ and counts this catalogue exactly once. It contains 160 mathematical content sources: 59 standalone documents and 101 included or generated fragments. The separate shared contract docs/fabius-notation.tex raises the total before this catalogue to 161 active TeX files. Including this standalone catalogue gives 162 active TeX files and 60 standalone documents. “Standalone” means that the uncommented source contains \documentclass; every other content source is a fragment inheriting its root preamble.

Manifest group	Standalone	Fragments	Total
Fabius_Function_and_Rvachev_Up/	1	0	1
fabius_lean_walkthrough/	1	0	1
FabiusFunction_Mathematical_Glossary/	1	0	1
papers/	7	0	7
Frontier synthesis root	1	0	1
drafts/exponents-and-q-series/	11	16	27
drafts/frontier-compilations/	3	1	4
drafts/integration-and-transforms/	1	7	8
drafts/inverse-and-sampling/	8	37	45
drafts/lambert-w/	1	4	5
drafts/representations/	12	26	38
drafts/rvachev_up_fourier_decay/	1	0	1
drafts/spectra-and-arithmetic/	10	10	20
drafts/thue-morse/	1	0	1
Mathematical content subtotal	59	101	160
Shared notation contract	0	1	1
This catalogue	1	0	1
Active TeX total including catalogue	60	102	162

The frontier paths in the table are all below semi-formalized-research-frontiers/; the root row is its standalone synthesis. Counts describe files, not distinct mathematical claims or publication units. Corrected and source-faithful historical-paper variants are counted separately because each is an active TeX source requiring an interpretable notation namespace.

19.2 Reproducible inventory and strict gate

The repository audit is invoked from the repository root as follows.

```
python Analysis/FabiusFunction/scripts/tex_notation_audit.py --strict --list
python Analysis/FabiusFunction/scripts/tex_notation_audit.py --semantic --strict --list
```

For each active source it records the relative path, byte and line counts, SHA-256 digest, standalone/fragment classification, canonical-input count, input targets, and lexical command counts. It excludes the archive by resolved path, not by a fragile filename pattern. A focused catalogue check supplies this file’s path as the positional argument.

The strict gate detects retired commands, local redefinitions of shared semantic commands, missing or duplicated canonical inputs, fragment imports of the shared input, selected literal bypasses of semantic operators, and known script-composition hazards. The opt-in semantic gate additionally requires classification of raw hats, delimiter spellings, number systems, indicator symbols, digit sums, asymptotic operators, sinc normalizations, and probability operators. Its sole raw-hat exemption is a

descriptive one-line local macro definition whose immediately preceding `LOCAL-NON-FOURIER-HAT:` comment names that macro and states its meaning; direct raw-hat use remains an error. The audit is deliberately not a TeX parser and cannot infer the meaning of every one-letter variable. The local declaration schema and collision register close that semantic gap. On any scan whose resolved file set is the complete active corpus, the audit also compares valid marker/declaration pairs with the reserved `\localaccentname` metadata in this catalogue and rejects missing, extra, or duplicate rows. The dedicated metadata token is not available for examples or literal listings. Rendered-PDF inspection later closes the composition and layout gap.

20 Completeness and maintenance contract

At shared-contract version 2026-08-31:

- `docs/fabius-notation.tex` defines exactly 91 shared commands.
- Every one of those 91 command names occurs in the canonical quick reference and in the alphabetical command index, with its exact arity.
- Every command belongs to a detailed normative entry defining its type-level role, normalization, exceptional cases, or the local data the command intentionally requires at each use.
- Both line-Fourier conventions, unit-period and explicit-period Fourier-series coefficients, both sinc normalizations, both Fabius inverses, the bounded/global/up-function triad, zero-inclusive/positive naturals, exact/linearized decay objectives, and pointwise/topological support have distinct printed forms.
- The retired-alias index is exhaustive with respect to the audit's retired-command registry. Ambiguous aliases map only after semantic classification.
- The active corpus has 35 validated `LOCAL-NON-FOURIER-HAT`: marker/declaration pairs representing 33 distinct document-local command names. Every distinct name occurs exactly once in the exhaustive accent register, and the whole-corpus audit rejects missing, extra, or duplicate registry metadata.
- Symbols not promoted to the shared API remain governed by the local declaration schema; they have no implicit corpus-wide type or normalization.

A future shared-command change is incomplete unless it updates, in the same logical change, the version stamp, quick reference, detailed entry, rendered-symbol index, alphabetical index, alias mapping if applicable, audit registry, and at least one focused strict-audit result. A future source that introduces a genuinely new mathematical convention updates the relevant encyclopedic entry even when no new shared command is needed. A convention found only in an explicit historical quotation is entered in the retired-alias register, not revived as normative project notation.

This catalogue is a notation contract and an interpretive reference. It does not convert frontier assertions into theorems, does not weaken hypotheses found in Lean, and does not substitute lexical or rendered checks for mathematical verification. Its promise is narrower and exact: every symbol has a declared semantic route, every global command has one normative meaning, and every unavoidable local ambiguity has an explicit scope rule.
